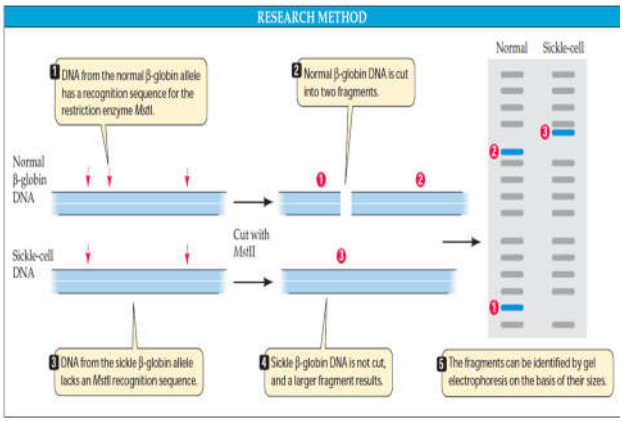
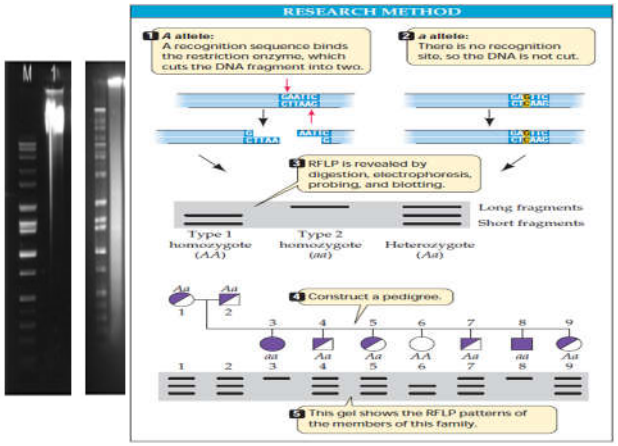
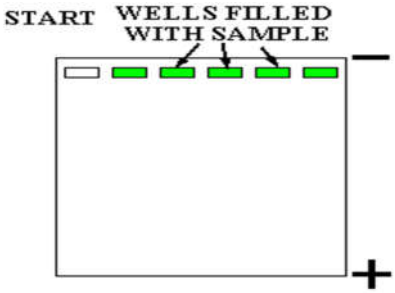
Lecture 81





Restriction Fragment Length polymorphisms (RFLP)
Can reveal mutations in DNA which may indicative of disease.

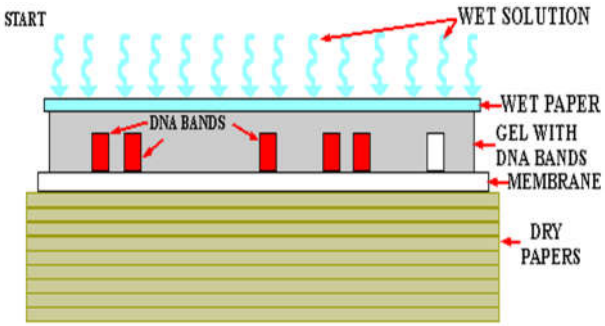


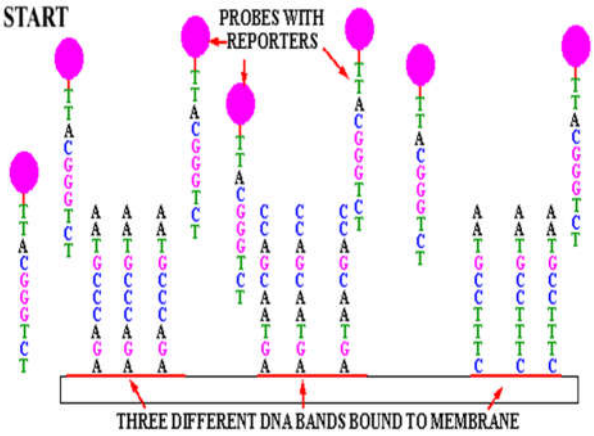
Cut Gel Pic

END

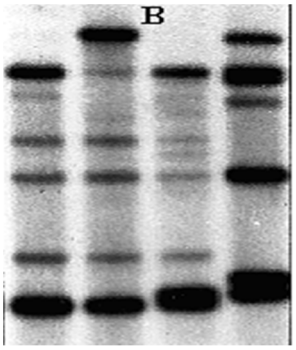
Lecture 82

DNA Transfer:





DNA Bound to Probe Exposed to X-ray film



END

Lecture 83

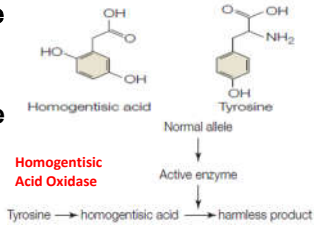
Genetic Information is Converted into Protein:

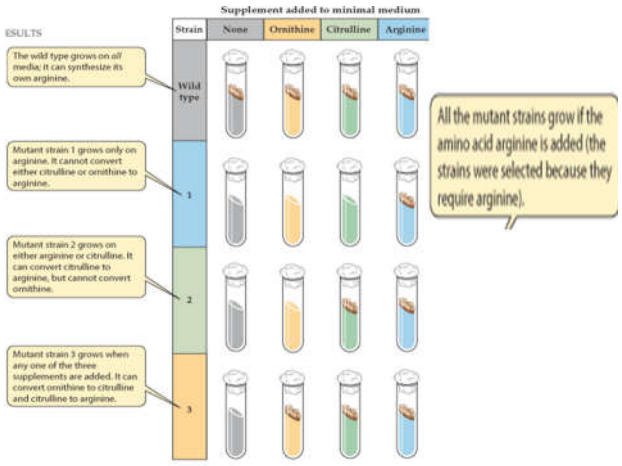
Alkaptonuria ("black urine")

The molecular basis of phenotypes was actually discovered before it was known that DNA was the genetic material.

He linked the biochemical phenotype of the disease to an abnormal gene and a missing enzyme.

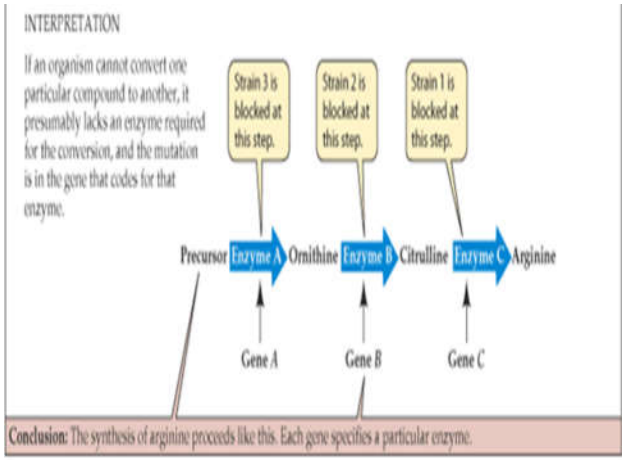
Most common in children whose parents were first cousins 12.5%



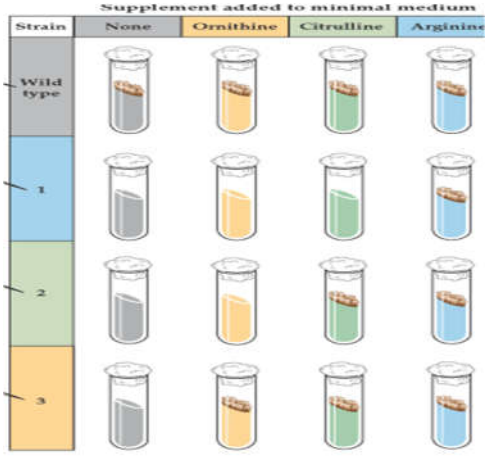


Neurospora:

- An altered gene resulted in an altered phenotype, associated with an altered enzyme.
- Neurospora haploid (n) recessive alleles.
- X-rays, which act as a mutagen prototrophs (original) converted to auxotrophs (increase).
- Some mutant strains could no longer grow on the minimal medium. One group could grow if supplemented with the arginine (*arg* mutants).



- *arg* mutants were grown in presence of various compounds *suspected intermediates* in the synthetic metabolic pathway for arginine,
- B & T classified each mutation as affecting one enzyme or another.
- wild-type and mutant cells examined for enzyme activities.
- results confirmed: Each mutant strain was indeed missing a single active enzyme in the pathway.

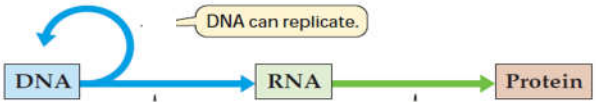


END

Lecture 84

The Dogma:

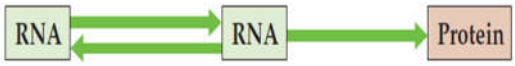
Info flows from DNA to RNA to proteins.



How does genetic information get from the nucleus to the cytoplasm?

What is the relationship between a specific nucleotide sequence in DNA and a specific amino acid sequence in a protein?)

Influenza virus, & poliovirus



Transcribing from RNA to RNA

HIV



make a DNA copy of their genome

END

THE MESSENGER HYPOTHESIS AND TRANSCRIPTION:

RNA molecule forms complementary copy of one DNA strand, moves to cytoplasm serves as template for protein synthesis.

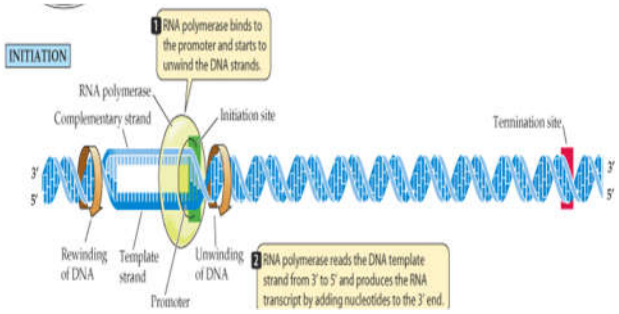
THE ADAPTER HYPOTHESIS AND TRANSLATION:

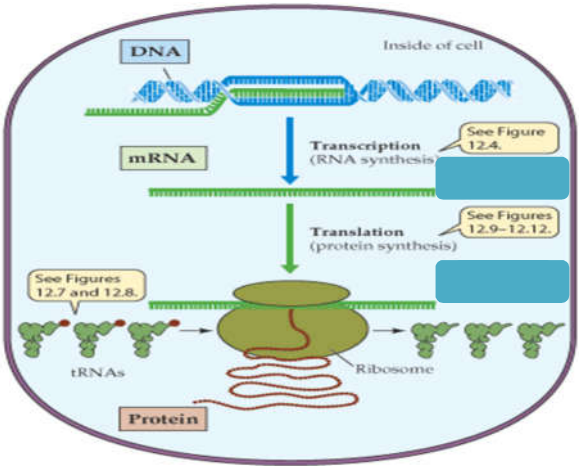
Adapter molecule binds a specific amino acid with one region and recognize a sequence of nucleotides with another region to translate the language of DNA into the language of proteins

Lecture 85

Transcription:

Initiation begins at promoter -tightly binds RNA Pol

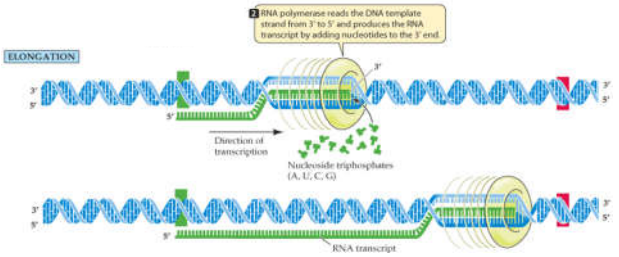




1) Where to start transcription.
2) Which strand to read.
3) The direction to take start from.
   Part of promoter is initiation site (transcription start).

RNA Pol moves 3' to 5', synthesis 5' to 3', ~~primer~~

Unwinds 20 bp at a time.

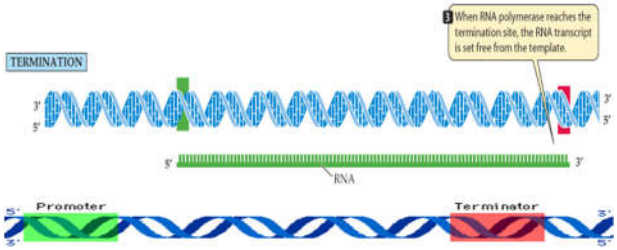


Transcript antiparallel to DNA template strand.

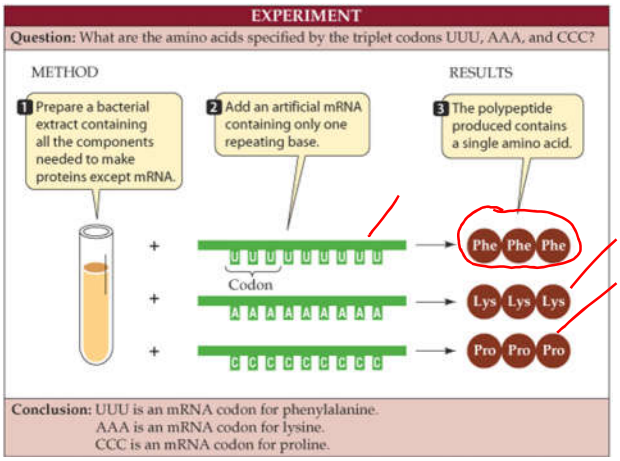
No proof reading error rate 1 in 10,000-100,000

Translation starts before termination in prokaryotes.

Pre-mRNA 1st product of transcription in eukaryotes.



END



Lecture 86

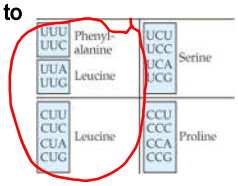
Genetic Code:

Three mRNA nucleotides form a codon:

which is complementary & antiparallel to corresponding DNA.

Genetic code relates codons to their specific AA.

~ Universal & Redundant but not ambiguous some exceptions.
Start & Stop



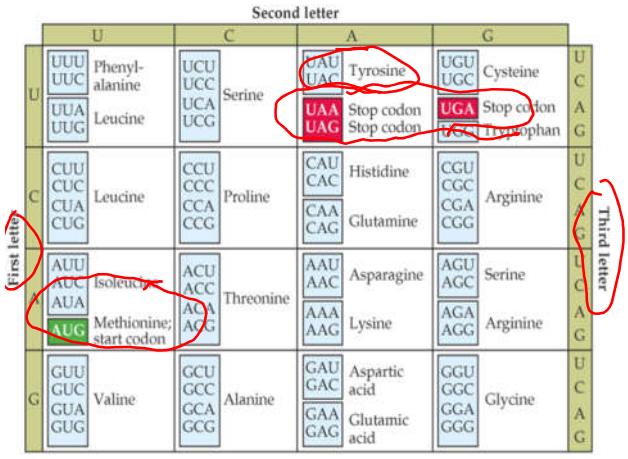
tRNA Translates Nucleotide Language into AA Language

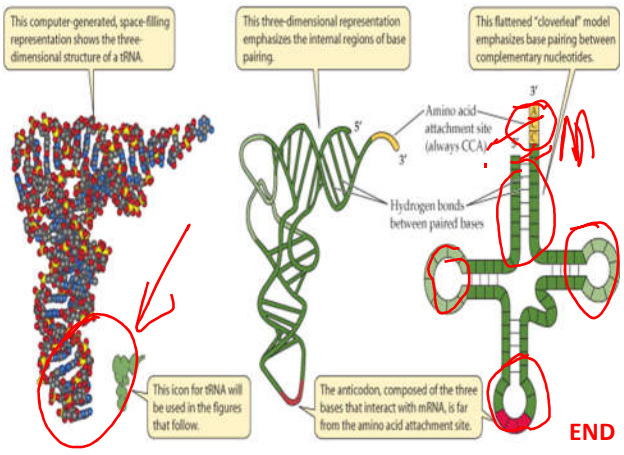
1) Carries AA
2) Associates with mRNA
3) Binds ribosome

tRNA has 75-80 nucleotides with internal H-bonding.

Conformation of tRNA allows it to bind ribosome.

3' end of tRNA (CCA) is attached to AA & mid point has anticodon (antiparallel).



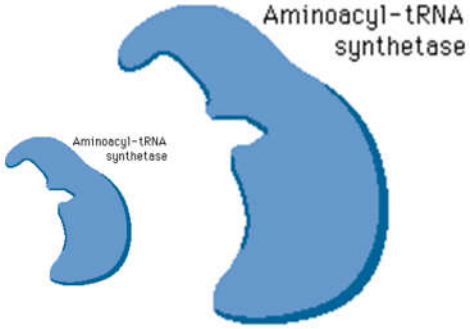


END

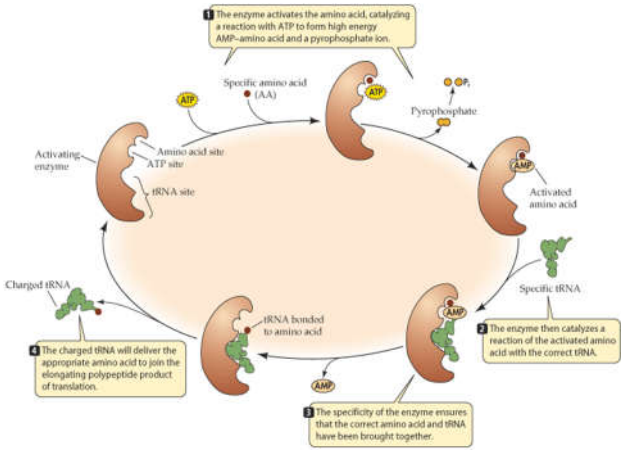
Lecture 87

Charging tRNA:

Specific for one AA & its corresponding tRNA.
Enzyme has three active sites for three molecules.
amino acid + ATP → aminoacyl-AMP + PPi
aminoacyl-AMP + tRNA → aminoacyl-tRNA + AMP
-COOH of AA is attached to 3' end of tRNA's free OH on ribose.
This high energy bond provides energy for peptide bond synthesis (2nd G code).



END

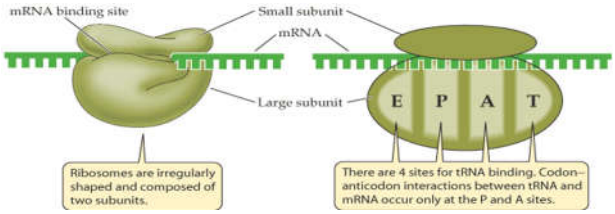


Lecture 88

Translation Workbench:

Two Subunits

Large subunit = 3 rRNAs + 45 poly peptides.
Small subunit = 1 rRNA + 33 poly peptides.
Ionic & hydrophobic forces hold complex together.
Complex self assembles.



Amino acid/tRNA recognized?

1) Took the cys-tRNA molecule and chemically modified the cysteine, converting it into alanine.
2) Hybrid charged tRNA was put into a protein synthesizing system.
3) Everywhere in the synthesized protein where cysteine was supposed to be, alanine appeared instead.
4) Protein synthesis machinery recognizes the anticodon of the charged tRNA, not the amino acid attached to it.



Ribosomes:

- A ribosome can use any mRNA and all species of charged tRNAs, and thus can be used to make many different polypeptide products.
- The ribosomes of prokaryotes Have different ribosomal proteins and RNAs.
- Antibiotics that target the prokaryotic ribosome (tetracycline) can be used to kill bacteria without harming a patient's cells.
- Mitochondria and chloroplasts also contain ribosomes, some of which are similar to those of prokaryotes

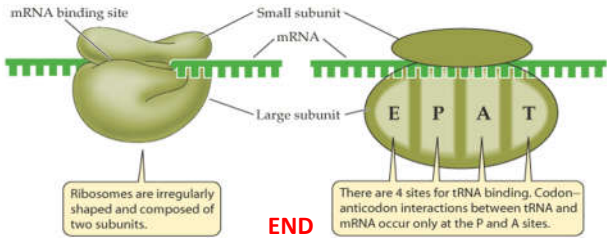
Large Subunit

T (transfer) site: first site where charged tRNA lands with the help of transfer factor

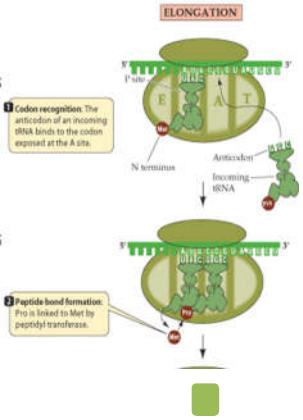
A (AA) site: Anticodon binds to mRNA codon,

P (polypeptide) site: tRNA adds AA to polypeptide ,

E (exit) site: after AA transfer tRNA exits here.



END



1) A charged tRNA with anticodon to next codon binds A site.

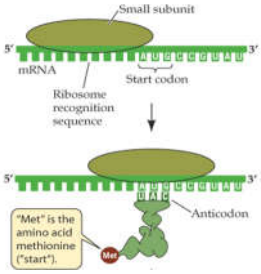
2) Large subunit breaks bond b/w tRNA & its AA at P site.

3) Assisted by elongation factors large subunit Catalyzes formation of peptide bond b/w the AA & the one attached to tRNA in A site (peptidyl transferase activity).

Lecture 89

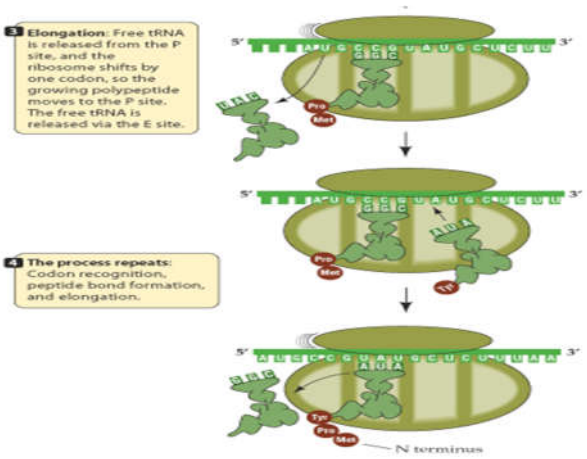
Translation:

1) In prokaryotes, the rRNA of the small ribosomal subunit first binds to a complementary ribosome binding site upstream start codon (AGGAGG; known as the Shine–Dalgarno sequence) on the mRNA.



2) Anticodon of methionine charged tRNA binds start the codon.

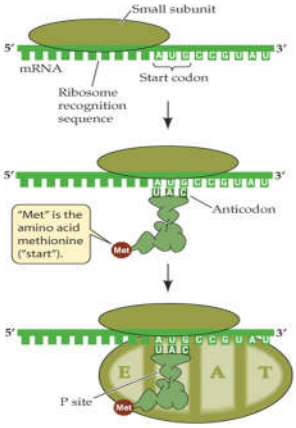
mRNA 5′ . . . A G G A G G. . . (start codon) . . . 3′
rRNA 3′ . . . U C C U C C. . . (P site) . . . 5′

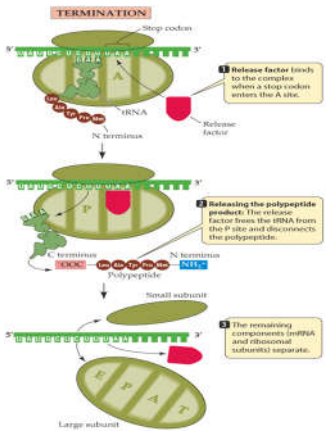


3) Mehtionine can be removed after translation.

4) rRNA in small subunit stabilizes H-bonding between t & m RNA.

5) Large subunit facilitated by initiation factors binds the complex with methionine charged tRNA at P site.

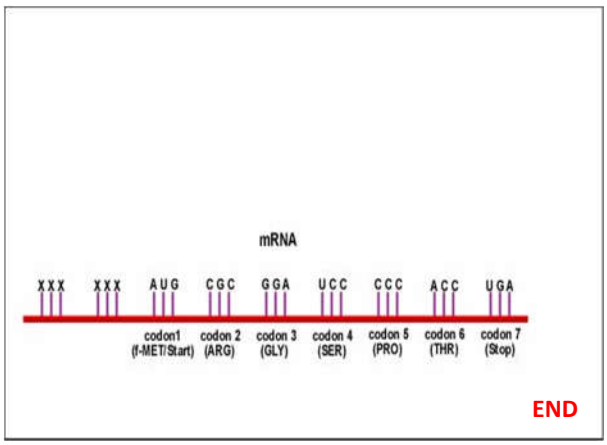




1) When stop codon enters A site elongation is terminates.

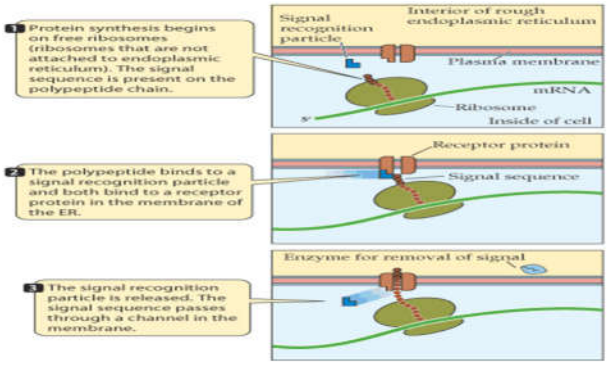
2) Stop codons bind release factor, which hydrolysis bond between polypeptide chain and tRNA at P site.

3) Protein folding information is a function of AA sequence.



END

If a polypeptide carries a particular signal of about 20 hydrophobic amino acids at its N terminus, it will be directed to the rough endoplasmic reticulum (ER)

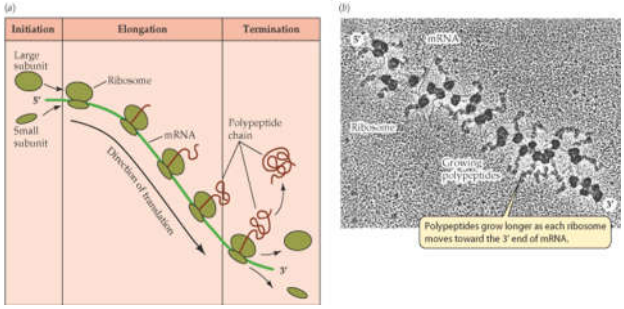


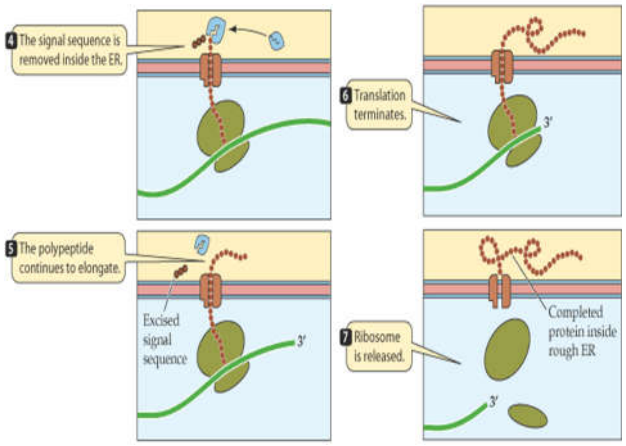
Lecture 90

Protein Targeting:

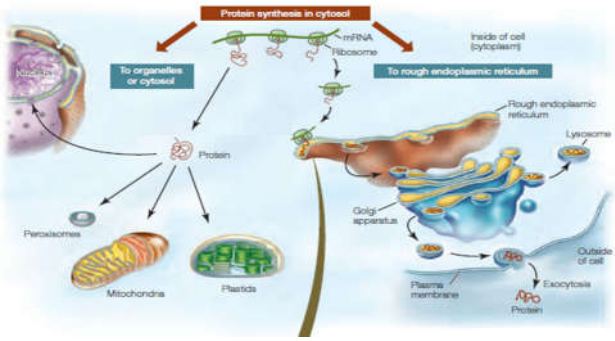
Polyribosome/ Polysome

Simultaneous Transcription and Translation

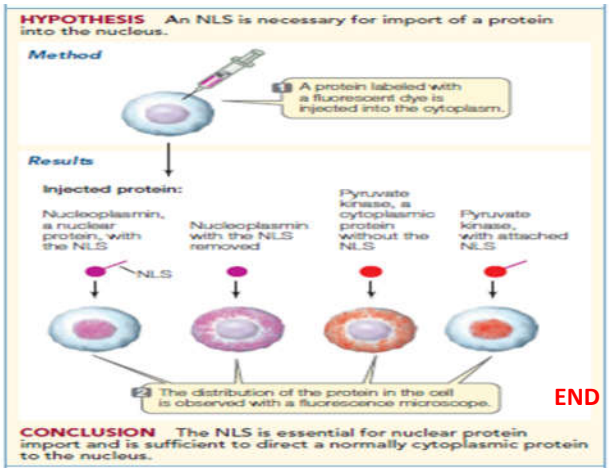




Compartmentalization



signal sequence—a short stretch of amino acids that indicates where in the cell the polypeptide belongs



END

Lecture 91

Mutations:

Heritable Changes in Genetic Information:

Somatic mutations : Occur in somatic (body) cells & passed on to the daughter cells after mitosis but not passed on to the children.

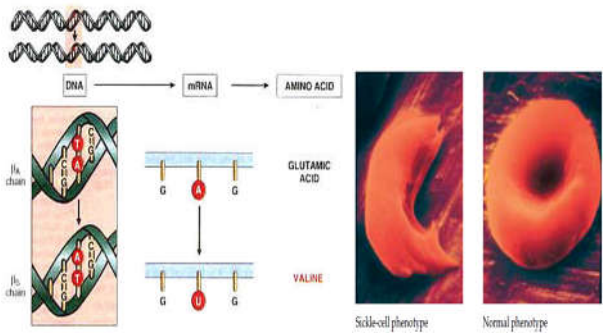
Germ line mutations: Occur the germ line cells that give rise to gametes. A gamete with the mutation passes it on to embryo after fertilization.

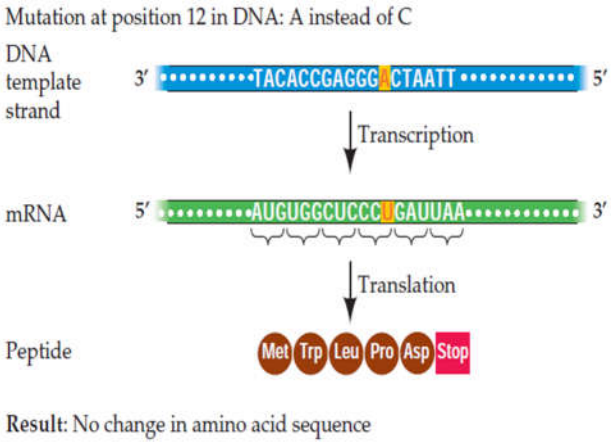
Point mutations: mutations of single base pairs.

Chromosomal mutations: change the position or orientation of a DNA segment.

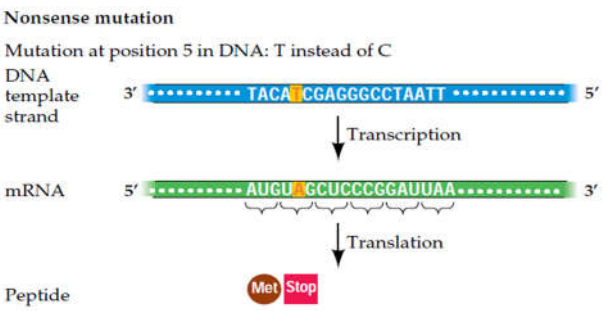
Silent Mutations: Mutations result in no change in amino acids due to redundancy genetic code.

Sickle-cell disease results from a defect in human β-globin gene, a protein in human red blood cells that carries oxygen.

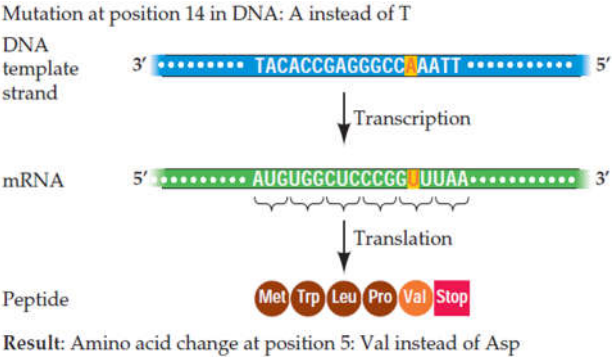




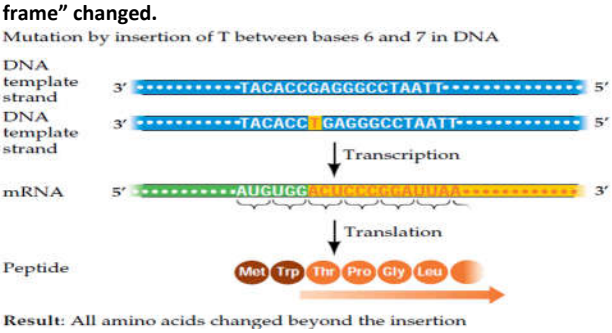
Nonsense mutations: Base substitution results in stop codon, (UAG) to form in the mRNA product:



Missense Mutations: Base substitution mutations that change the genetic message i.e. one amino acid substitutes for another in the protein.



Frame-Shift Mutations: No substitutions. Single base inserted or deleted from DNA. Genetic message out of register "reading frame" changed.



Good Mutations , Advantages (βs),Diversity , Evolution

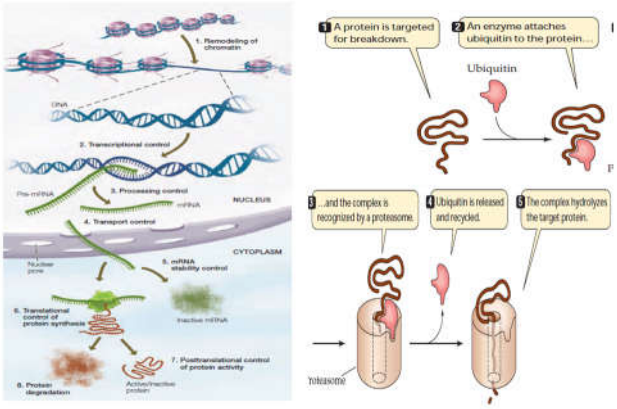
Lecture 92

Transcription Regulation:

Gene Categories

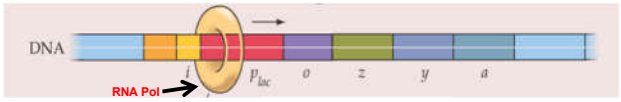
- Some of genes are expressed in all cells all the time. Housekeeping genes are responsible for the routine metabolic functions (e.g. respiration).
- Some are expressed as a cell enters a particular pathway of differentiation.
- Some are expressed all the time in only those cells that have differentiated in a particular way.
- Some are expressed only as conditions around and in the cell change. For example, the arrival of a hormone may turn on/off certain genes in that cell.

Epigenetics is the study of heritable changes in gene expression that do not involve changes in the DNA sequence.

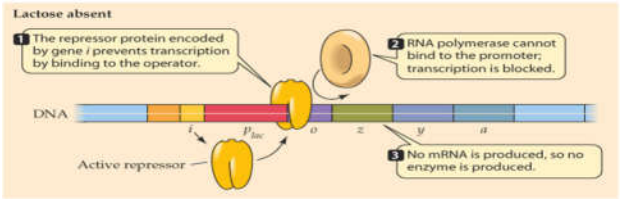


14.22 The Proteasome Breaks Down Proteins Proteins targeted for breakdown are bound to ubiquitin, which "leads" them to the proteasome, a complex composed of many polypeptides.

The lac operon contains a promoter, to which RNA polymerase binds to initiate transcription, and an operator, to which a repressor protein can bind.

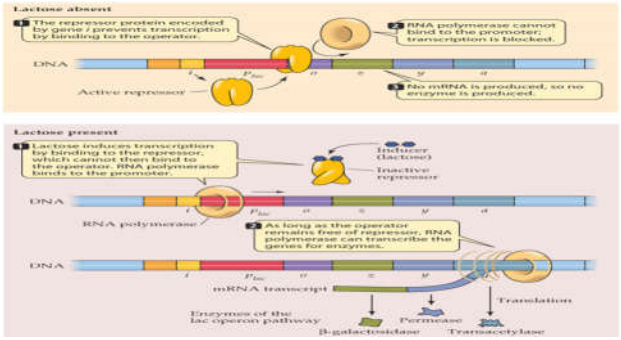


Repressor is bound, transcription is blocked.



E. Coli:

The lac operon contains a promoter, to which RNA polymerase binds to initiate transcription, and an operator, to which a repressor protein can bind.

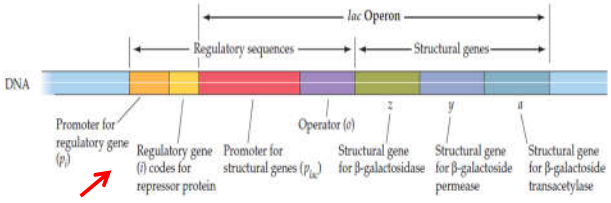


Lecture 93 & 94

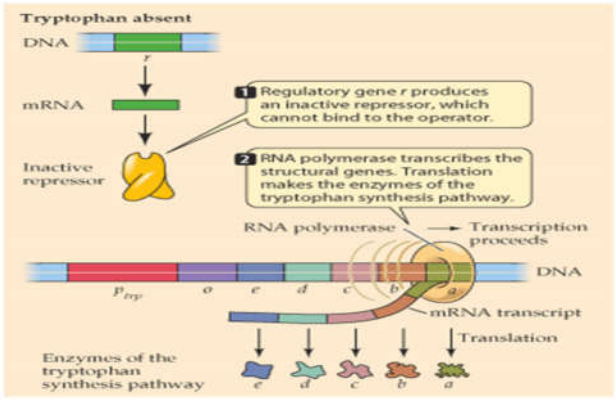
Transcription Regulation in E. Coli:

François Jacob & Jacques Monod (Nobel Prise)

Cluster of genes with 1 promoter is called operon
Operon consists of a promoter, an operator, and two or more structural genes.
The operator lies between the promoter & structural genes binds regulatory proteins that either activate or repress transcription.

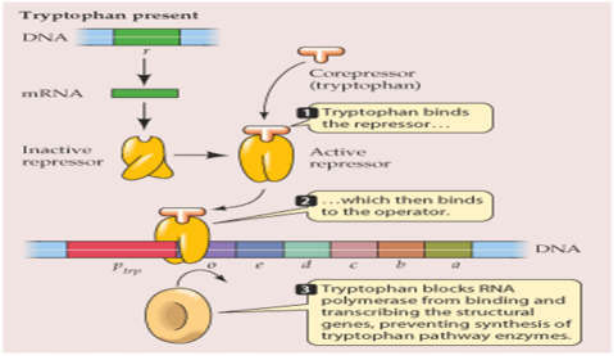


Repressible System

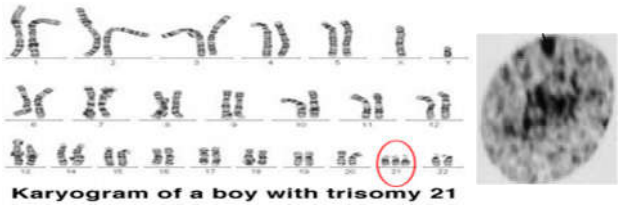


Negative Control of Transcription:

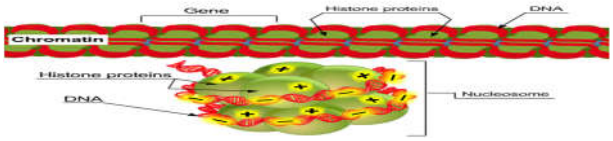
The regulatory molecule (the repressor) prevents transcription.



Eu & Hetero Chromatin

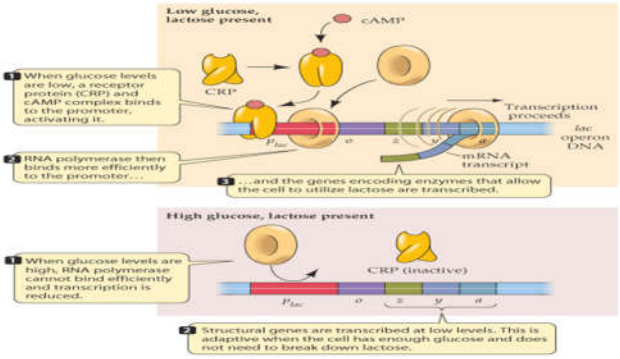


Most of DNA is highly ordered & tightly coiled around positively charged histone proteins & is virtually inaccessible to RNA pol.

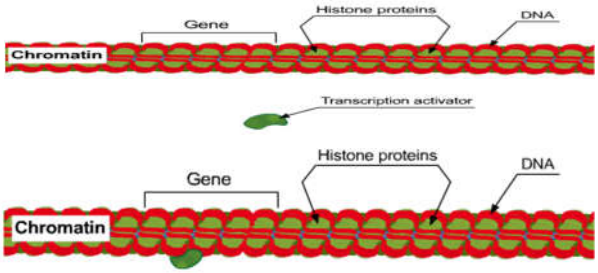


Positive Control of Transcription:

the regulatory molecule (the CRP–cAMP complex) enhances transcription.



- In response to specific signal a cell may produce or activate a transcription activator.
- Transcription activator recognizes and binds a specific sequence of a specific gene and work in conjunction with other proteins that can alter chromatin structure



The Three Systems:

- An inducible operon regulated by a repressor protein
- A repressible operon regulated by a repressor protein
- An operon regulated by an activator protein

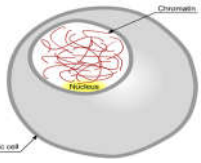
Lecture 95

Transcription Regulation In Eukaryotes:

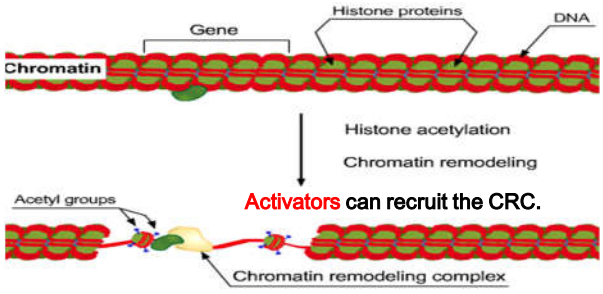
The Eukaryotic Dilemma

- For a cell to develop and function properly it must turn on the right genes and right time.
- Genes are located in the nucleus
- in a tangled chromatin material
- wrapped around histones.

The structure of chromatin dictates whether a gene can be transcribed not.



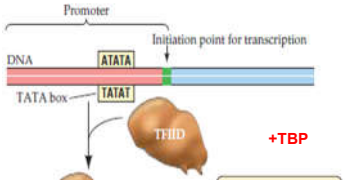
- The alteration may occur by addition of acetyl groups to histones which reduce positive charge.
- Chromatin remodeling complex can also alter the structure of chromatin to make it accessible to transcription factors that attract RNA Pol.



Lecture 96

Eukaryotic Transcription:
Initiation:

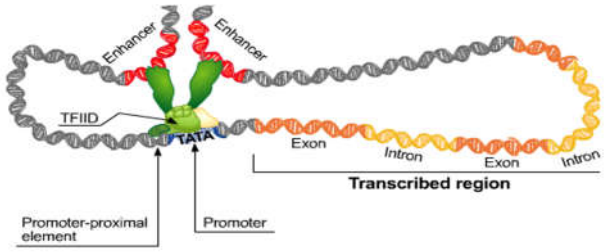
Promoter has two essential sequences.
1) Recognition sequence (RNA polymerase).
2) TATA box where DNA begins to denature
Eukaryotic RNA polymerase II cannot simply bind to the promoter and initiate transcription. It needs help from transcription factors. First, the protein TFIID



Most eukaryote promoters have TATA box, which is located 25bp upstream of start site.

Detailed View

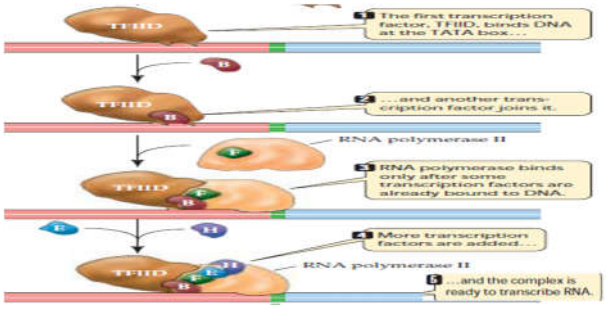
Many genes also have enhancer (1000 nucleotide away) where specific activators only made by certain cells can bind.

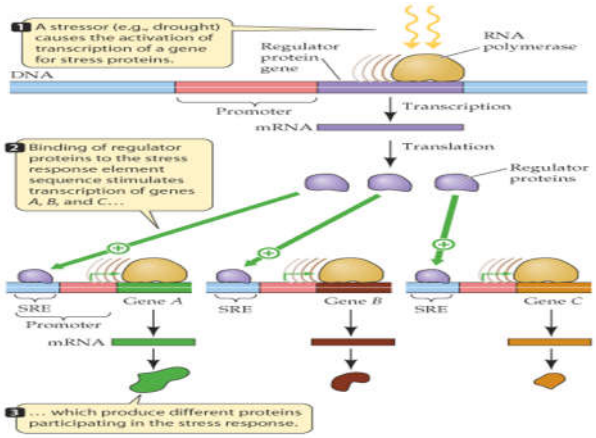


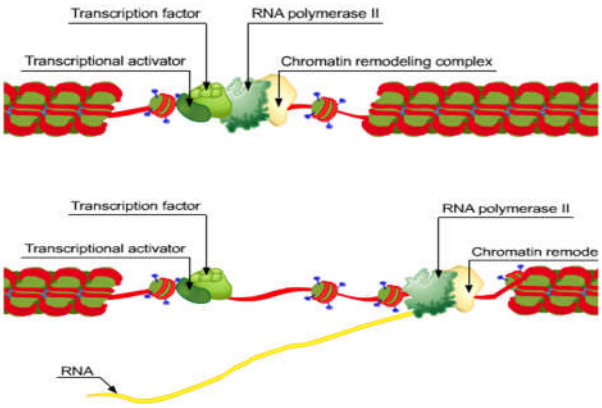
The simultaneous regulation of widely separated genes is possible through common sequences in their promoters, to which the same regulatory proteins bind.

Transcription factors:

TFIID binding changes its own shape and that of the DNA, presenting a new surface that attracts the binding of other transcription factors to form a transcription complex.
RNA Pol last to join



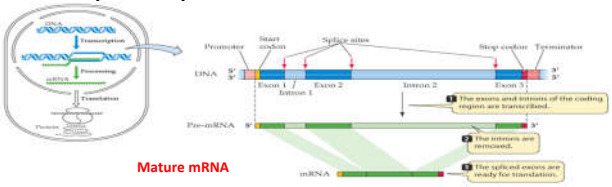


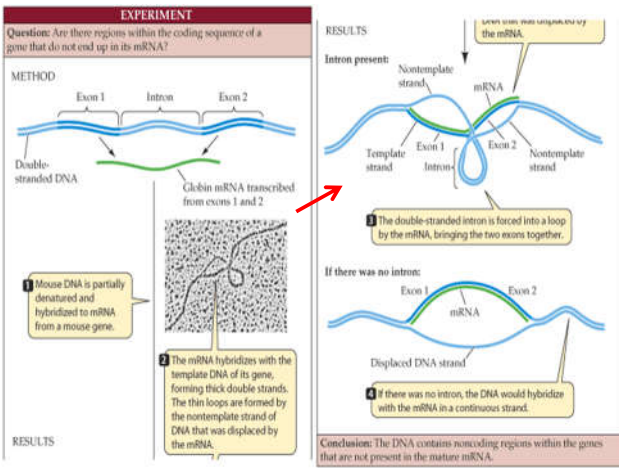


Lecture 97

RNA Processing:
pre-mRNA (freshly transcribed) is processed to make it functional.:

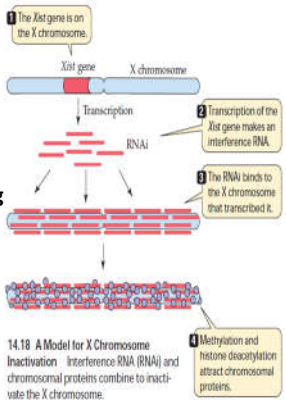
1-Introns are removed during RNA maturation
2) The transcribed pre-mRNA is altered by the addition of a G cap modified GTP at the 5' end to protect RNA & facilitate translation.
3) After the last codon a poly A tail (100-300) is added at the 3' end for export and protection





The Mechanism:

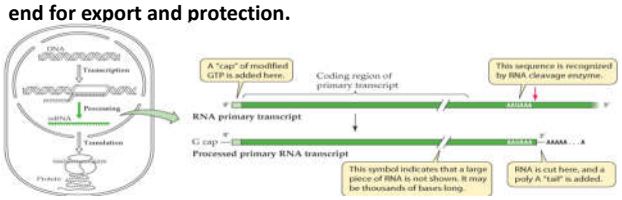
X chromosomes have a special gene Xist (X inactivation-specific transcript).
RNA transcribed from it does not leave nucleus.
It bind to the X chromosome from which it is transcribed, & this leads to a spreading of inactivation along the chromosome.
The active X makes interference RNA (RNAi) anti-Xist gene, appropriately called Tsix. This gene codes for an RNAi that binds by complementary base pairing to Xist RNA at the active X chromosome.



14.18 A Model for X Chromosome Inactivation Interference RNA (RNAi) and chromosomal proteins combine to inactivate the X chromosome.

The Mechanism

snRNPs bind Intron-exon bounderies consensus sequences
RNA of U1 has complementary bases that bind CS at 5' exon-intron boundery.
RNA of U2 has complementary bases that bind CS near 3' intron-exon boundery.
2) The transcribed pre-mRNA is altered by the addition of a G cap modified GTP at the 5' end to protect RNA & facilitate translation.
3) After the last codon a poly A tail (100-300) is added at the 3' end for export and protection.



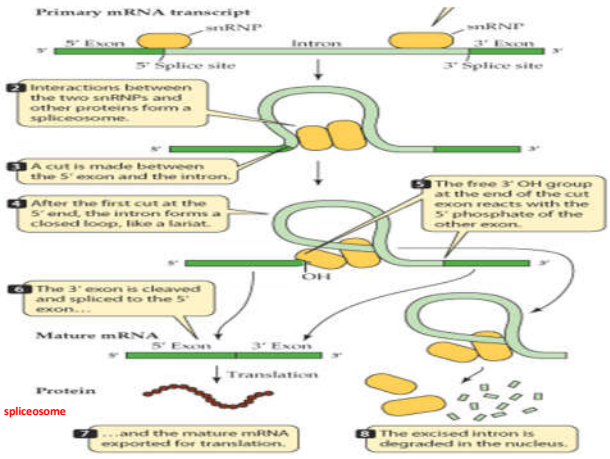
Lecture 98

Cell Signaling & Prehistoric world

The Prehistoric world:

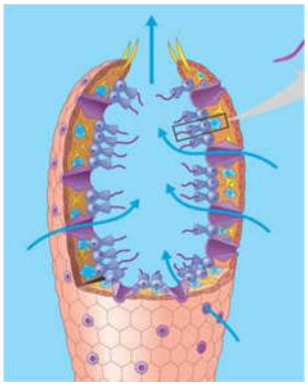
Earths changing atmosphere ,Ability to adapt key to survival and continuity, Ability to sense changes quintessential, Nutrition, Safety & Mate.





spliceosome





Multicellular = Specialization = Coordination
Ability to sense & respond to external and internal environment

The Necessity:

- In order to survive even simplest organisms need to sense and respond to their environment.
- It is critical that the cells of multicellular organisms communicate to coordinate their efforts (Running).
- Multicellular organism cells are specialized and rely on each other for the support (brain sugar).
- During development there have to be checks balances on differentiation (analogy society).

Signals:

- Our sense organs allow us to respond to light, odors and tastes (chemical), temperature, touch, and sound.
- Magnetism can be a signal: some bacteria and birds orient themselves to Earth's magnetic poles.

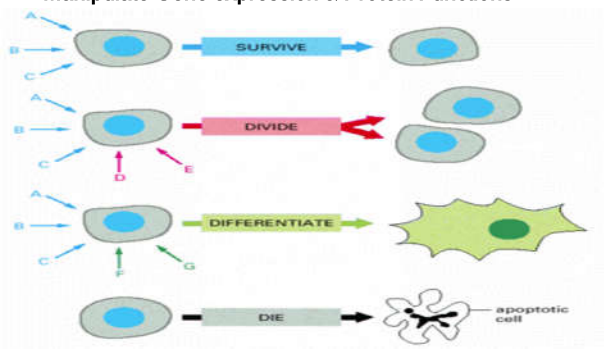
Gene Gategories:

- Some of genes are expressed in all cells all the time. Housekeeping genes are responsible for the routine metabolic functions (e.g. respiration).
- Some are expressed as a cell enters a particular pathway of differentiation.
- Some are expressed all the time in only those cells that have differentiated in a particular way.
- Some are expressed only as conditions around and in the cell change. For example, the arrival of a hormone may turn on/off certain genes in that cell.

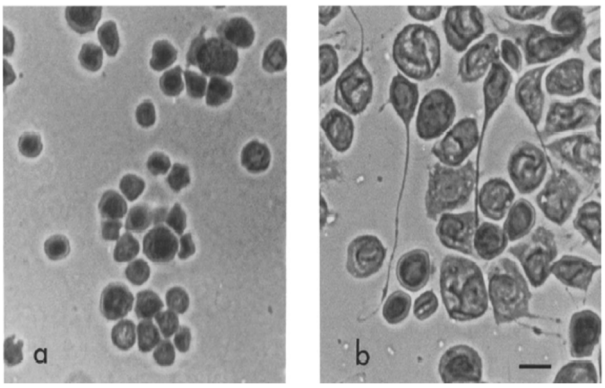
Lecture 99

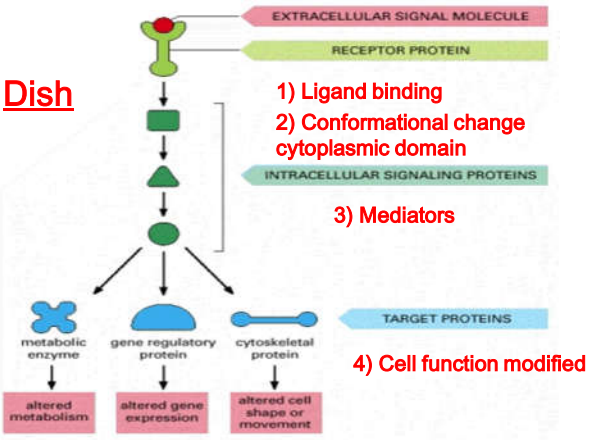
The Instructions & The Design:

Manipulate Gene expression & Protein Functions

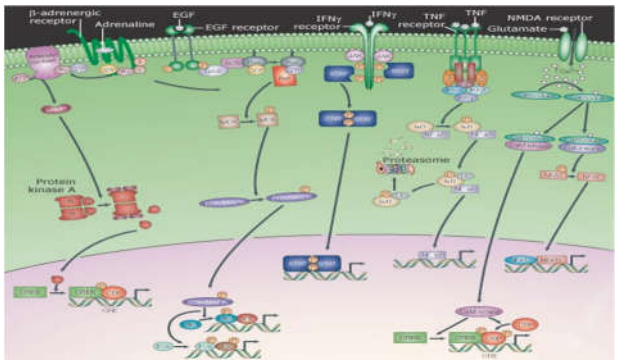


Alters Phenotype/Physiology of Cells



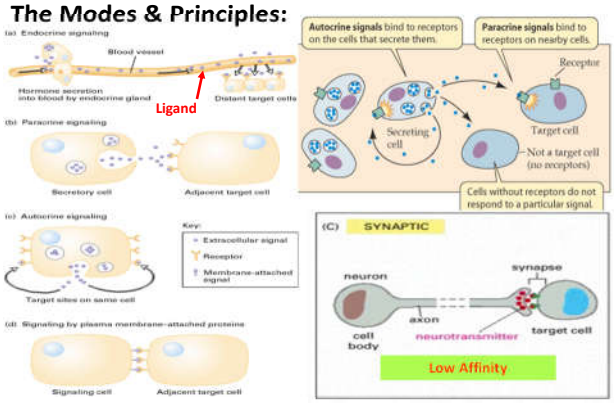


The entire signaling process, from signal detection to final response is called signal transduction pathway/signaling cascade .

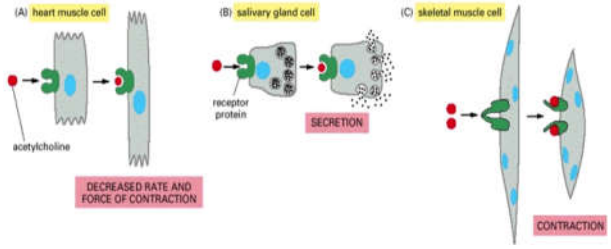


Lecture 100

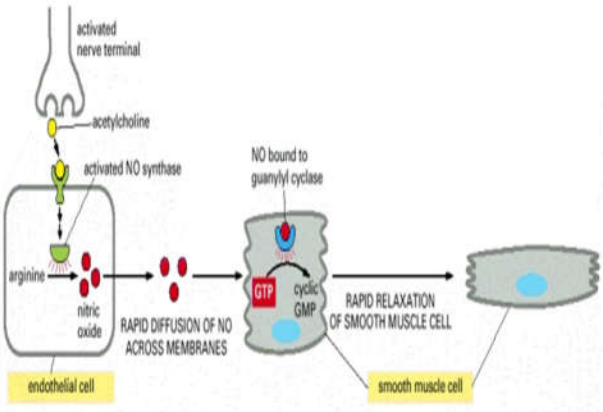
The Modes & Principles:



- Receptor combinations confer cell behavior in an environment flooded with hundreds of ligands.
- Cellular machinery specifies cell response to a particular ligand.

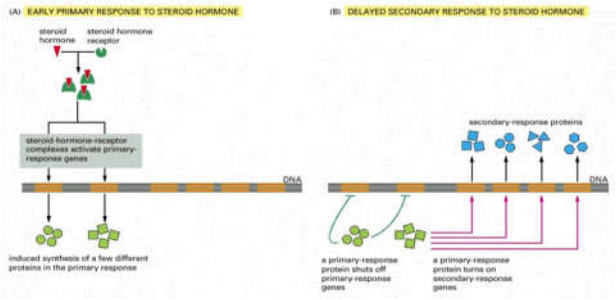


Ach → Receptor-ACh → NO synthase → Deamination
NO → Diffusion → G-cyclase → cGMP → Relaxation
NO half life 5 seconds



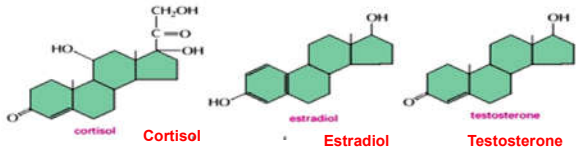
Lecture 101

The Modes & Principles II:



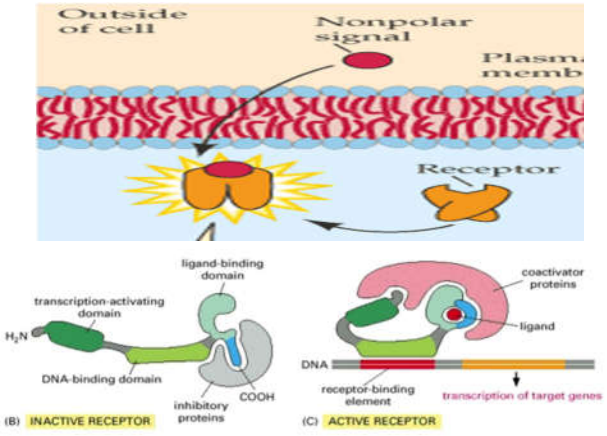
Ligand binding removes inhibitory proteins and facilitates binding of transcription activators

Small Hydrophobic Lipid soluble molecules eg steroid & thyroid hormones, retinoids & Vit D etc.
Blood transport via carrier proteins longer life (thy days Ach ms)
Carrier left outside.



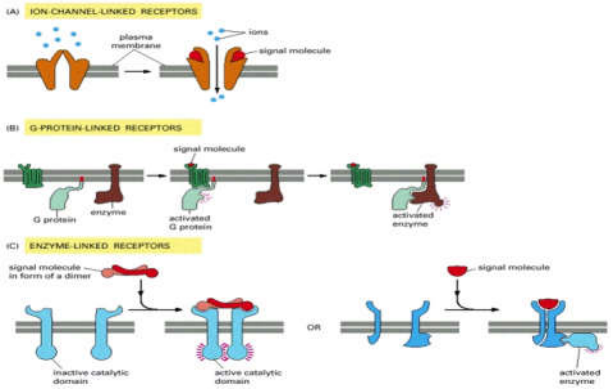
Cortisol Estradiol Testosterone

Inactive ICR may be DNA bound or in cytoplasm (NLS nonfunctional).
Activated receptor binds DNA induces gene transcription.
Different cells with identical ICRs regulate different genes due to other cell specific mediators.
Right combination of co-activators/gene regulators required to transcribe specific genes. (testosterone)



Lecture 102

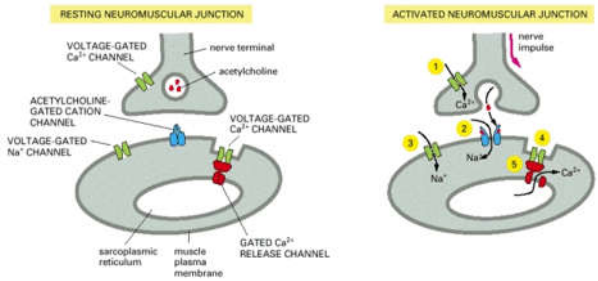
The Modes & Principles III



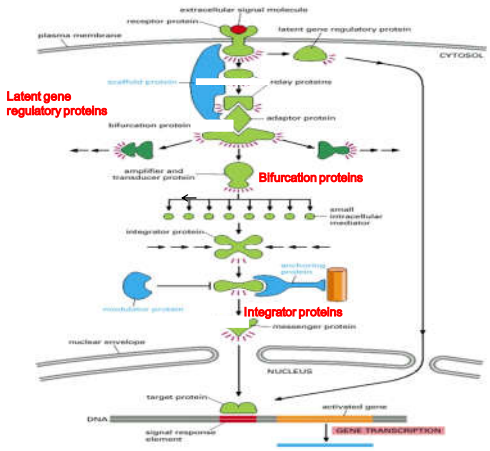
4) Amplifier proteins either enzymes or ion channels, increase the signal they receive, (Produce large amounts of small intracellular mediators/activate signaling proteins (adenylyl cyclase).
5 Transducer proteins convert the signal into a different form (adenylyl cyclase).

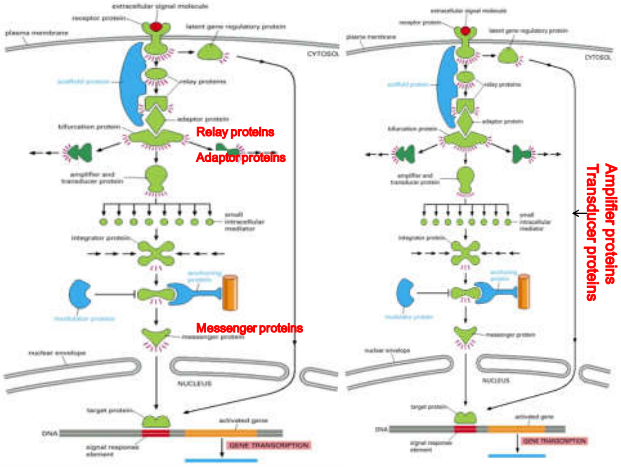
6 Bifurcation proteins spread signal from one pathway to another.
7 Integrator proteins receive signals from 2++ pathways integrate them before relaying a signal onward.
8 Latent gene regulatory proteins activated at the cell surface then migrate to nucleus to stimulate gene transcription.

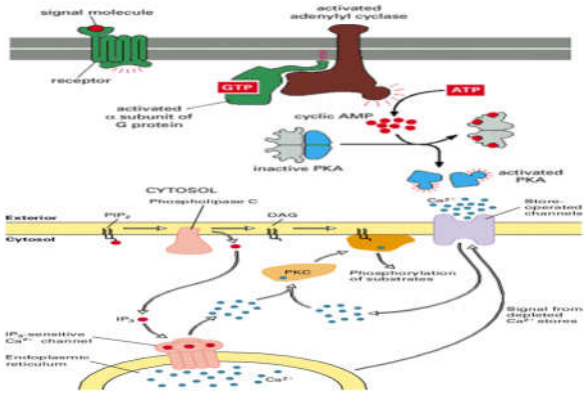
SICMs are produced/released in response to signal received by the receptor.
SICMs donot have an enzymatic activity of their own however they modify the function of other molecules.

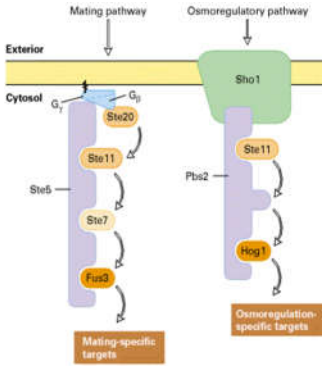


Neurotransmitters produce all or noting response.
1) Relay proteins pass the message to the next signaling component in the chain.
2) Messenger proteins carry the signal from one part of the cell to another (cytosol to nucleus).
3 Adaptor proteins link one signaling protein to another, without themselves conveying a signal.









Same ligand-receptor complexes may produce different responses.
Many cells use scaffold proteins to enhance speed efficiency & specificity of response.

Lecture 104

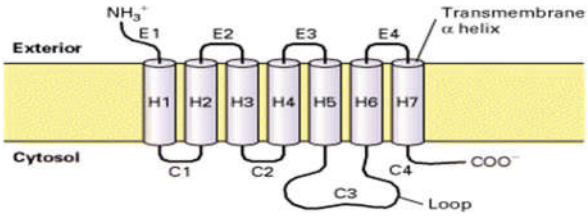
G Protein-Linked Receptors.

Helices 3, 5 & 6, that participate in agonist binding

Ligand binding moves helices 5 and 6, closer

C3 loop conformation & binds/activates G protein

Helix 7 functions in determining ligand specificity



Lecture 103

Signaling in E. coli.

After ligand binding change in tertiary structure of extra cellular part of EnvZ leads to structural change in its cytoplasmic domain making it a kinase (auto..).

EnvZ-P can now phosphor-ilate OmpR (responder) outside signal in and amplified.

Receptor conformational change after ligand binding which activates kinase activity.

Phosphorilation alters responder function.

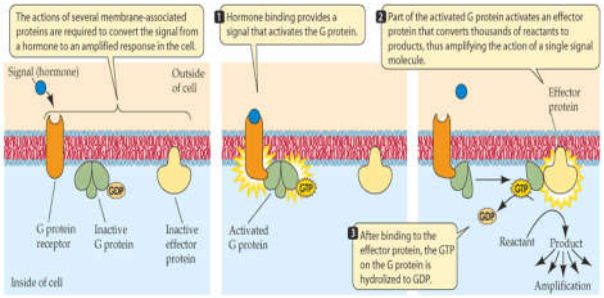
Signal amplified.

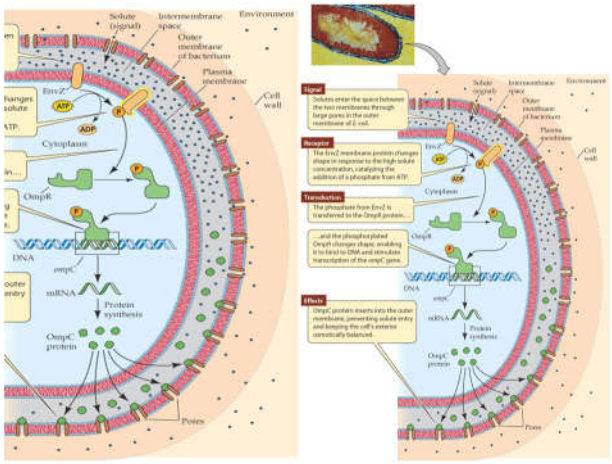
Transcription factor activated.

Protein synthesis results in altered cell activity.

Ligand binding causes a structural change permitting G protein to bind receptor.

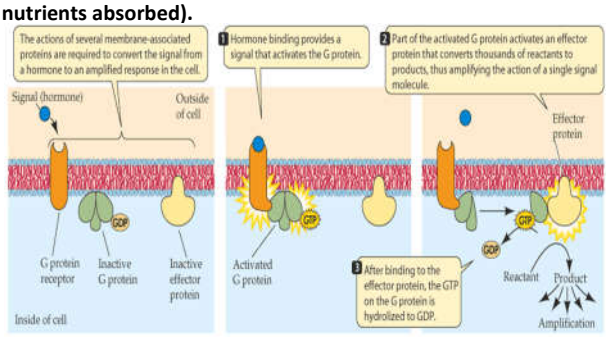
Binding of G protein to activated receptor causes it to exchange GDP for GTP (ligand releases).

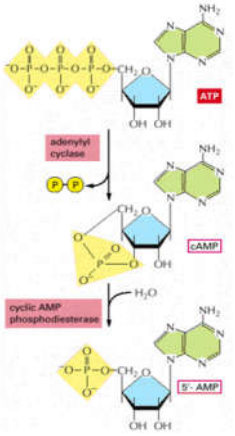




Subunit of G protein separates and activates an effector molecule (causing a functional change).

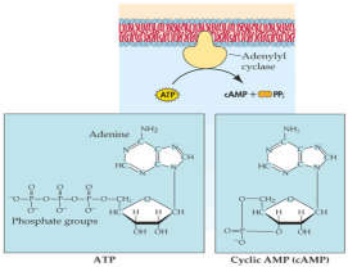
Epinephrine effects different cells differently (heart muscle contracts, intestinal vascular smooth muscle relaxes more nutrients absorbed).





Second messengers are allosteric regulators and do not have enzymatic activity.

Cyclic AMP (cAMP) can bind ion channels to open them or bind enzymes to exposing their active sites.



Lecture 105

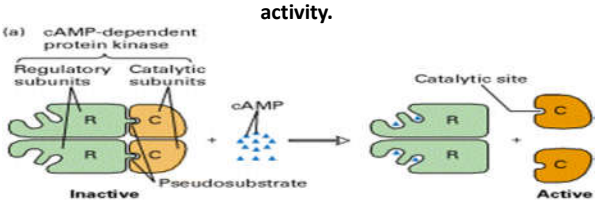
Protein Kinases A.

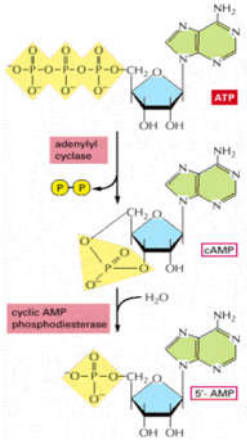
AKA cAMP-dependent protein kinase (cAPK)

Diverse effects of cAMP are mediated through the action of cAMP-dependent protein kinases.

Phosphorylate specific serine & threonine residues.

Phosphorylation of many enzymes increases their catalytic activity, whereas phosphorylation of others decreases their activity.





Adenylyl cyclase

one receptor 100 active Gs

G-GTP, activates a single AC

One Gs bound AC catalyzes synthesis of many cAMP

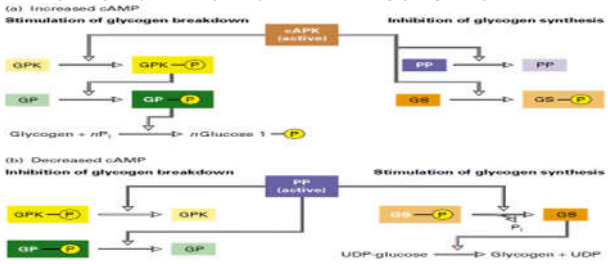
Several hundred cAMP molecules are produced per receptor-hormone complex .

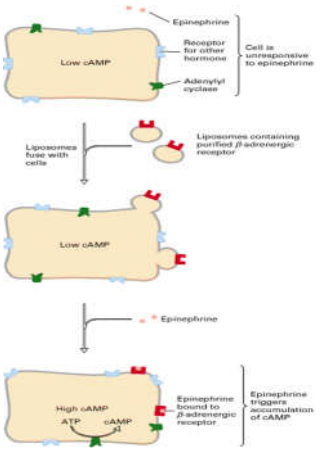
The PKAs are tetramers, two regulatory (R) subunits and two catalytic (C) subunits.

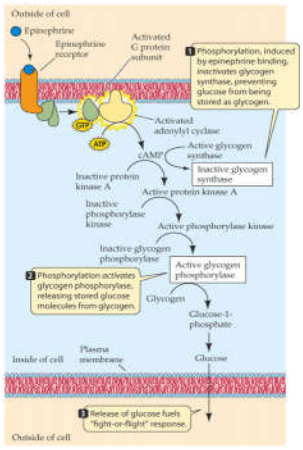
cAMP binding to the R subunits (pseudosubstrate) causes dissociation of the two C subunits

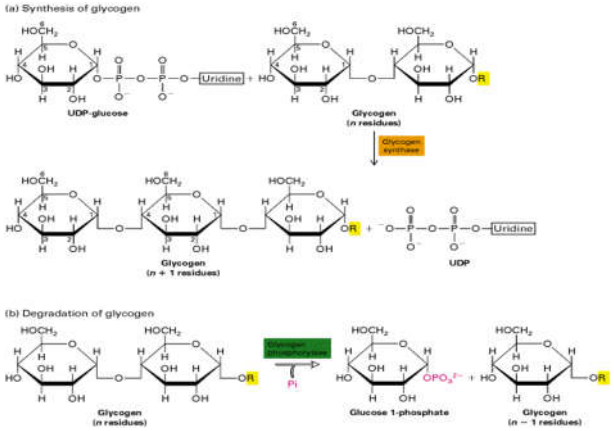
Cooperative binding

glycogen phosphorylase kinase (GPK), glycogen phosphorylase (GP) phosphoprotein phosphatase (PP), glycogen synthase (GS)



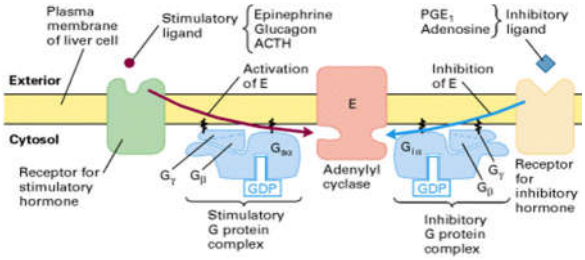




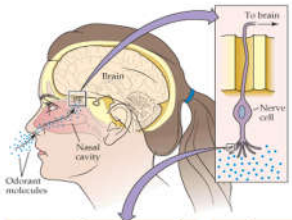


Lecture 106

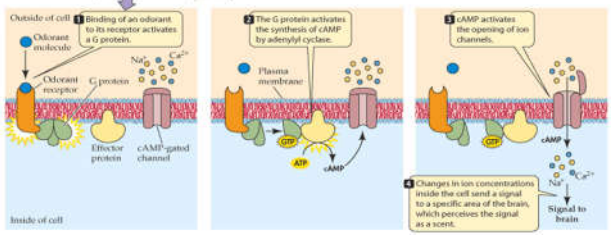
G Proteins.

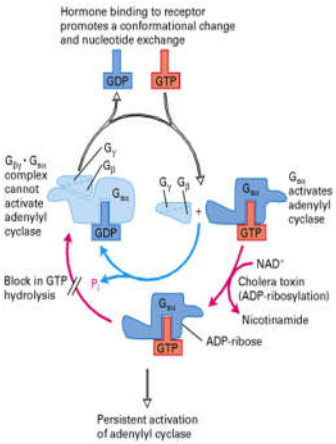


PGE$_1$/adenosine receptors interact with G_i,
Active $G_{i\alpha}$ (same β and γ different α) inhibits AC
$G_{\beta\gamma}$ also can directly inhibit some isoforms of AC
$G_{\beta\gamma}$ open K channels in heart cells (inh contract)



1000 odor signal receptor genes





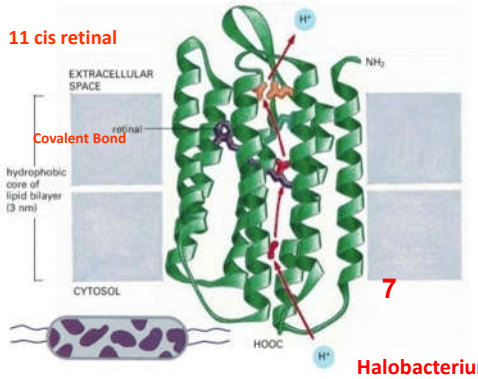
The α subunit of cholera toxin is an enzyme penetrates the plasma membrane prevents bound GTP to GDP hydrolysis.

cAMP rise causes membrane proteins to permit a massive flow of water from the blood into the intestinal lumen, causing diarrhea (water loss) resulting in death from dehydration.

Lecture 107

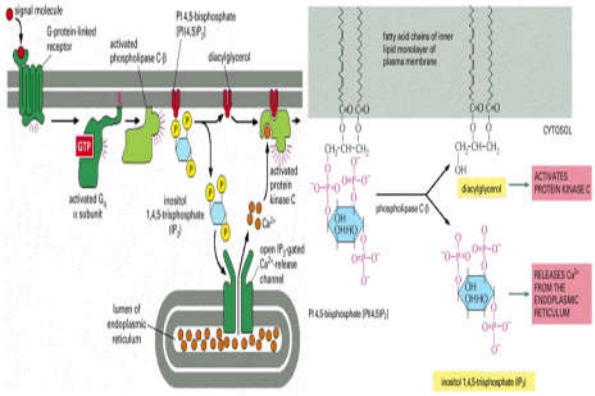
G Proteins Vision.

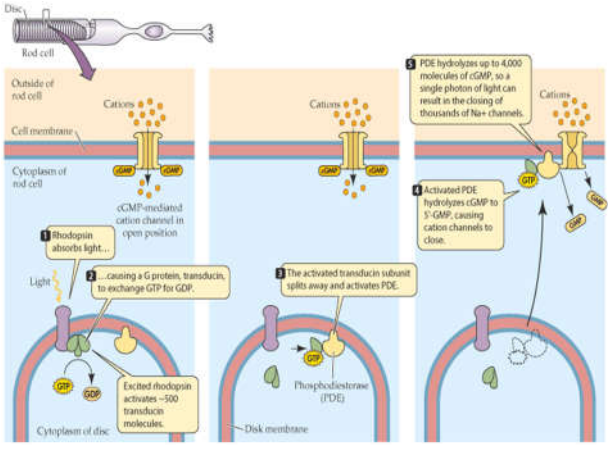
11 cis retinal

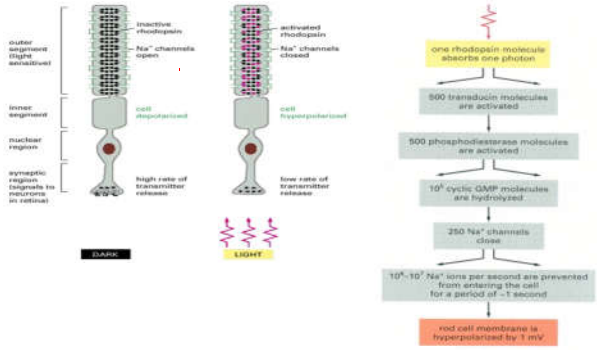


Halobacterium Salinarum

DAG→ Arachidonic acid→ Eicosanoids (PG)







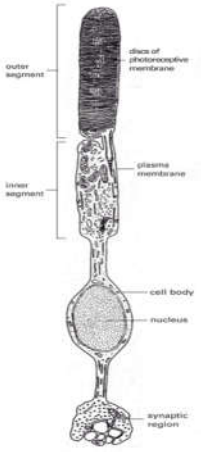
Photon→ trans retinal→ opsin conformation→ $G_{t\alpha}$ (tranducin)→ cGMP phosphodiesterase→ Na channel close→ hyper polarization

Bind to the receptor but do not activate hormone induced effects.
An antagonist acts as an inhibitor of the natural hormone (or agonist) by competing for binding sites on the receptor, thereby blocking the physiological activity of the hormone.

Tamoxifen

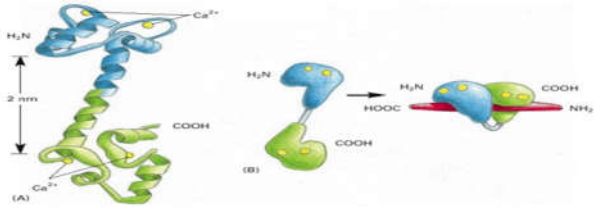
END

SOME FAMILY MEMBERS	ACTION MEDIATED BY	FUNCTIONS
G_s	α	activates adenylyl cyclase; activates Ca^{2+} channels
G_{olf}	α	activates adenylyl cyclase in olfactory sensory neurons
G_i	α	inhibits adenylyl cyclase
	βγ	activates K^+ channels
G_o	βγ	activates K^+ channels; inactivates Ca^{2+} channels
	α and βγ	activates phospholipase C-β
G_t (transducin)	α	activates cyclic GMP phosphodiesterase in vertebrate rod photoreceptors
G_q	α	activates phospholipase C-β



Lecture 109

Cell Signaling & Memory.



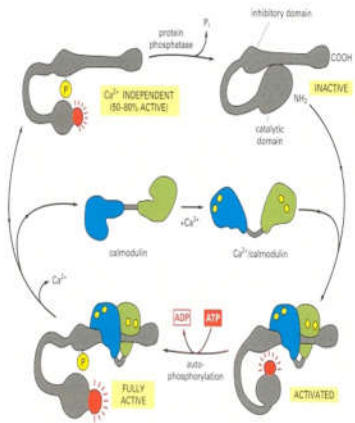
Troponin in muscle contraction (1st Ca^{2+}binding)
Calmodulin binds 4 Ca^{2+} allosteric (PKA) (1%)
Calmodulin-dependent Protein Kinases (CaM-Kinases) (serine/threonine)
CaM-Kinases phosphorilate myosin LT chain SM

Lecture 108

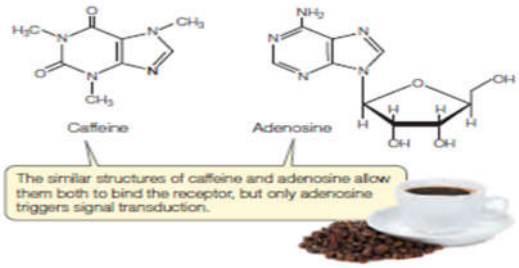
Cell Signaling & Us.

β1- and β2-adrenergic receptors (Heart & Lungs Respectively) are coupled to G proteins (Gs), which activate adenylyl cyclase. α1 and α2 receptors are coupled to two other G proteins, Gq and Gi, respectively. adenylyl cyclase Inhibition & phospholipase C activation.
Epinephrine bound to α2 receptors causes the arteries to constrict, cutting off circulation to the blood vessels in the intestinal tract, skin, and kidneys. (peripheral organs)
Agonists (terbutaline)

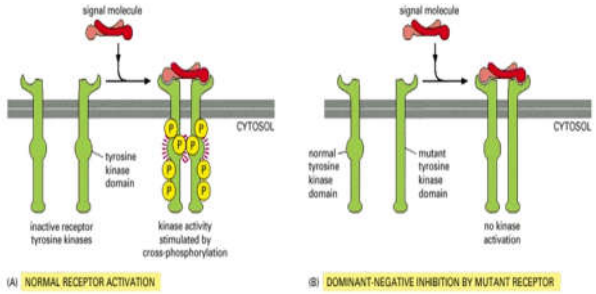
Mimic the function of a hormone by binding to its receptor and causing the normal response.
Antagonists (β-blockers)



Calmodulin binding → conformation catalytic domain.
phosphorylates inhibitory domain prevents disociation.
CaM-kinase II active w/o signal.
mutants poor memory.



Tired brain produces **adenosine** that binds specific receptor, decreases brain activity → drowsiness.
Caffeine's structure is similar to **adenosine**, it occupies adenosine receptors without inhibiting brain cell function, and alertness is restored.



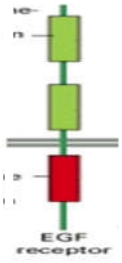
Ephrins & Eph Receptors .
Regulate the cell adhesion & repulsion responses .
Migration of cells and axons development .
Most numerous receptor tyrosine kinases.
Act as ligand & receptor
bidirectional reciprocal signaling.

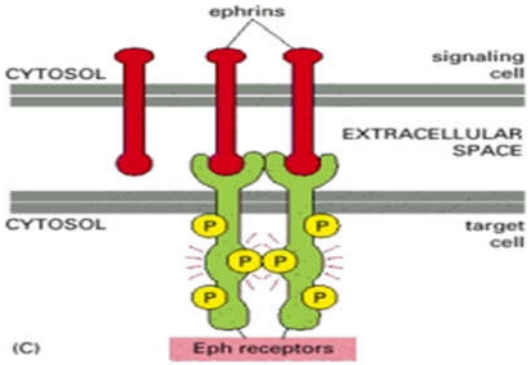
Lecture 110

Enzyme Linked Receptors.

Growth Factors

Extracellular signal promote growth, proliferation, differentiation, or survival of cells.
Usually act as local mediators at very low concentrations (10^{-9}-10^{-11} M).
The responses slow (hours) eventually lead to changes in gene expression.
Extracellular ligand-binding domainon 1 transmembrane segment.
Cytosolic domain either has an intrinsic enzyme activity or associates directly with an enzyme.





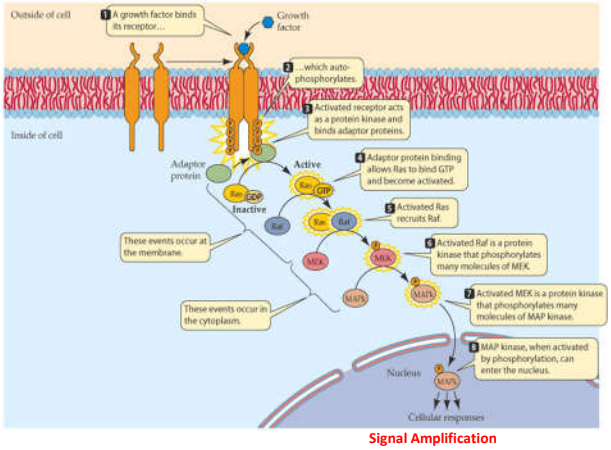
SIGNALING LIGAND	RECEPTORS	SOME RESPONSES
Epidermal growth factor (EGF)	EGF receptor	stimulates proliferation of various cell types
Insulin	insulin receptor	stimulates carbohydrate utilization and protein synthesis
Insulin-like growth factors (IGF-1 and IGF-2)	IGF receptor-1	stimulate cell growth and survival
Nerve growth factor (NGF)	Trk A	stimulates survival and growth of some neurons
Platelet-derived growth factors (PDGF AA, BB, AB)	PDGF receptors (α and β)	stimulate survival, growth, and proliferation of various cell types

2 receptor come together, form dimer.
Neighbouring kinase domains cross-phosphorylate each other on multiple tyrosines, autophosphorylation.
Mutant receptor that dimerizes normally inactive kinase domain dominant-negative.

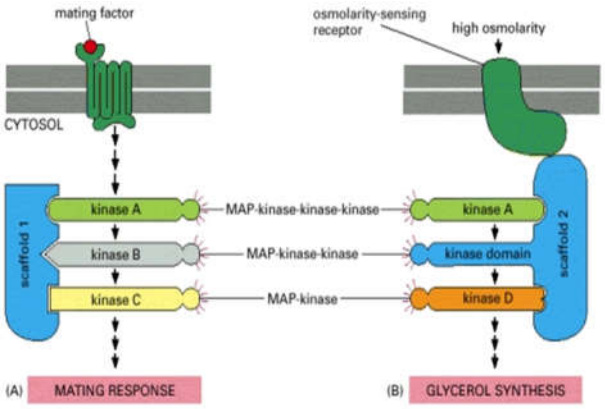
Lecture 111 & 112

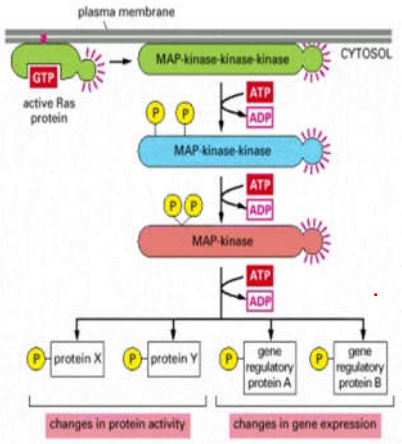
Enzyme Linked Receptors 2.

Fibroblast cells can be induced to proliferate by treatment with a mixture of platelet-derived growth factor (PDGF) and epidermal growth factor (EGF).
Microinjection of anti-Ras antibodies into these cells blocks cell proliferation.
Conversely, injection of a constitutively active mutant Ras protein (i.e., Ras^{D}), which hydrolyzes GTP very inefficiently and thus persists in the active state, caused the cells to proliferate in the absence of the growth factors.



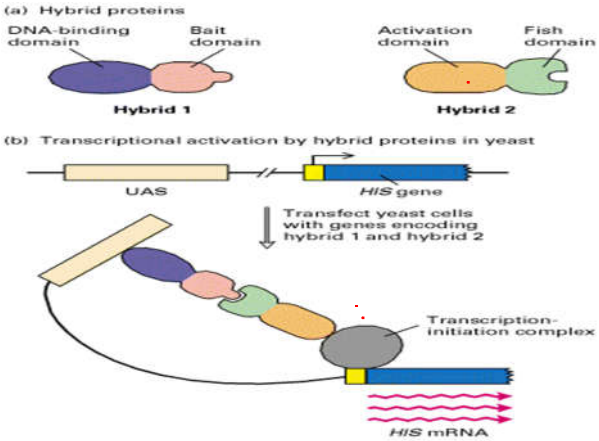
Signal Amplification

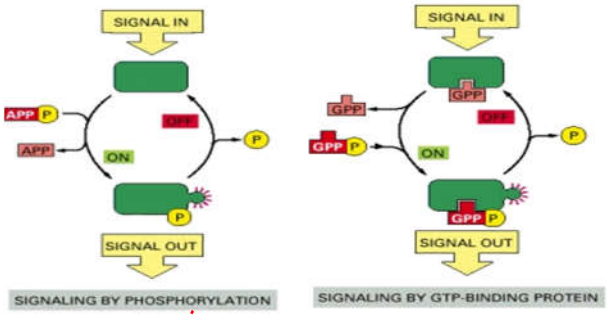




EGF activated MAPK stays active for 5 min, NGF activated MAPK is active for several hours.

5MAPK,7MEK & 7 Raf, specificity via scaffolds.



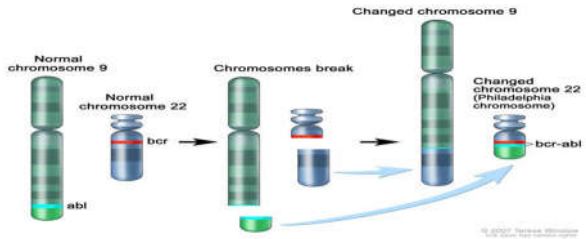


Phosphorilation: Kinases & Phosphatases
1/3 proteins are phosphorilated
2% genes encode Kinases
G-Proteins:Trimeric Bind GTP/GDP

Lecture 113

RTK Signaling Gone Wrong .

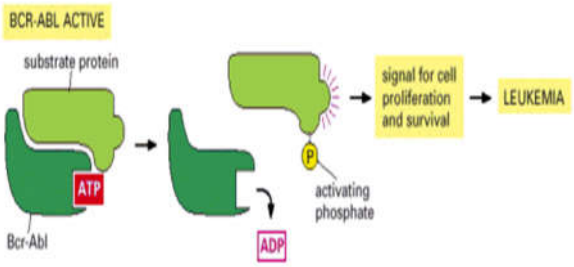
Philadelphia chromosome

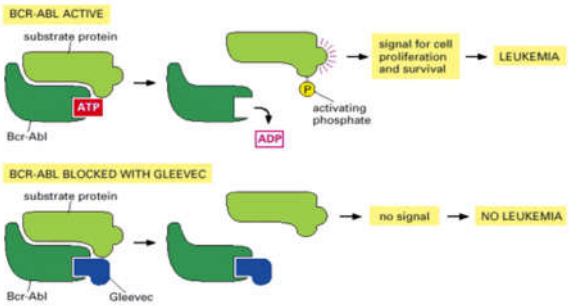


CML results from a chromosomal translocation between chromosome 9 and 22.

Abelson (Abl) tyrosine kinase gene

BCR/ABL fusion protein produced by the Philadelphia chromosome activates constitutively the cytosolic tyrosine kinase ABL that normally would be activated only when the cell is stimulated by a growth factor (PDGF).
The result: chronic myelogenous leukemia

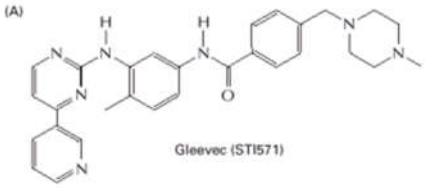




Gleevec prevents Bcr-Abl from transferring a phosphate group from ATP onto a tyrosine residue in a substrate protein. This blocks onward transmission of a signal for cell proliferation and survival.

Imatinib mesylate (Gleevec® AKA STI$_{571}$)

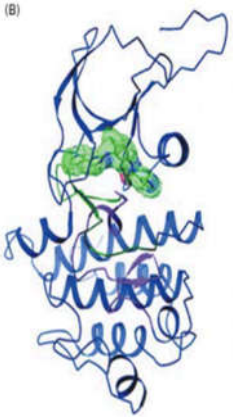
The drug can be given orally.





Neurofibromatosis causes tumors to grow on nerves abnormalities of the skin, nervous system, bones, endocrine organs and blood vessels.

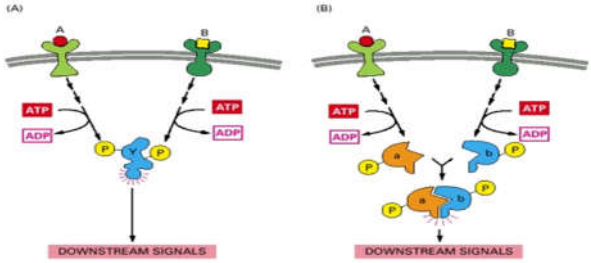
Neurofibromin a negative regulator of RAS (OP) is mutated.



This molecule fits into the active site of the ABL protein preventing ATP from binding .

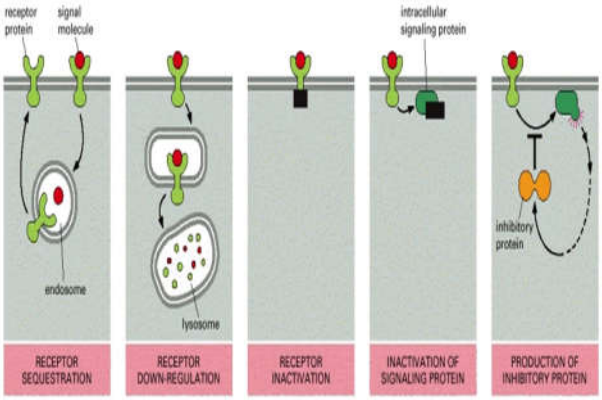
Without ATP as a phosphate donor, the ABL protein cannot phosphorylate its substrate(s).

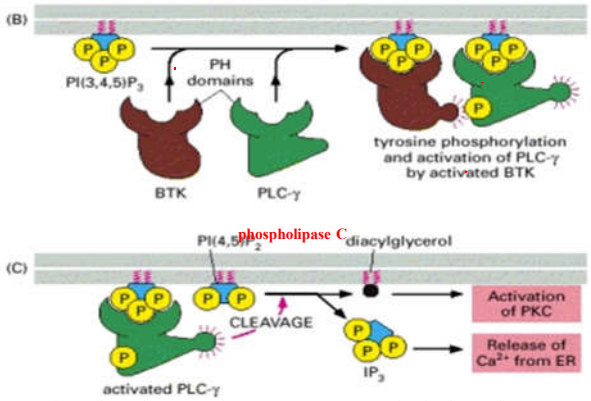
Almost 90% of the CML patients treated with the drug showed no further progression of the disease



Complex cellular behavior (survival, differentiation & proliferation) result of specific combination of signals.

Response dependent on integrator proteins .(coincidence)

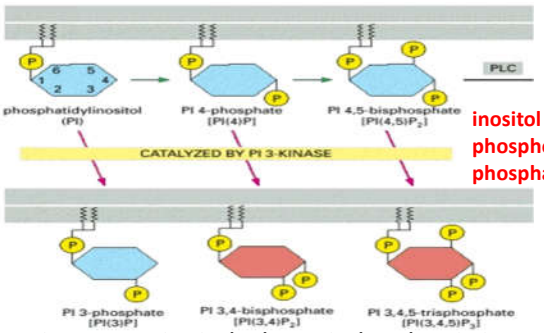




BTK phosphorylates and activates PLC-γ, which then cleaves PIP2 to generate IP3 and diacylglycerol to relay the signal onward

Lecture 114

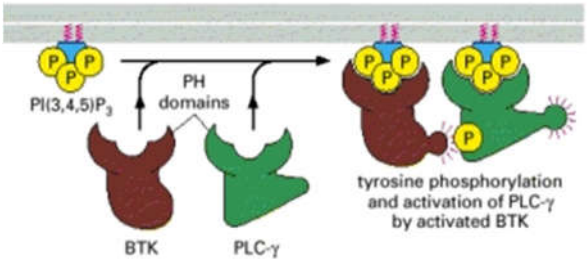
PI3-kinase .



inositol phospholipid phosphatases

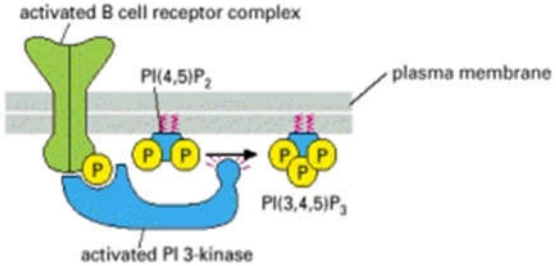
Signaling proteins bind PI(3,4)P2 and PI(3,4,5)P3 Pleckstrin homology (PH) domain 2

PI(4,5)P2 is cleaved by PLC-β (gpcr) or PLC-γ (RTK) 200 proteins



Bruton's tyrosine kinase, a protein whose mutation is associated with X-linked agammaglobulinemia.

platelet protein Pleckstrin

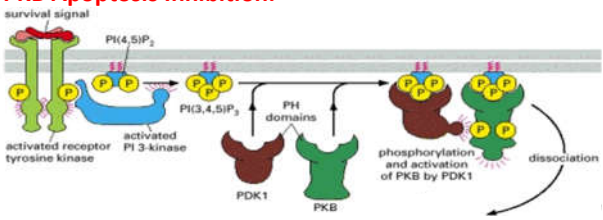


Antigen receptors on B lymphocytes (B cells) activate PI 3-kinase, the resulting inositol lipid docking sites recruit both BTK and PLC-γ to the cytoplasmic face of the plasma membrane.

platelet protein Pleckstrin

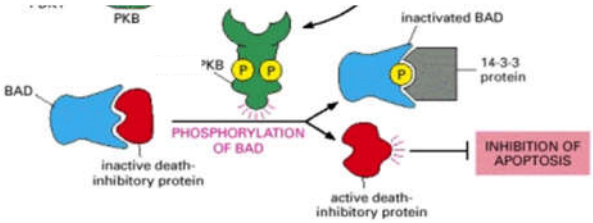
Lecture 115

PKB Apoptosis Inhibition.



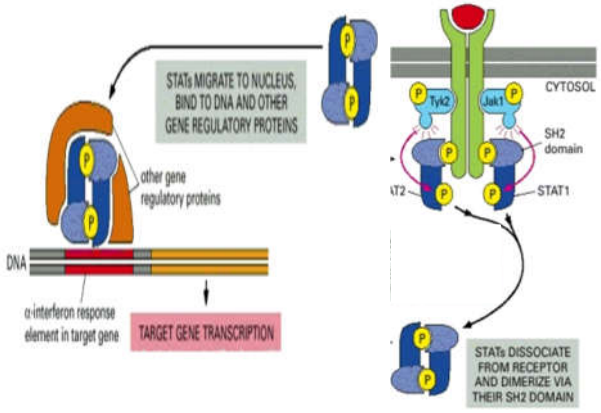
PI 3-kinase signals cells to survive, by indirectly activating protein kinase B (PKB) (Akt). PDK1 PKB PH domains direct them to the plasma membrane where PDK1 phosphorilates PKB .

PI Dependent Kinase



Activated PKB returns to the cytoplasm and phosphorylates BAD. BAD encourages cells to undergo apoptosis, promoting cell survival.

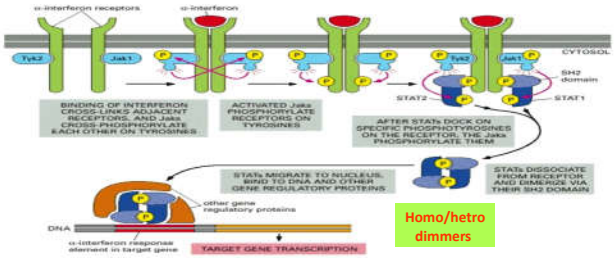
PKB also promotes cell survival by inhibiting other cell death activators, in some cases by inhibiting the transcription of the genes that encode them.



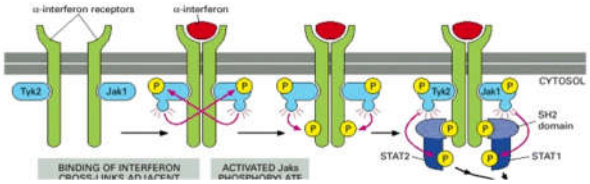
Lecture 116

Jak-STAT Signaling.

Increase antigen presentation of macrophages.
Activate & increase lysosome activity macrophages.
Cells to increase expression of MHC class I .
Promotes adhesion and binding required for leukocyte migration.
Promotes NK cell activity.
Activate inducible Nitric Oxide Synthase.
Jak-STAT one of the most direct signaling.
Ligand binding induces receptor to dimerize.
Binding brings Jaks close to cross-phosphorylate each other, increasing kinase activity.
The Jaks then phosphorylate receptors, creating STAT docking sites



Erythropoietin stimulates erythrocyte precursor cells to survive, proliferate, and differentiate, is associated with Jak2.
Jak2-deficient mice erythrocyte development fails, and the mice die early in development.
Prolactin stimulates breast cells to produce milk Activated STAT5 stimulates the transcription of genes that encode milk proteins.
α-interferon, for example, recruit both STAT1 & STAT2, γ-interferon receptors recruit only STAT1



1) STAT binds receptor
Jaks phosphorylate STAT forming SH2 domain & dissociation from the receptor.
2) STAT SH2 domain mediates binding to another STAT molecule, forming STAT homo/heterodimer

STAT dimer moves to the nucleus, where, in combination with other gene regulatory proteins, it binds to a specific DNA response element in various genes and stimulates their transcription

SIGNALING LIGAND	RECEPTOR-ASSOCIATED JAKS	STATS ACTIVATED	SOME RESPONSES
γ-interferon	Jak1 and Jak2	STAT1	activates macrophages; increases MHC protein expression
α-interferon	Tyk2 and Jak2	STAT1 and STAT2	increases cell resistance to viral infection
Erythropoietin	Jak2	STAT5	stimulates production of erythrocytes
Prolactin	Jak1 and Jak2	STAT5	stimulates milk production
Growth hormone	Jak2	STAT1 and STAT5	stimulates growth by inducing IGF-1 production
GM-CSF	Jak2	STAT5	stimulates production of granulocytes and macrophages
IL-3	Jak2	STAT5	stimulates early blood cell production

Lecture 117

TGF-β Receptor Family.

Consists of a large number of structurally related, secreted, dimeric proteins.

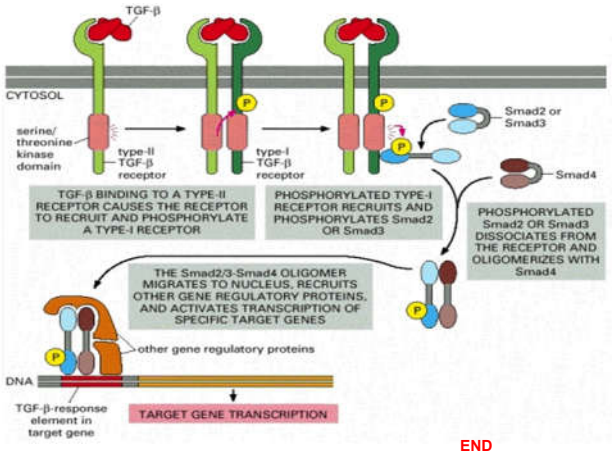
During development, they regulate pattern formation and influence various cell behaviors, ie proliferation, differentiation, extracellular matrix production, and cell death.

In adults, tissue repair and in immune regulation.

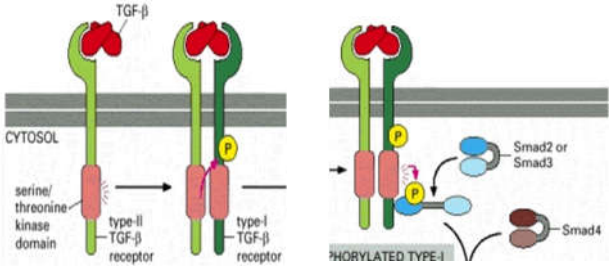
Act through enzyme-linked receptors that are single-pass transmembrane proteins.

Ligand first binds and activates a type-II receptor homodimer, which recruits, phosphorylates, and activates a type-I receptor homodimer, forming an active tetrameric receptor complex

Activated TGF-β receptors phosphorylate Smad2 activated BMP receptors phosphorylate Smad1, Smad5, or Smad8.One of these Smads then dissociates from the receptor and binds to Smad4



END



Both type-I and type-II receptors, required for signaling.

The type-I receptor directly binds and phosphorylates a latent gene regulatory protein of Smad family .

Lecture 118

Proteolysis Signaling Pathways.

Each cell in the embryo has to be guided along one developmental pathway according to its history, position & the character of its neighbors.

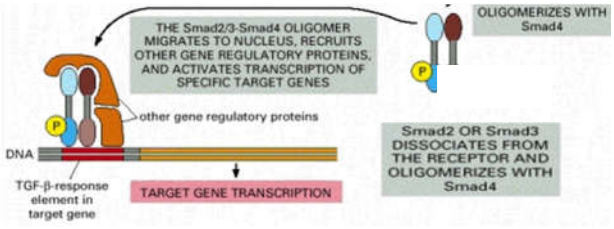
Cells must exchange signals with their neighbors to coordinate theirs behavior with others.

Developmental processes depend on signaling pathways that involve regulated proteolysis.

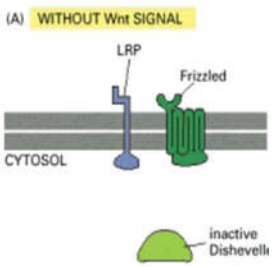
Defective Notch, Wnt, and Hedgehog signalling results in seriously disturbed development, the mouse dies as an embryo or at birth.

Co-receptor LDL-receptor-related protein (LRP).

Cytoplasmic signaling protein called Dishevelled is activated when Wnt is present.



Smad complex moves into the nucleus, it associates with other gene regulatory proteins, binds to specific sites in DNA, and activates a particular set of target genes.



Wnt proteins are secreted signal molecules.

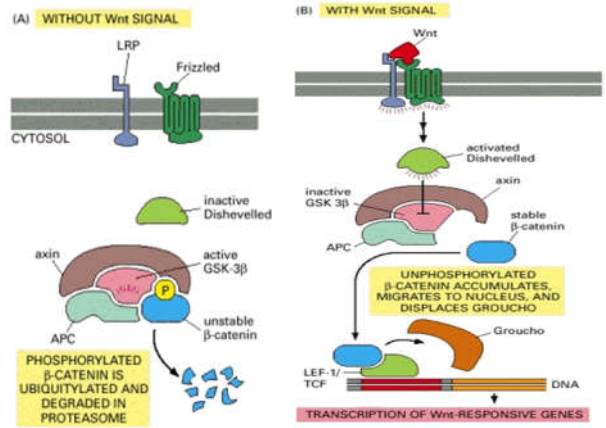
Frizzled (receptor) family of seven-pass transmembrane proteins resemble GPCR.

β-catenin functions both in cell-cell adhesion and as a latent gene regulatory protein.

β-catenin links the cadherins to the actin cytoskeleton .

β-catenin not associated with cadherins is rapidly degraded.

glycogen synthase kinase-3β (GSK-3β) phosphorylates β-catenin, thereby marking the protein for ubiquitylation.



adenomatous polyposis coli (APC) adherens junctions scaffold protein called axin

Intracellular protease cleaves off the cytoplasmic tail of Notch when it binds Delta,
Tail moves into the nucleus to activate the transcription of a set of Notch-response genes.
3 proteolytic cleavages last 2 depend on Delta.

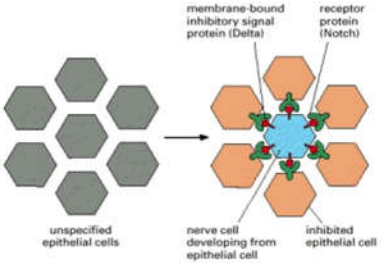
Heterodimer

Intracellular protease cleaves off the cytoplasmic tail of Notch when it binds Delta,.
Tail moves into the nucleus to activate the transcription of a set of Notch-response genes.
The products of the main genes directly activated by Notch signaling are themselves gene regulatory proteins, but with an inhibitory action: they block the expression of genes required for neural differentiation



β-amyloid precursor protein (APP)
Alzheimer's

Wnt binding to Frizzled and LRP inhibits of β-catenin phosphorylation/degradation.
β-catenin is a coactivator & induces transcription of Wnt target genes.
APC gene mutations occur in 80% of human colon cancers.
APC binding to β-catenin, inhibited so that β-catenin accumulates.



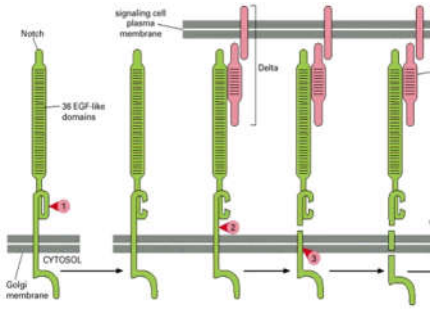
Delta, on the surface of future neural cell binds Notch on a neighboring cell, Delta signals to the neighbor not to become neural

Lecture 119

Immune System.

Most organisms are under constant attack from pathogens.
Over the course of evolution both the warring parties develop several strategies for the conquest and defense.
Since survival is at stake this is at stake new offensive and defensive weapons are being constantly developed by both to overcome the other.
There are several features of our bodies that fall in the category of defensive system that we generally do not give the due credit these are NON SPECIFIC.
Our skin is a defensive barrier that hinders pathogen invasion.
When skin is lost due to injury infection is one of the primary concerns.
Gastric juice in our stomachs can destroy pathogens that may cause infection of the digestive system.

Nerve cells arise as isolated single cells within an epithelial sheet of precursor cells.
Future nerve cell signal its immediate neighbors not to develop in the same way at the same time, (lateral inhibition) contact-dependent signaling.



single-pass trans-membrane proteins.

Our tears and saliva have enzymes that can destroy several microorganisms.
When dogs and cats lick their wounds they not only clean and remover dead cells they are also applying antibiotics.
Sneezing removes potential pathogens.
Normal flora of the gut restricts growth of harmful organisms.

Diversity

Specificity

Memory

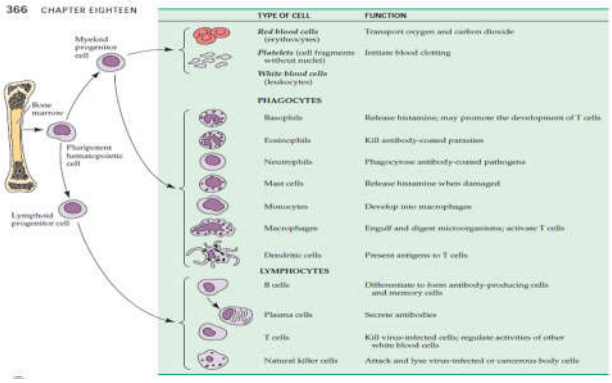
Self/non-self distinction

The Attack

The best way to understand immune response is by drawing any analogy with a war enemy forces advancing on the territory of the opponent.
I don't know how many of you have lived through a war having witnessed a couple my self I can tell you with certitude that the first response to an attack is blowing of the sirens to alert to rest of forces of an imminent attack and alerting the civilian population to take cover.
The siren is the first response to an attack whether it comes from sea, land or air. After an attack has been detected and announced the first line of defense is moved forward.
The first line of defense is generally highly mobile light weight combat units that have one goal which is to stall the enemy so that the heavy armament has time to reach the conflict zone and the war council has enough time to evaluate the situation and devise a tactical strategy to neutralize the hostile forces

Lecture 121

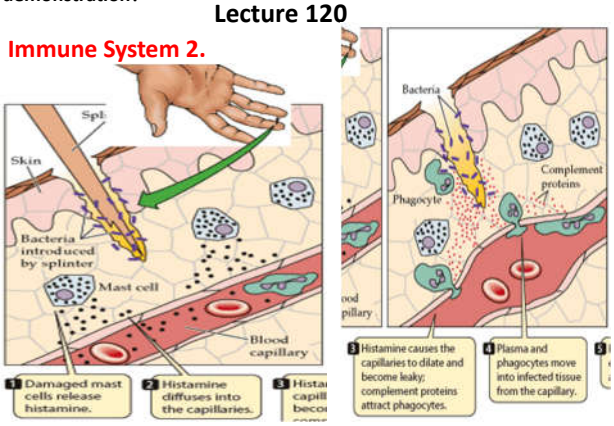
Immune System 3.



What happens when some one gets pricked by a thorn demonstration?

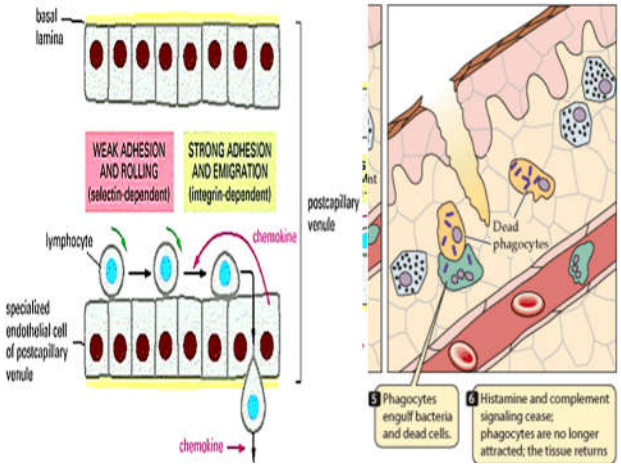
Lecture 120

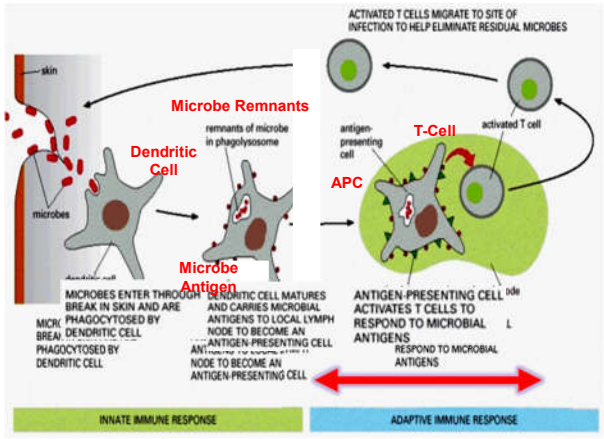
Immune System 2.

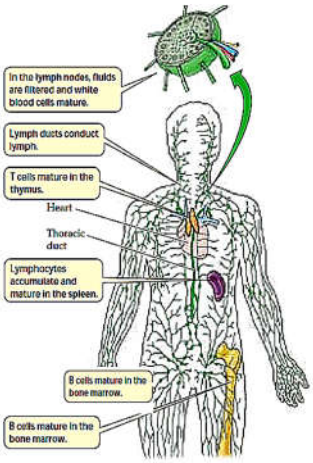


without nuclei → Red blood cells RBC (erythrocytes)
Transport oxygen and carbon dioxide
Platelets (cell fragments Initiate blood clotting)

Basophils Release histamine.
Eosinophils Kill antibody-coated parasites.
Neutrophils Phagocytose antibody-coated pathogen.
Mast cells Release histamine when damaged.
Monocytes Develop into macrophages.
Macrophages Engulf and digest microorganisms.
Dendritic cells Present antigens to T cells.
B cells Differentiate form antibody-producing cells and memory cells Plasma cells Secrete antibodies.
T cells Kill virus-infected cells; regulate activities of other white blood cells.
Natural killer lyse virus-infected/cancerous cells.

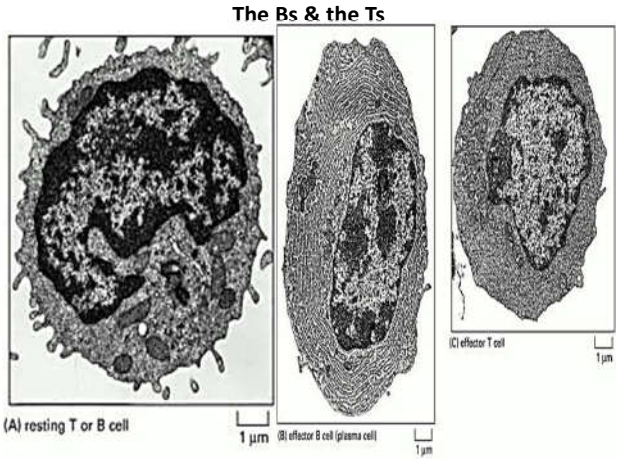


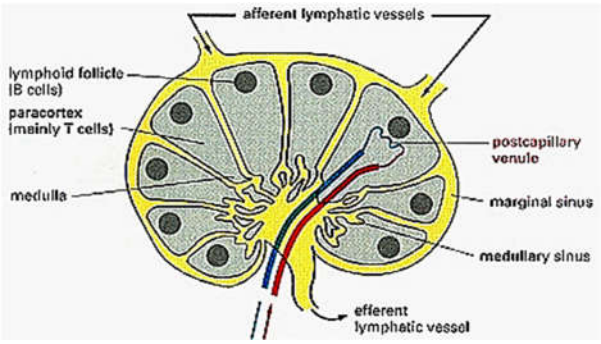




Lymph is a fluid derived from blood and other tissues that accumulates in intercellular spaces throughout the body.

A network of ducts and vessels collects lymph (w/o RBCs) from the body's tissues and carries it toward the heart.





Lymph Nodes, contain a variety of WBCs. As fluid passes through a node, it is filtered and inspected for non-self materials by these defensive cells.

IIS Phagocytic cells have receptors that have affinity for certain pathogen-associated molecules.

IIS programmed to be activated by specific pathogen molecules. (Toll Receptors)

IIS activates AIS.

Adaptive Immune Response AIR is highly specific to the pathogen that induced it.

There are 2×10^{12} lymphocytes in the human body, making the immune system comparable in cell mass to the liver or brain.

IIS is activated at sites of infection, whereas AIS are activated in peripheral lymphoid organs.

AIR can distinguish self from non-self.

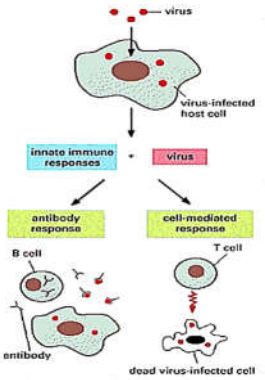
Long term protection. (measles)AIR activated by self is harmful.

Substances that activate AIS are called antigens

AIR can distinguish even optical isomers

Lecture 122

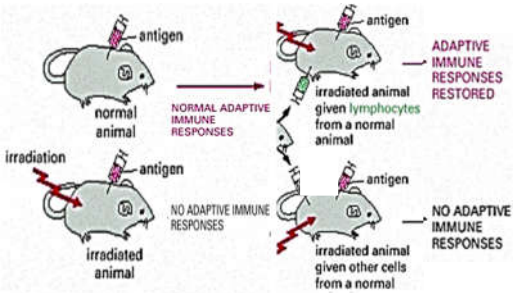
AIS.



AIS has two components.

Humoral immunity is mediated by soluble antibody proteins produced by B lymphocytes that neutralize and kill pathogens/toxins (20%-30%).

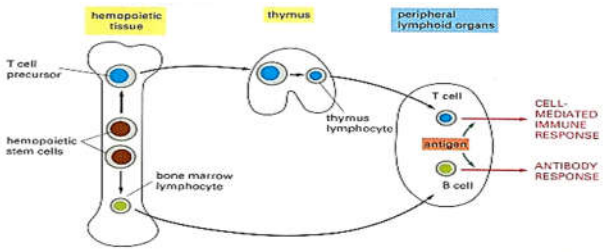
Cellular immunity is mediated by T lymphocytes which target cells infected by a pathogen (60%-70%).



Rats were heavily irradiated to kill most of their WBCs including lymphocytes. These rats were unable to mount AIR . Various types of cells were transferred into the animals . Only lymphocytes restored the AIR of irradiated animals.

Lecture 123

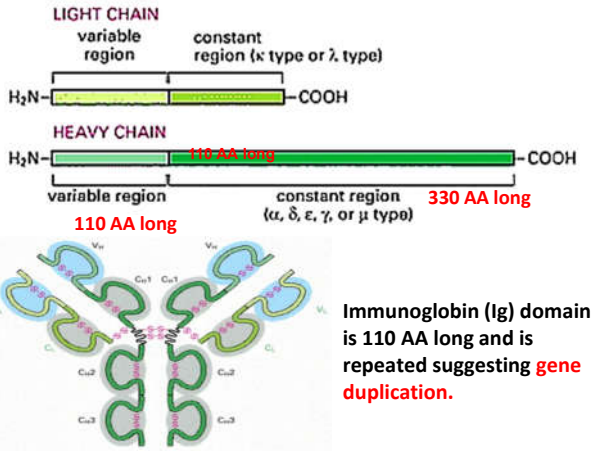
The B-Cells.



T cells develop in the thymus, and B cells, in mammals, develop in the bone marrow .

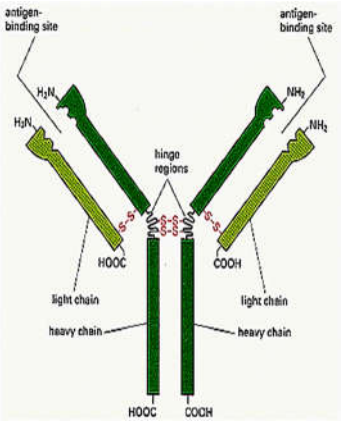
Most lymphocytes die in central lymphoid organ.

B-Cell Maturation Antigen Independent



Immunoglobin (Ig) domain is 110 AA long and is repeated suggesting gene duplication.

Antibody Structure

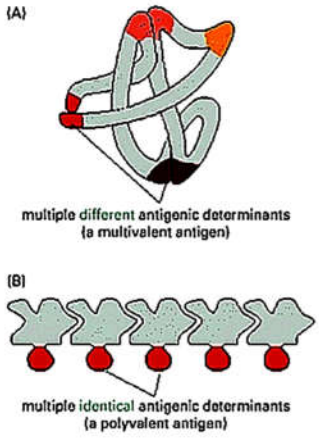


It is composed of four polypeptide chains 2 identical heavy chains & 2 identical light chains.

Tail (Fc) & hinge region are formed by the two heavy chains.

2 identical Binding Sites.

Flexible hinge region improves efficiency of antigen binding and cross-linking.

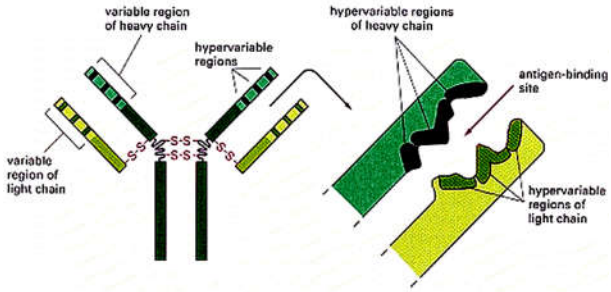


Affinity

Strength of a single anitgenic determinat with a single Ab binding site (# independent)

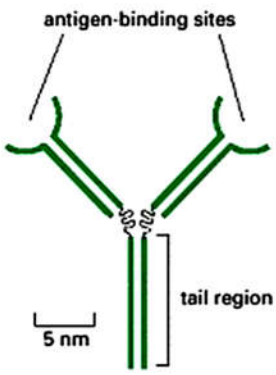
total binding strength: Avidity.

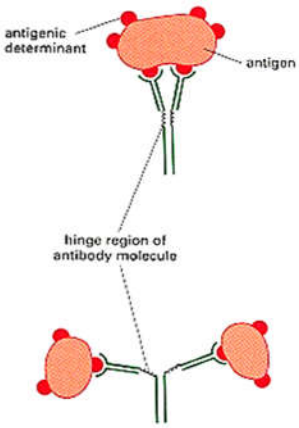
IgG binds Polyvalent 100 times stronger than monovalant.

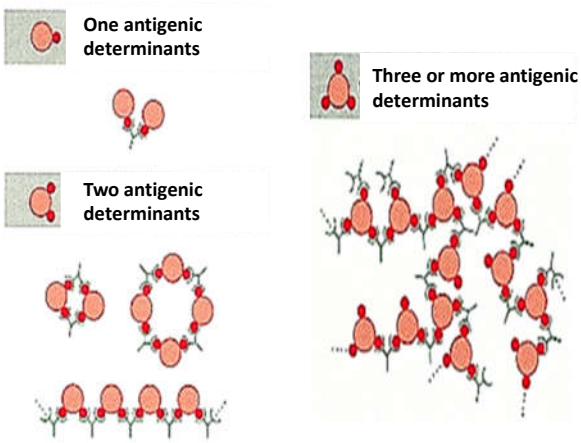


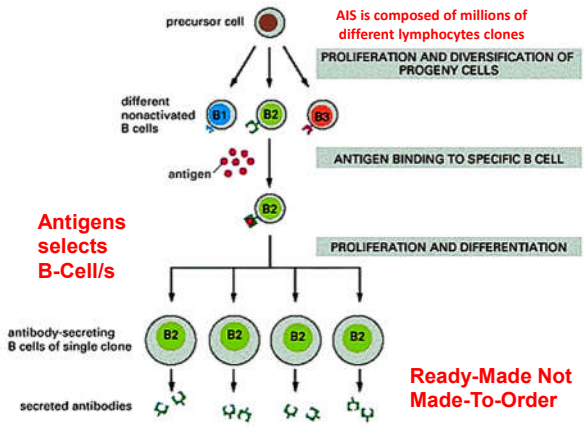
Framework Regions

The two antigen-binding sites are identical, each formed by the N-terminal region of a light chain and the N-terminal region of a heavy chain.







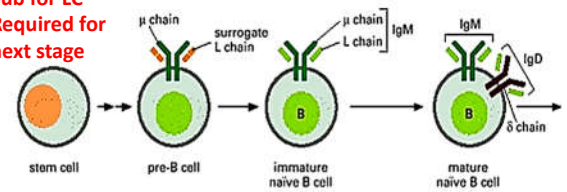


Lecture 124

The B-Cells 2.

B-Cell Maturation Antigen Independent

Sub for LC Required for next stage

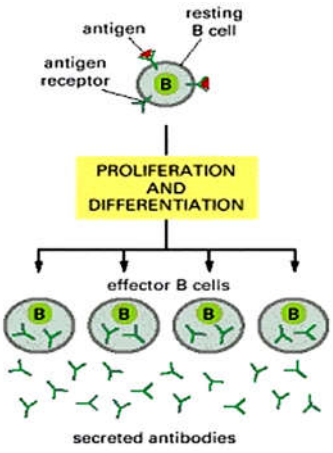


1st cell to respond to antigen

Development in bone marrow

Circulation through peripheral lymphoid organs

IgM is the 1st class of Ab to appear on cell surface and secreted B cells can then switch Ab class



AIS can respond to millions of different foreign antigens in a highly specific way. Animals randomly generate a vast diversity of lymphocytes. Lymphocytes that react against foreign antigens encountered by animal are selected for action.

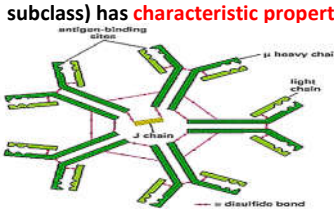
Lecture 125

The Antibodies.

5 classes of antibodies, IgA, IgD, IgE, IgG, & IgM, class of heavy chain—α, δ, ε, γ, & μ, respectively.

Four human IgG subclasses (IgG1, IgG2, IgG3, and IgG4), having γ1, γ2, γ3, andγ4 heavy chains, respectively.

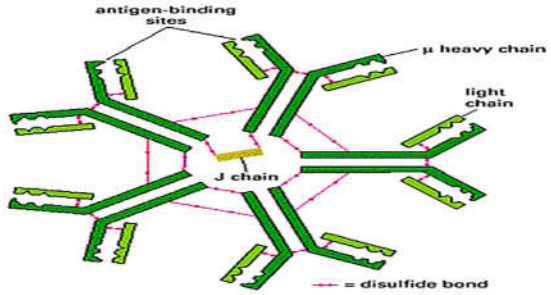
The various heavy chains give a distinctive conformation to the hinge and tail regions of antibodies, so that each class (and subclass) has characteristic properties of its own.



IgM first class of antibody to appear on the surface of a developing B cell.

The major class secreted in the early stages of a primary antibody response.

Secreted form, IgM is a pentamer, J chain is required for pentamer formation.



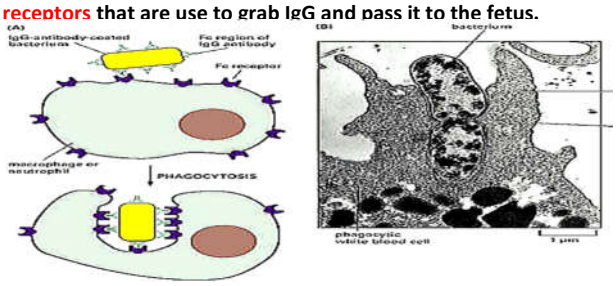
The binding of an antigen to a single secreted pentameric IgM molecule can activate the complement system, wich can either mark the pathogen for phagocytosis or kill it directly.

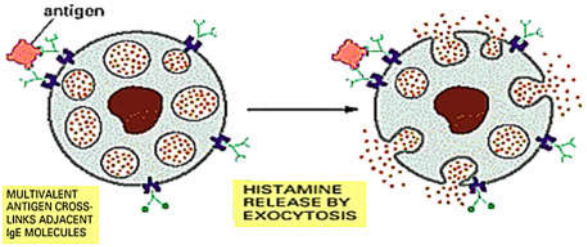
IgG is a monomer.

Phagocytic cells bind, ingest, and destroy infecting microorganisms that have become coated with the IgG antibodies produced in response to the infection.

IgG only Ab that can pass from mother to fetus

placenta cells that are in contact with mothers blood have Fc receptors that are use to grab IgG and pass it to the fetus.



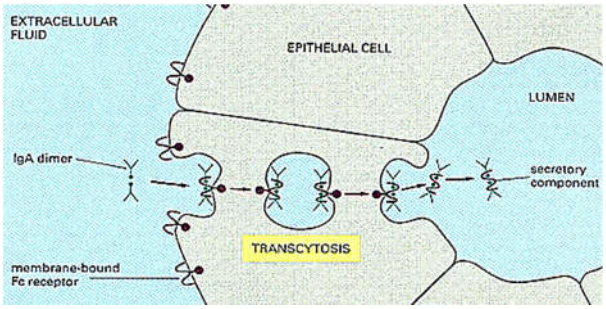


Antigen binding triggers the mast cell or basophil to secrete a variety of cytokines and biologically active amines, especially histamine.

These molecules cause blood vessels to dilate & become leaky, which helps WBCs, antibodies, complement sys to enter sites of infection.

These molecules cause allergic reactions as hay fever, asthma, and hives.

END



IgA is transported through secretary epithelial cells from the extracellular fluid into the secreted fluid by another type of Fc receptor that is unique to secretory epithelia.

This Fc receptor can also transport IgM into secretions less efficiently

Lecture 126

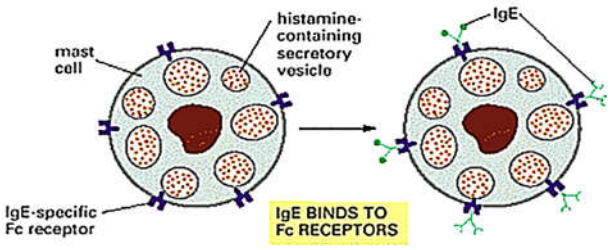
Immunological Memory.

The adaptive immune system, like the nervous system, can remember prior experiences.

First exposure to an antigen A: Immune response appears after several days, rises rapidly and exponentially, and then, more gradually, declines.

Secondary immune response: Lag period is shorter, and the response is greater indicating animal remembers its first exposure to antigen A.

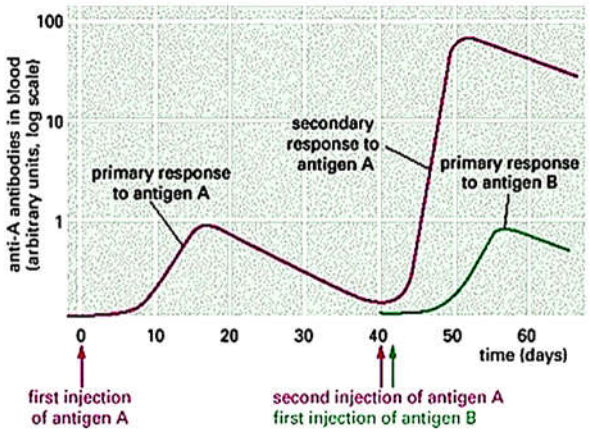
If animal is given a different antigen B instead of second injection of antigen A the response is typical primary not secondary, immune response.

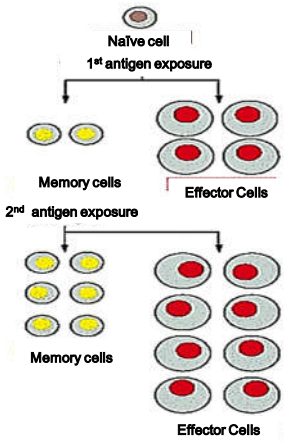


Tail region of IgE molecules four-chain monomers, binds (high affinity) another class of Fc receptors.

These receptors are located on the surface of mast cells in tissues and of basophils in the blood.

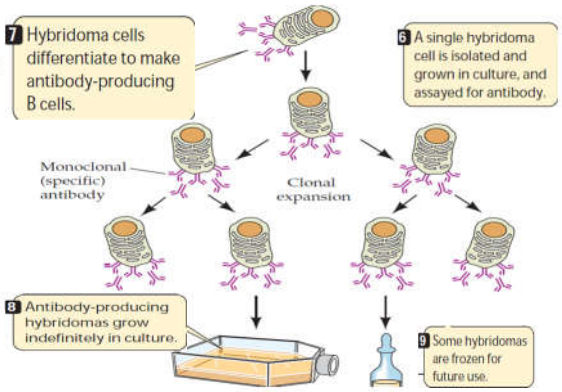
The IgE molecules bound to them function as passively acquired receptors for antigen.





Cellular Basis of Immunological Memory

When naïve lymphocytes are stimulated by their specific antigen, they proliferate and differentiate.
Most become effector cells which function & then die, while others become long-lived memory cells.



Vaccination

INFECTIOUS AGENT	DISEASE	VACCINATED POPULATION
Bacteria		
Bacillus anthracis	Anthrax	Exposed in biological w
Bordetella pertussis	Whooping cough	Children and adults
Clostridium tetani	Tetanus	Children
Corynebacterium diphtheriae	Diphtheria	Children
Haemophilus influenzae	Meningitis	Children
Mycobacterium tuberculosis	Tuberculosis	All people
Salmonella typhi	Typhoid fever	Areas exposed to agent
Streptococcus pneumoniae	Pneumonia	Elderly

Lecture 127

Antibody Diversity.

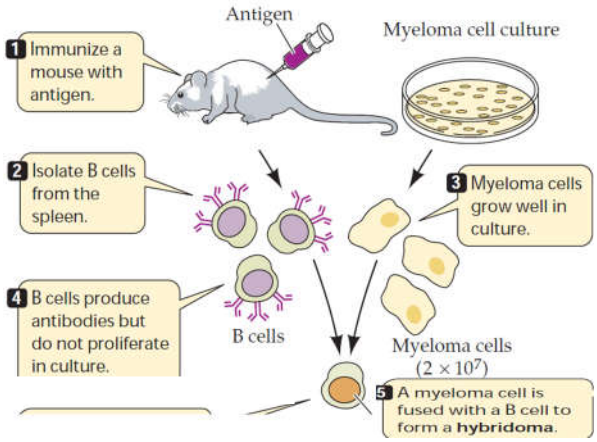
The Dilemma

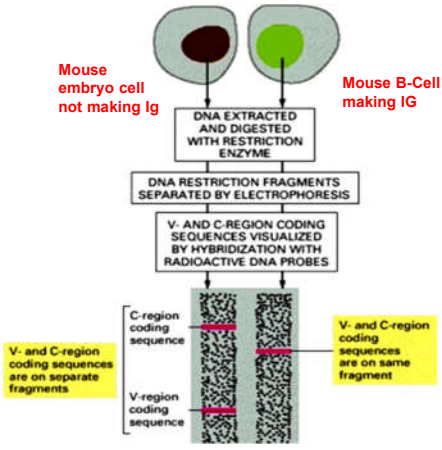
Even in the absence of antigen humans can make more than 10^{12} different antibody molecules.

Estimated number of human genes is 30,000

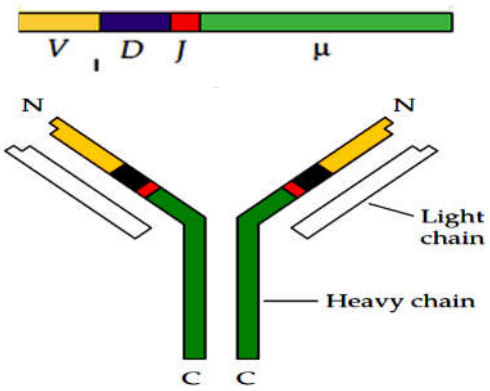
Mammalian immune system has evolved unique genetic mechanisms that enables it to generate an almost unlimited number of different light & heavy chains in a remarkably economical way, by joining separate gene segments before TRANSCRIPTION.

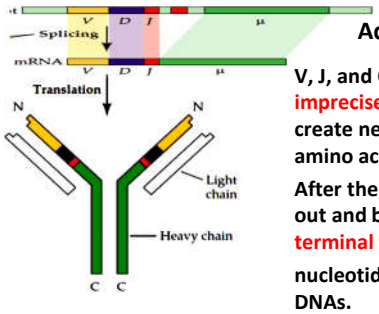
Birds and fish sheep and rabbits





The First Evidence
1970





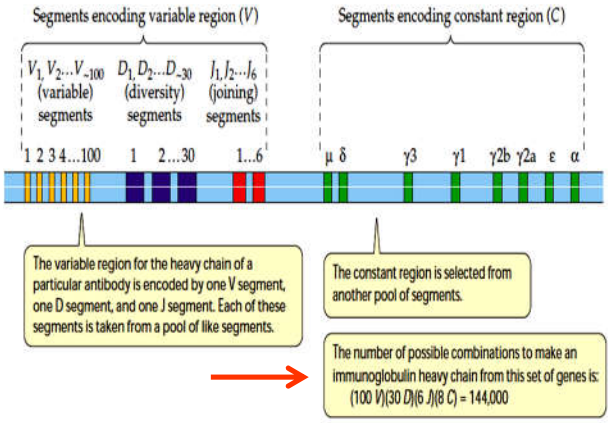
Additional Mechanisms

V, J, and C regions are rearranged imprecisely, errors at junctions create new codons resulting in amino acid changes.

After the DNA sequences are cut out and before they are joined, terminal transferase, adds some nucleotides to the free ends of the DNAs.

Immunoglobulin genes have relatively high mutation rate which creates new alleles & adds to antibody diversity.

Birds and fish sheep and rabbits.



Lecture 128

Antibody Diversity.

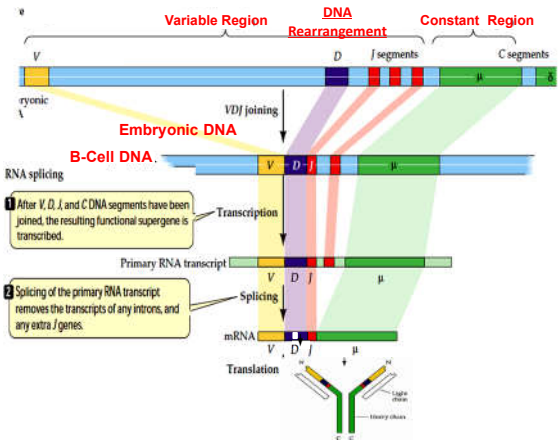
The Dilemma

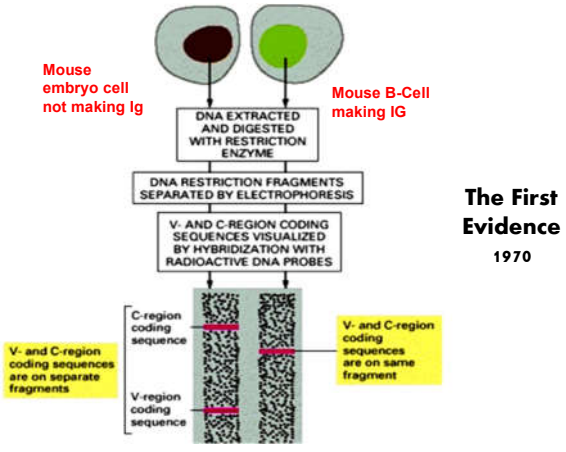
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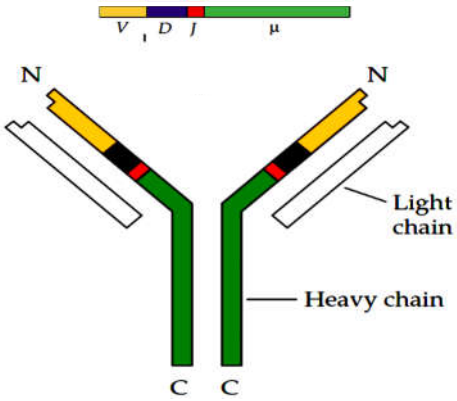
Birds and fish sheep and rabbits .

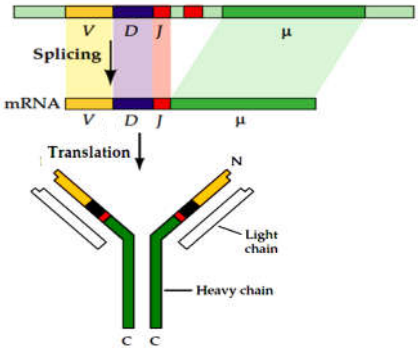


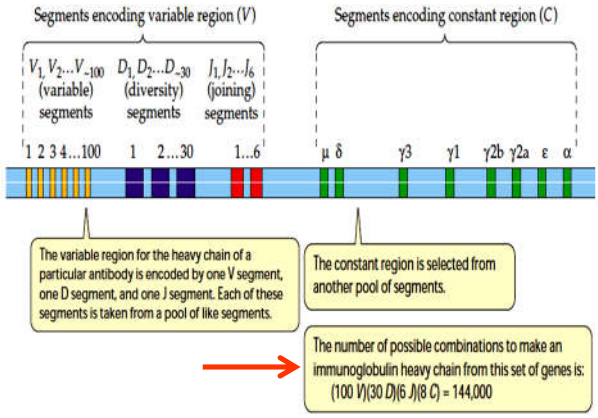


The First Evidence

1970







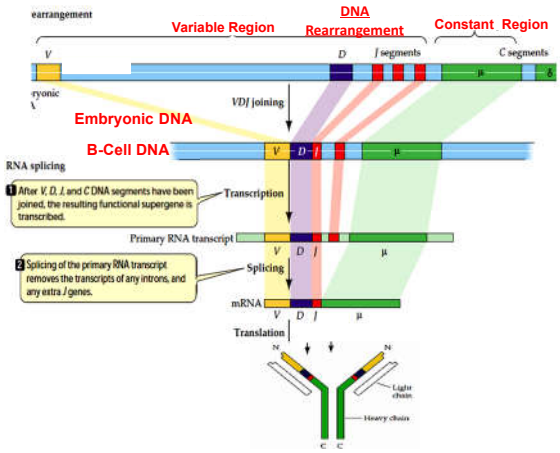
Additional Mechanisms.

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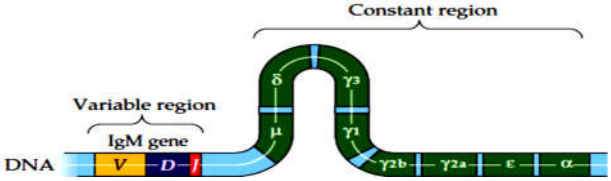
Birds and fish sheep and rabbits.



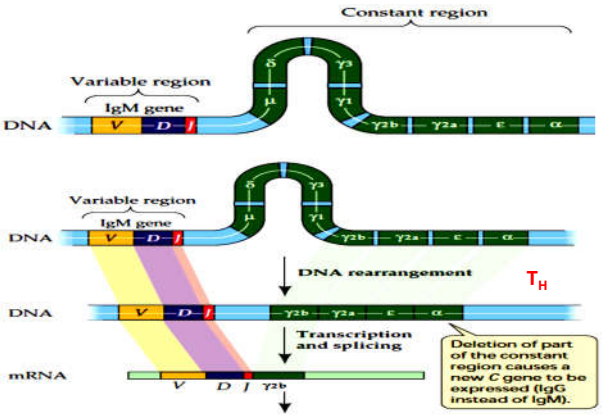
Lecture 129

Class Switching.

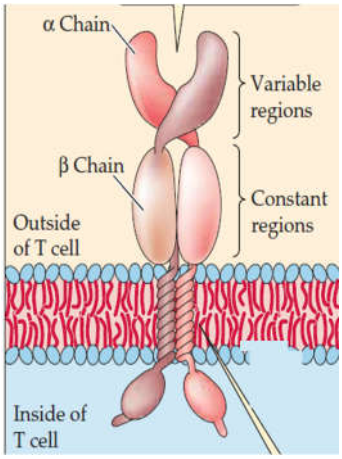
B-Cell initially produces IgM (antibody's heavy chain is encoded by the first constant region gene, the μ segment) molecules, which recognise specific antigenic determinant.



When B cell becomes a plasma cell another deletion occurs in the cell's DNA, which position heavy-chain variable region gene next to a different constant region (γ, ε, or α)



same variable regions of the light and heavy chains, and tantigen specificity



Heterodimer

Glycoproteins not immunoglobulins .

Two polypeptide chains, each encoded separate genes, homologous to BCR genes.

Source may be evolutionarily more ancient group of genes.

Lecture 130

T cells & TCR.

2 types : Cytotoxic T cells & Helper T cells.

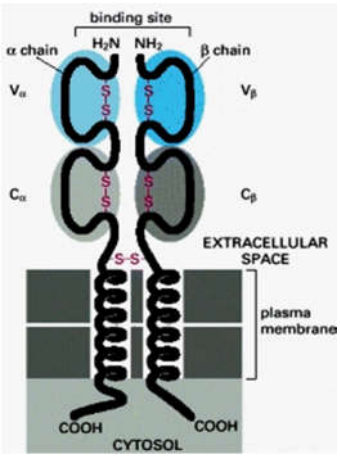
T cells eliminate intracellular pathogens.

T cells can only recognize antigen partly degraded fragments associated with MHC.

Effector cytotoxic T cells directly kill cells infected with a virus or some other intracellular pathogen.

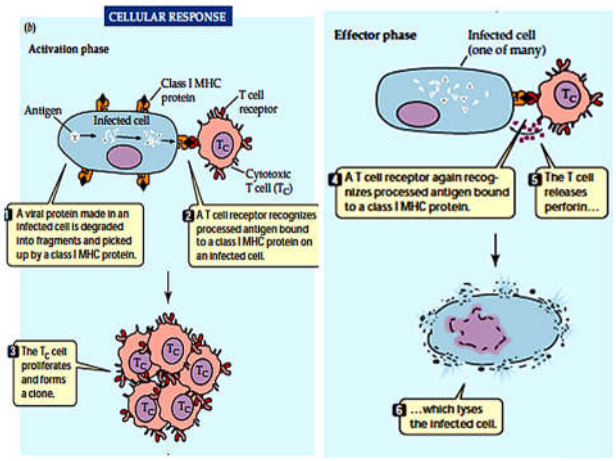
Effector helper T cells stimulate the responses of other cells phagocitic macrophages, B cells, and cytotoxic T cells.

Humoral Immunity.



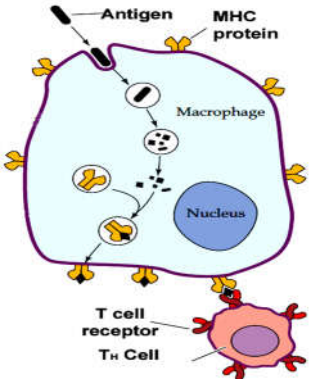
T cell receptor pools contain separate V, D, and J gene segments, which are brought together by site-specific recombination during T cell development in the thymus.

T cells recognize protein fragments of antigen that have been partly degraded inside the antigen-presenting cell.



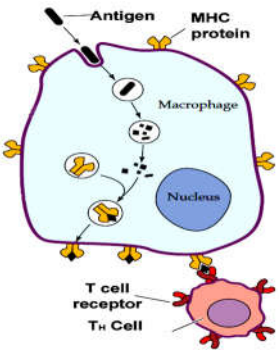
Lecture 131

APCs Activate T Cells.



T cells are initially activated in peripheral lymphoid organs by recognizing foreign antigen on the surface of an APC (dendritic cell, macrophage & B cell).

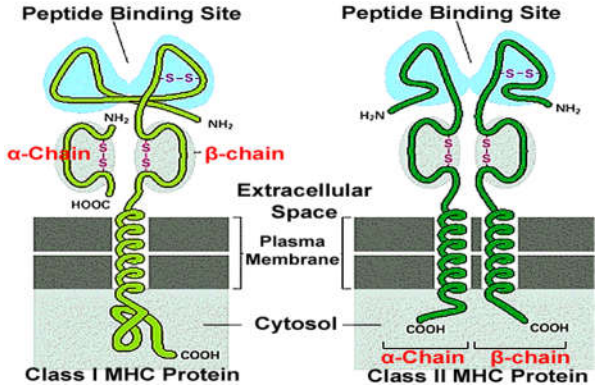
Antigen is peptide fragments that are generated by the degradation of foreign protein antigens inside the antigen-presenting cell.



The recognition process depends on the presence of MHC proteins on the antigen-presenting cell, MHC bind these fragments, carry them to the cell surface, and present them.

Activated, effector T cells then recognize the same peptide-MHC complex on the surface of the target cell they influence.

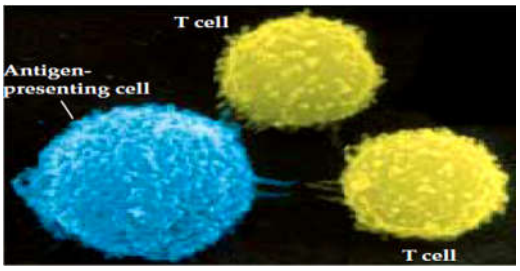
B cell, a cytotoxic T cell, or an infected macrophage in the case of a helper T cell, or a virus-infected cell in the case of a cytotoxic T cell.



both transmembrane heterodimers extracellular N-terminal domains that bind antigen for presentation to T

APC express major histocompatibility complex (MHC) genes which encode proteins that present antigens to the immune system.

There are two types of MHC molecules MHC class I and MHC class II with distinct functions.



Genetic Diversity

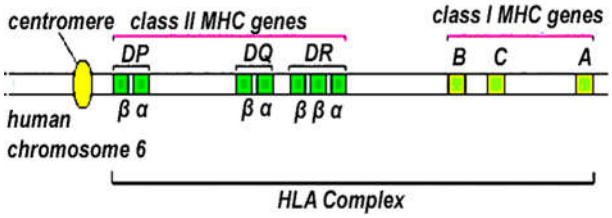
3 genes for class I proteins (HLA-A, B &C)

3 genes for class II proteins (HLA-DP, DQ &DR)

6 genes each for both the classes

Class II can α & β maternal & paternal can pair up

Polymorphism VS Gene Rearrangement.

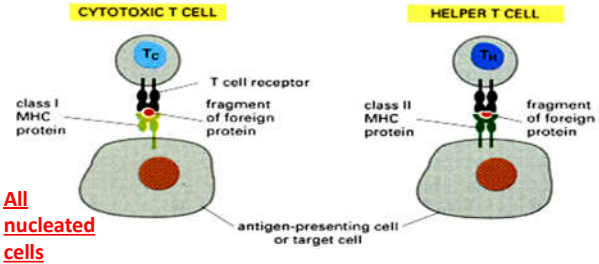


Rare for two unrelated individuals to have an identical set of MHC proteins.

Lecture 132

MHC.

2 main structurally & functionally distinct classes of MHC proteins: class I MHC proteins, present foreign peptides to cytotoxic T cells, & class II present foreign peptides to helper T cells.



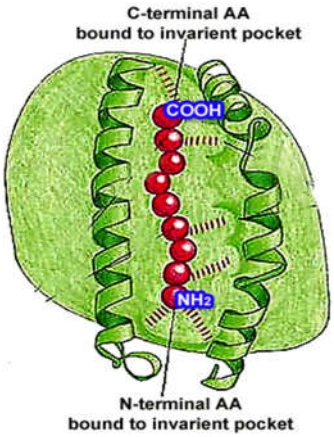
All nucleated cells

A person makes a small number of MHC proteins MHC should be able to present peptide fragments from almost any foreign protein to T cells.

MHC groove can accommodate an extended peptide about 8–10 amino acids long.

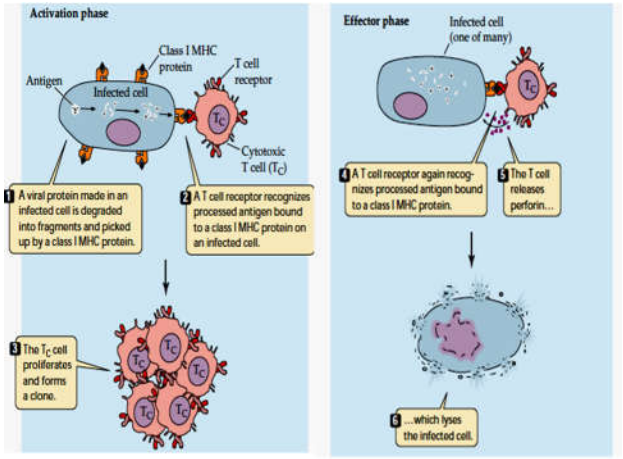
In extended conformation terminal amino group binds to an invariant pocket at one end of the groove and its terminal carboxyl group bound to an invariant pocket at the other end of the groove.

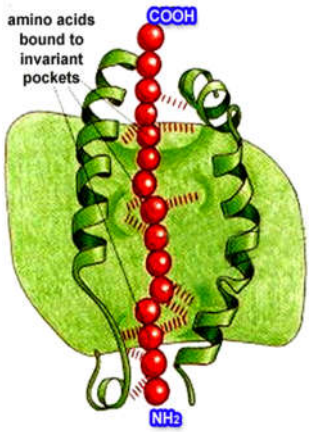
Pockets recognize peptide backbone features common to all peptides, each class I MHC protein can bind a peptides of diverse sequence.



Antigen Binding

The side chains of other amino acids of the peptide point outward, in a position to be recognized by receptors on cytotoxic T cells.

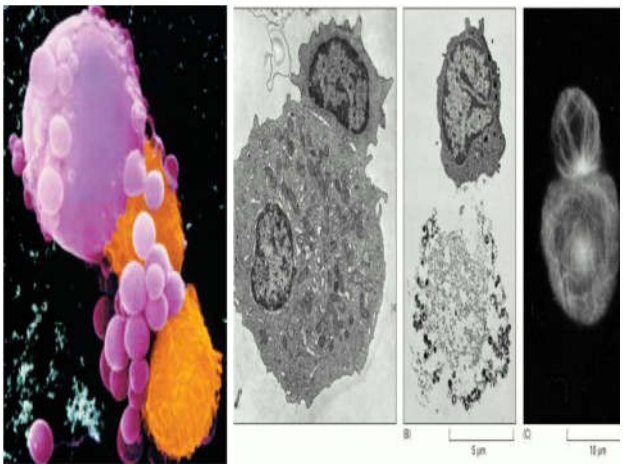




Class II can accommodate longer peptides, which are usually 13–17 amino acids.

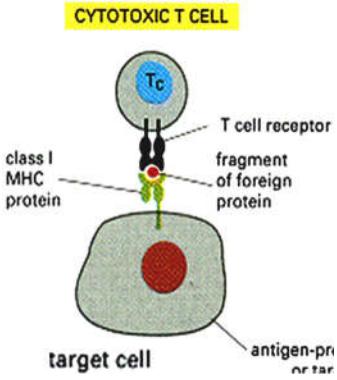
Peptide not bound by ends.

It is held in the groove by parts of its peptide backbone that bind to invariant pockets formed by conserved amino acids that line all class II MHC peptide-binding grooves

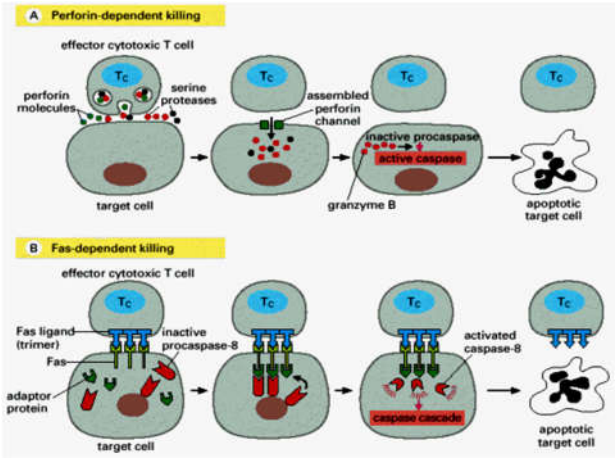


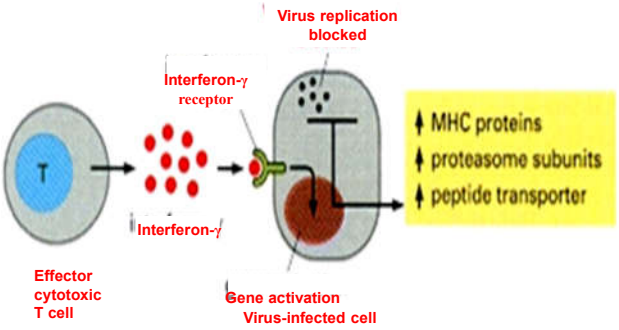
Lecture 133

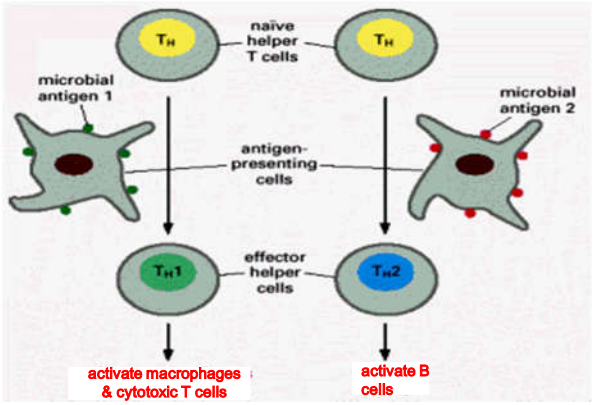
Cytotoxic T cells.



Cytotoxic T cells provide protection against intracellular pathogens such as viruses and some bacteria and parasites that multiply in the host-cell cytoplasm, where they are sheltered from attack by antibodies.



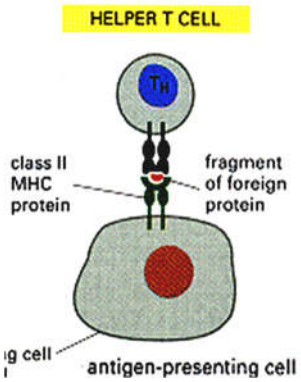




nature of the invading pathogen and the types of innate immune

Lecture 134

Helper T cells.



Helper T cells are crucial for defense against both extracellular and intracellular pathogens.

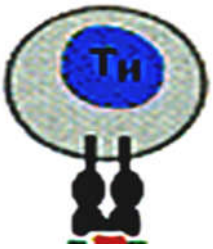
Once a helper T cell is activated by an antigen-presenting cell to become an effector cell, it then help activate other cells.

APCs provides two kinds of signals.

Signal 1 is provided by a foreign peptide bound to an MHC protein on the surface of the presenting cell, peptide-MHC complex signals through the T cell receptor and its associated proteins.

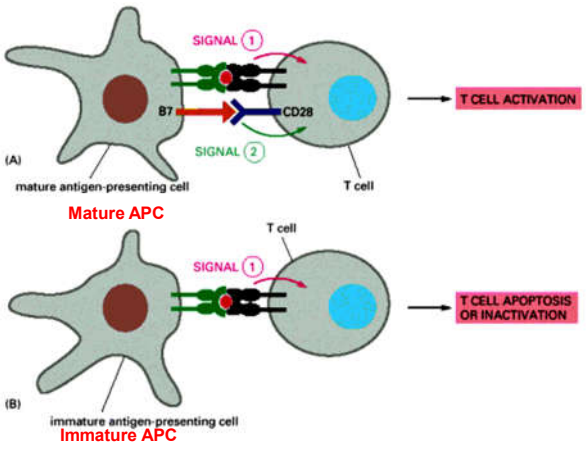
Signal 2 is provided by costimulatory proteins, especially the B7 proteins (CD80 and CD86), which are recognized by the co-receptor protein CD28 on the surface of the T cell.

The expression of B7 proteins on an antigen-presenting cell is induced by pathogens during the innate response to an infection.



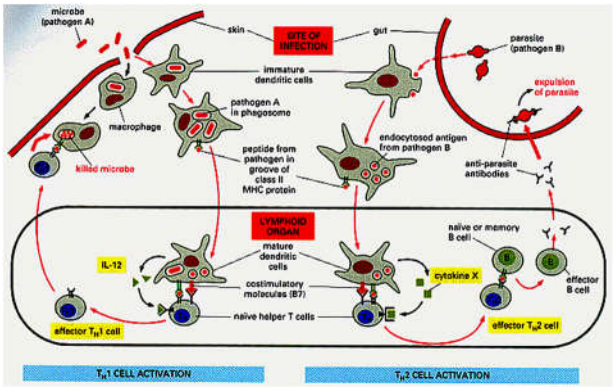
Helper T cells stimulate B cells to make antibodies that inactivate or eliminate extracellular pathogen & their toxic products.

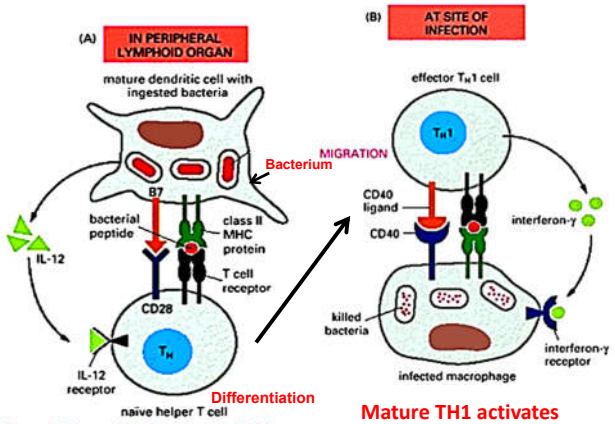
When activated by an antigen-presenting cell, a naïve helper T cell can differentiate into either of two distinct types of effector helper cell, called TH1 and TH2. TH1 cells mainly help activate macrophages and cytotoxic T cells, whereas TH2 cells mainly help activate B cells



Lecture 135 & 136

T_H1 & T_H2.



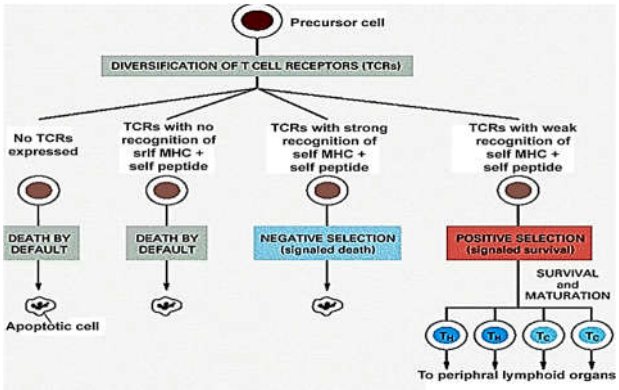


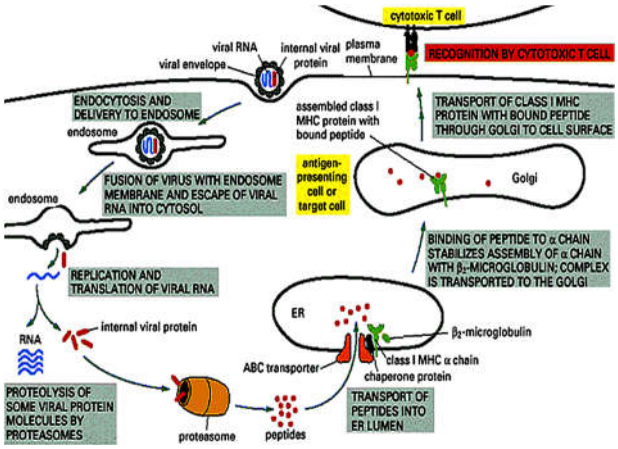
Dendritic cell Stimulates TH1

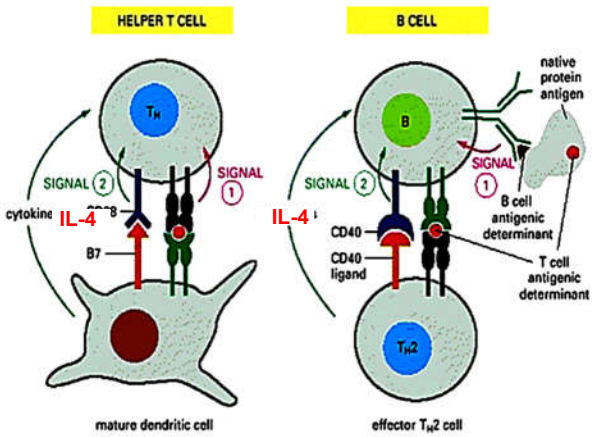
Mature TH1 activates Macrophage to kill

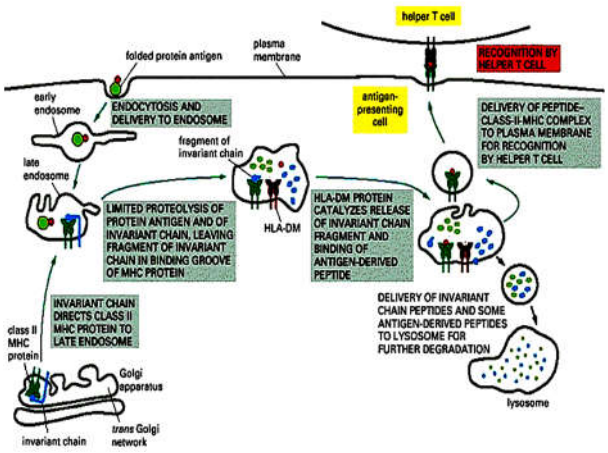
Lecture 137

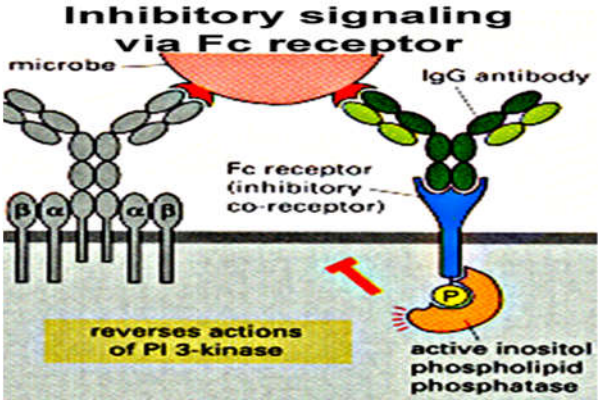
Checks & Balances.



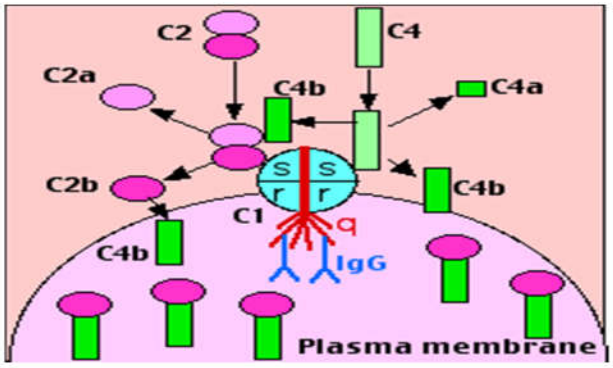








END



Lecture 138

Complement System .

Most of the time, the binding of antibodies to antigen performs no useful function until and unless it can activate an effector mechanism.Complement system serves several effector roles.

Complement system provides the actual protection from the response while the interaction of antibodies and antigen provides the specificity of the response.Put another way, antibodies "finger" the target, complement destroys it

The complement system consists of some 30 proteins circulating in blood plasma.These proteins are synthesized mainly in the liver, and they account for about 5% of the globulin fraction of blood serum.Most of these are inactive until .they are cleaved by a protease which, in turn, converts them into a protease.Thus many components of the system serve as the substrate of a prior component and then as an enzyme to activate a subsequent component

This binding leads to conformational changes in C1q molecule, which leads to the activation of two C1r (serine protease) molecules. Then they cleave C1s. (another serine protease)

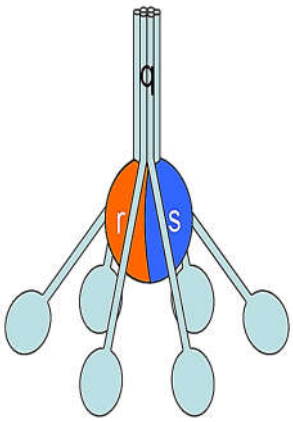
The C1-complex now binds to and splits C2 and C4, producing C2a and C4b

C4b, which binds covalently to sugar residues on cell-surface glycoproteins,& a smaller, inactive, fragment of C4a which diffuses away C2 is cleaved into C2b, which binds non-covalently .

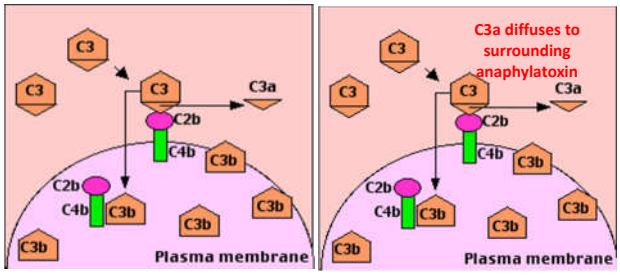
to a site on C4b, leaving a smaller, inactive, fragment of C2a which diffuses away

The complex of C4b•2b is called "C3 convertase" because it catalyzes the cleavage of C3

C3b binds covalently to glycoproteins scattered across the cell surface. Macrophages and neutrophils have receptors for C3b and can bind the C3b-coated cell or particle



The classical pathway is triggered by activation of the C1-complex (which consists of six molecule C1q and two molecules C1r and C1s), either by C1q's binding to antibodies from classes M and G, complexed with antigens, or by its binding C1q to the surface of the pathogen



The complex of C4b•2b is called "C3 convertase" it catalyzes the cleavage of C3. C3b binds covalently to glycoproteins scattered across the cell surface. Macrophages and neutrophils have receptors for C3b and can bind the C3b-coated cell or particle

This effect qualifies C3b as an opsonin

Histamine release from mast cells vascular permeability

Cleavage of C5:

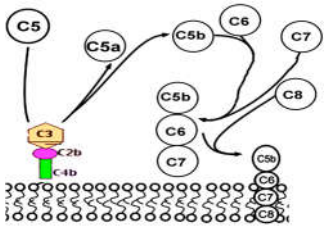
Some of the C3b binds to molecules of C5 creating an allosteric change that exposes them to cleavage by C4b•2b (which is thus a "C3/C5 convertase") Cleavage of C5 by the C3/C5 convertase initiates the assembly of a set of complement proteins that make up the membrane attack complex.

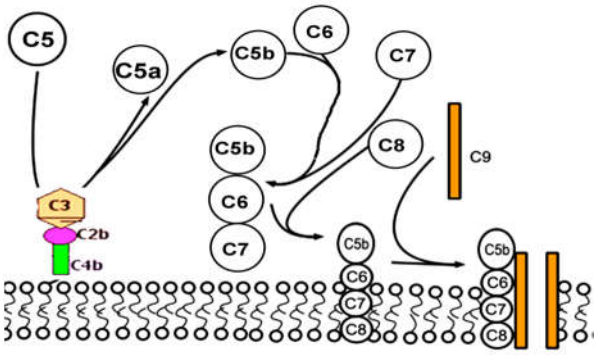
The membrane attack complex can also be formed by another C5 convertase produce by the "alternative pathway"

Cleavage of C5 produces:

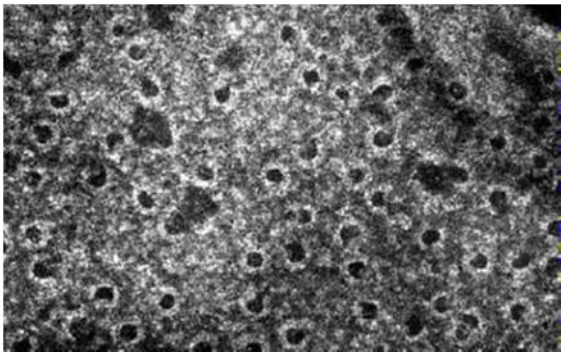
C5a, which is released into the fluid surroundings where it is a potent anaphylatoxin.

C5b, which serves as the anchor for the assembly of a single molecule each of C6, C7 and C8





C5b•6•7•8 guides the polymerization of 18 molecules of C9 inserted into the lipid bilayer passage H2O causing cell lyses.



electron micrograph shows holes punched through the cell wall of the bacterium components of the complement system

Lecture 139

Complement System 2:

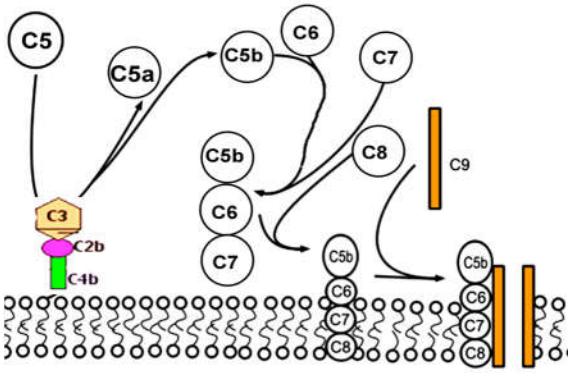
Cleavage of C5 :

Some of the C3b binds to molecules of C5 creating an allosteric change that exposes them to cleavage by C4b•2b (which is thus a "C3/C5 convertase") Cleavage of C5 by the C3/C5 convertase initiates the assembly of a set of complement proteins that make up the membrane attack complex.

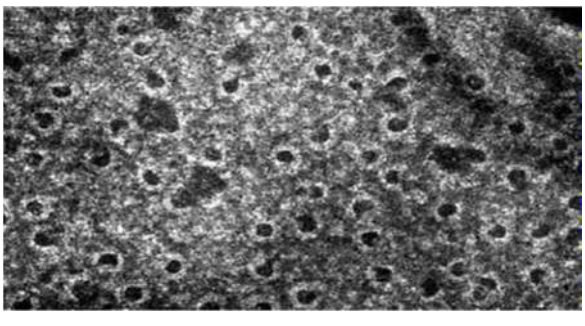
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Cleavage of C5 produces:C5a, which is released into the fluid surroundings where it is a potent anaphylatoxin

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C5b•6•7•8 guides the polymerization of 18 molecules of C9 inserted into the lipid bilayer passage H2O causing cell lyses.



electron micrograph shows holes punched through the cell wall of the bacterium components of the complement system

Lecture 140

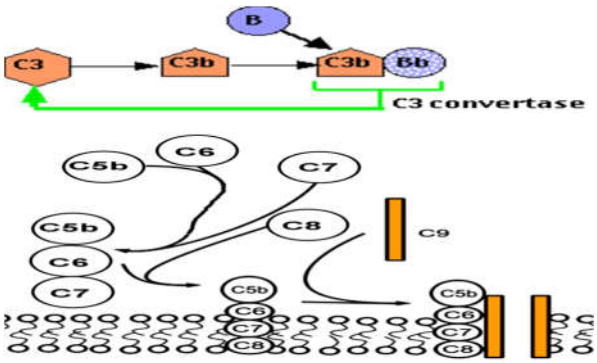
The Alternative Pathway:

The complement system can also be triggered without antigen-antibody complexes, Even in their absence, there is a spontaneous conversion of C3 to C3b,

Ordinarily the C3b is quickly inactivated: the C3b binds to inhibitory proteins and sialic acid present on the surface of the body's own cells, and the process is aborted.

However, bacteria and other foreign materials that may get into the body lack these proteins.

C3b binds a protein called Factor B forming a complex of C3b•Bb which is a C3 convertase, acts on more C3 to form: C3b•Bb•C3b, which is a C5 convertase and can start the assembly of the membrane attack complex & more C3b (amplification massive production of C3b)



The lectin pathway is homologous to the classical pathway, but with the opsonin, mannan-binding lectin (MBL) & ficolins, instead of C1q.

This pathway is activated by binding mannan-binding lectin to mannose residues on the pathogen surface, which activates the MBL-associated serine proteases, MASP-1, MASP-2, MASP-3, which can split C4 into C4a and C4b & C2 into C2a and C2b,C4b and C2a then bind together to form C3-convertase, as in the classical pathway.

The explosive potential of the complement system requires that it be kept under tight control, at least 12 proteins are known that do this Factor H removes Bb from the alternative pathway C3 convertase breaking the positive feedback loop. Factor I inactivates C3b,C1 inhibitor (C1INH) binds to sites on activated C1r and C1s shutting down their proteolytic activity, when C1 is activated by antigen-antibody complexes, there is only a brief interval during which it can cleave C4 and C2 before it is deactivated by C1INH.

Lecture 141

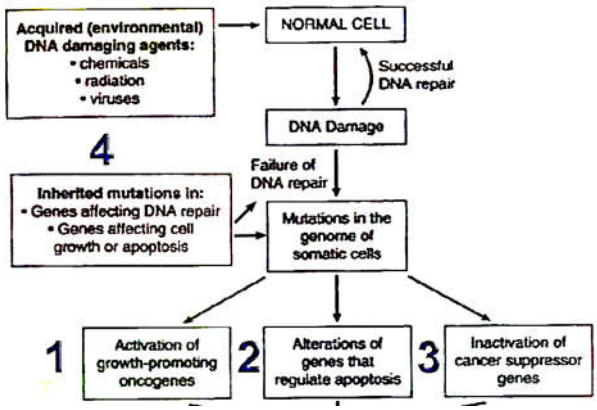
Neoplastic Transformations:

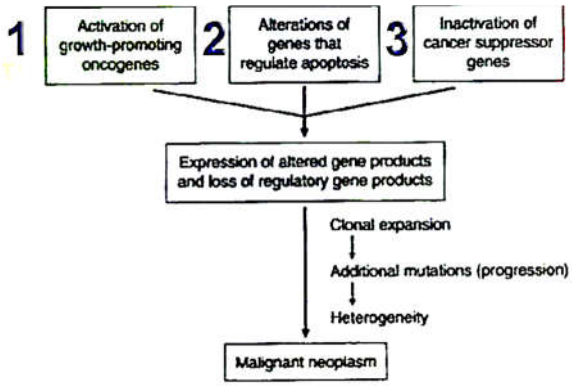
Proto-oncogene:

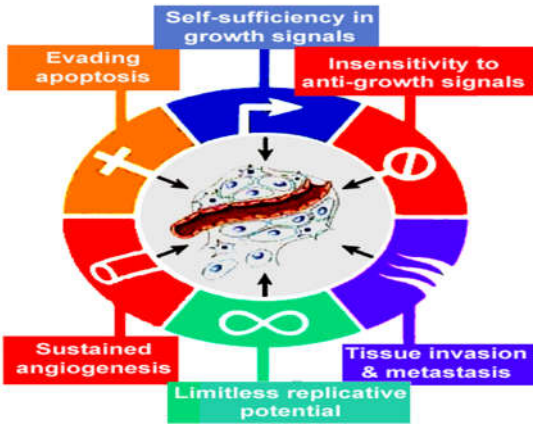
A proto-oncogene is a normal gene that can become an oncogene, either after mutation or increased expression. Proto-oncogenes code for proteins that help to regulate cell growth& differentiation.

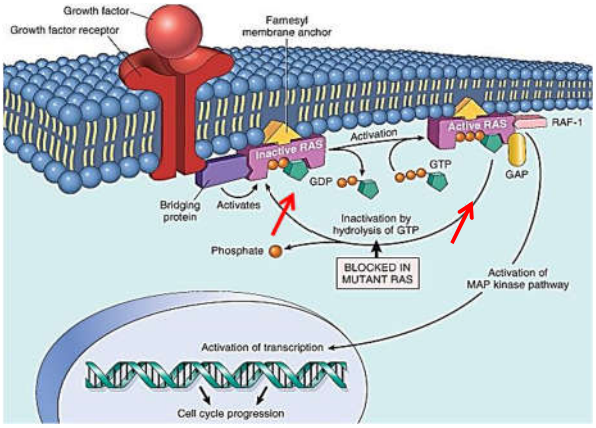
Proto-oncogenes are often involved in signal transduction and execution of mitogenic signals, usually through their protein products. Upon activation, a proto-oncogene (or its product) becomes a tumor inducing agent, an oncogene.
Some oncogenes, usually involved in early stages of cancer development, increase the chance that a normal cell develops into a tumor cell, possibly resulting in cancer.

Classes of Oncogenes: Growth Promoting Oncogenes, Growth-Inhibiting Cancer Suppressor Genes, DNA Repairing Genes, Apoptosis Genes









Lecture 142

Growth Promoting Oncogenes:

Autonomous Cell Growth In Cancer Cells Is Promoted By Following Elements

a) Growth Factors

b) Growth Factor Receptors

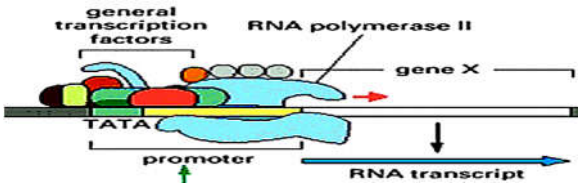
c) Signal-Transducing Proteins (secondary messengers)

d) Nuclear Transcription Factors

e) Cyclins and Cyclin Dependent Kinases (CDKs)

Some cancer cells acquire the ability through mutations to produce growth factors to which they are responsive (autocrine signaling).

Many glioblastomas secrete PDGF and sarcomas make TGF- α. Similar loops are fairly common many types of cancers



MYC and the polymerase assemble on a specific DNA sequence and drive transcription of genes.

MYC protooncogene is expressed in virtually all the cells, & the MYC protein is rapidly induced when cells receive a signal to divide.

MYC binds DNA and causes transcriptional activation of many growth related proteins.

Dysregulation of of MYC gene occurs in Burkitt Lymphoma (B-cell tumor); MYC is also amplified in many other cancers.

Mutated Growth Factor Receptors continue to deliver continues signals in the absence of the ligand.

EGF receptor ERBB1 is over expressed in 80% of SCC of lung.

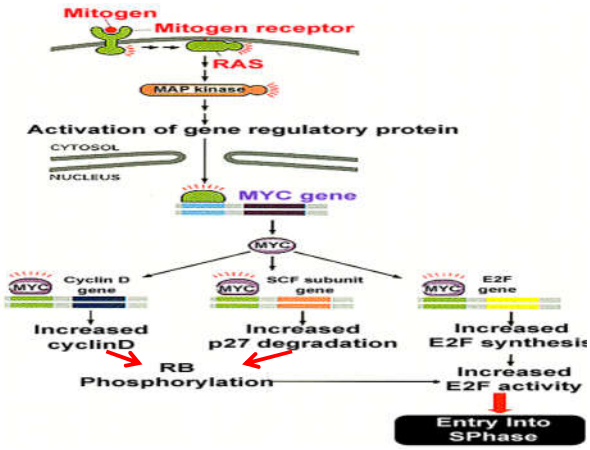
ERBB2/HER2 is amplified in 25%-30% of breast cancers and adenocarcinomas of lung , ovary and salivary glands.

These tumours are sensitive to small amounts of growth factors.

Mutaion of RAS is the most common oncogene abnormality, approximately 30% of all human tumors contain mutated version of RAS gene.

GAP/neurofibramin 1 (a RAS deactivating protein) is mutated in neurofibramatosis type 1.

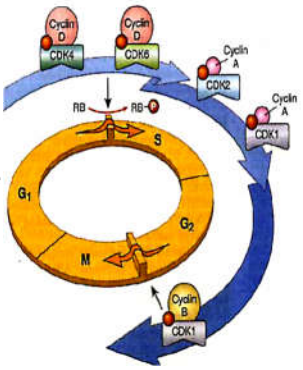
BCR-ABL hybrid gene activates RAS pathways (like p53 Normal ABL gene also promotes apoptosis via NLS domain).



The binding of mitogens to cell-surface receptors leads to the activation of Ras and a MAP kinase cascade.

One effect of this pathway is the increased production of the gene regulatory protein Myc.

Myc increases the transcription of several genes, including the gene encoding cyclin D



Lecture 144

Growth-Inhibiting Cancer Suppressor Genes

Cancer Suppressor Genes

The signals and the signal-tranducing pathways for growth inhibition are less well understood than those for growth promotion.

The antigrowth signals may cause dividing cells to go into G0 (dormancy) where they remain signalled not to divide or they may cause cells to enter postmitotic differentiation.

Activated retinoblastoma (RB) protein prevents cells from advancing from G1 to S phase.

Lecture 143

Growth Promoting Oncogenes Cyclins & CDKs:

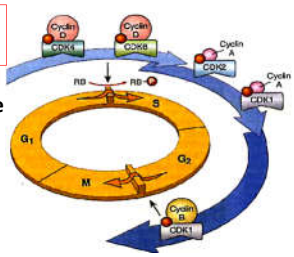
CDKs Phosphorilate crucial target proteins & are expressed constitutively during cell cycle in inactive form however cyclins are synthesized during specific phases of cell cycle.

Activated CDK4 and CDK6 phosphorylate RB which over come the hurdle of G1 to S hurdle

The activity of CDKs is regulated by CDK inhibitors (CDKIs)

Cyclin D genes are over expressed in many cancers, e.g. affecting the breast, esophagus, liver & subset of lymphomas

Mutations affecting other cyclins and CDKs also occur in some neoplasms.



RB First Tumor Suppressor Gene:

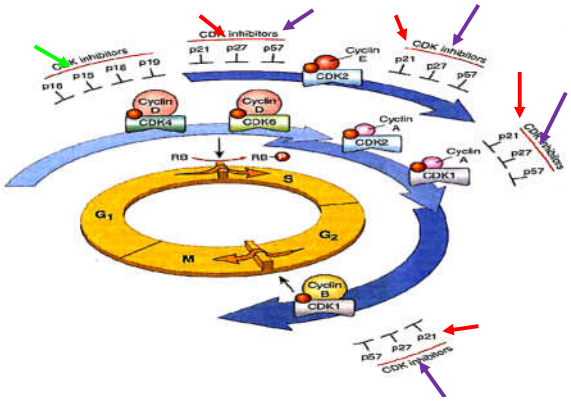
RB gene product is a DNA binding protein and is ubiquitously expressed.

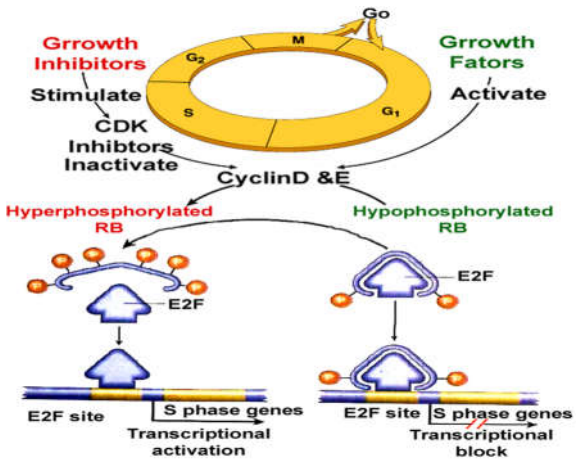
Hypophosphorilated (active) RB serves as a break for G1 to S phase transition, Hyperphosphorilated (inactive) RB allows cells to divide.

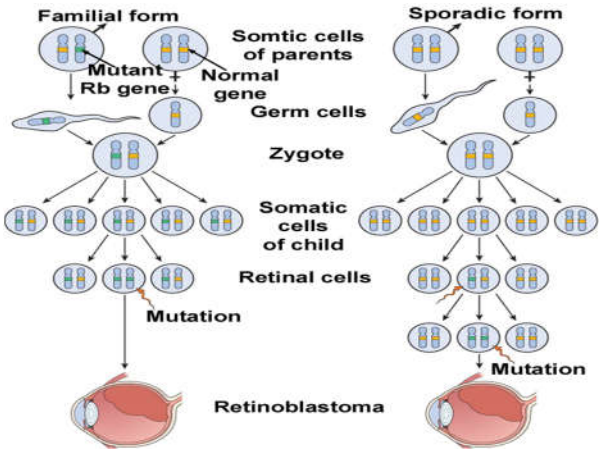
Hypophosphorilated RB binds transcriptional factor E2F when RB is phosphorilated by CDK4, CDK6 and CDK2 it releases E2F.

RB target of some oncogenic viruses neutralizing

TGF- β stimulates production of p15 and inhibits the transcription of CDK2, CDK4 and cyclin A

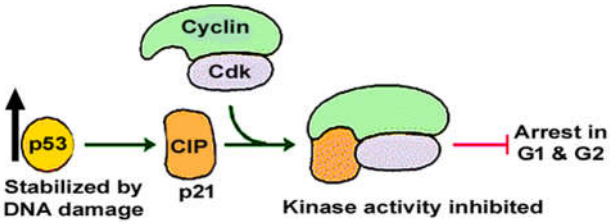




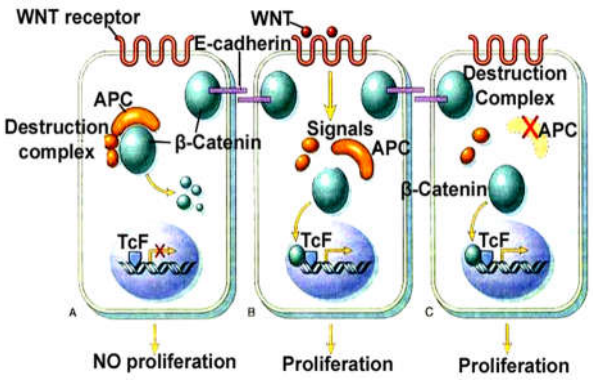


The normally unstable p53 protein is stabilized by damaged DNA, so its concentration increases.

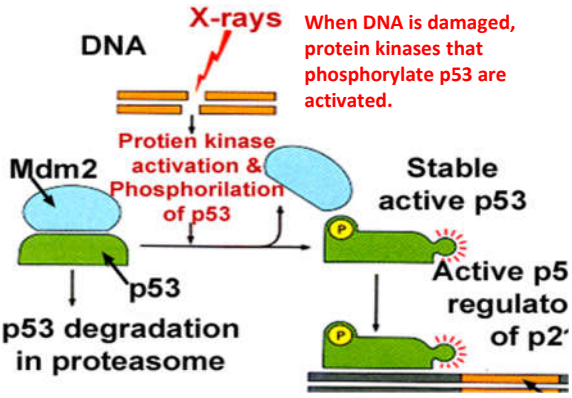
Acting as a transcription factor, p53 induces expression of p21, a cyclin-kinase inhibitor that inhibits all Cdks (Cdk1-, Cdk2-, Cdk4-, and Cdk6-cyclin complexes).



Binding of p21 to these Cdk-cyclin complexes leads to cell cycle arrest in G1 and G2.



APC Suppresses Cell Division by Degrading β-Catenin; a Transcription Factor Activator 70-80%



When DNA is damaged, protein kinases that phosphorylate p53 are activated.

Mdm2 normally binds to p53 and promotes its ubiquitylation and destruction in proteasomes.

Lecture 145

DNA Repairing Genes:

p53 has many anti-cancer mechanisms: It can activate DNA repair proteins when DNA has sustained damage.

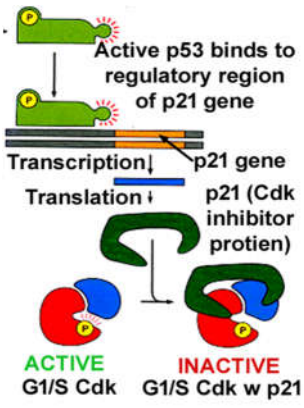
p53 can also arret the cell cycle at the G1/S regulation point on DNA damage recognition (if it holds the cell here for long enough, the DNA repair proteins will have time to fix the damage & the cell will be allowed to continue the cell cycle.)

It can initiate apoptosis, the programmed cell death, if the DNA damage proves to be irreparable.

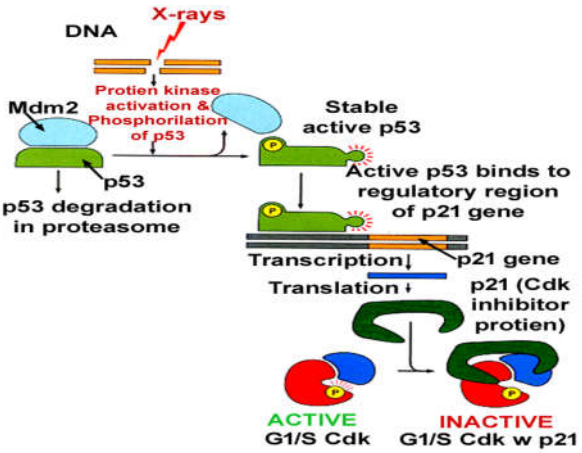
p53 is one of the most commonly mutated genes in human cancers (it can sense DNA damage via other proteins e.g. ATM).

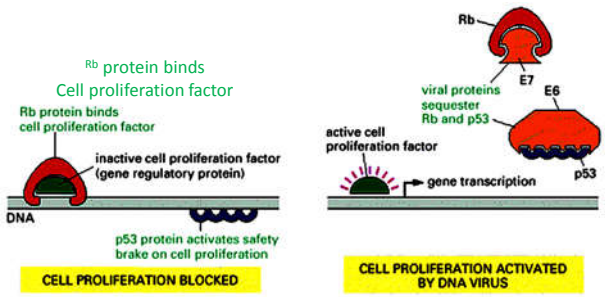
p53 has an anti-proliferative effects and also regulates apoptosis.

Phosphorylation of p53 blocks its binding to Mdm2; as a result, p53 accumulates to high levels and stimulates transcription of the gene that encodes the CKI protein p21.



The p21 binds and inactivates G1/S-Cdk and S-Cdk complexes, arresting the cell in G1. In some cases, DNA damage also induces either the phosphorylation of Mdm2 or a decrease in Mdm2 production, which causes an increase in p53 .

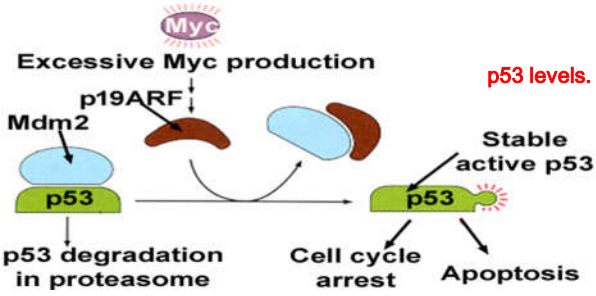




Papillomavirus uses two viral proteins, E6 and E7, to sequester the host cell's p53 and Rb. The SV40 virus uses a single dual-purpose protein called large T antigen, for the same purpose.

(a related virus that infects monkeys)

Abnormally high levels of Myc cause the activation of p19ARF, which binds and inhibits Mdm2 and thereby causes increased

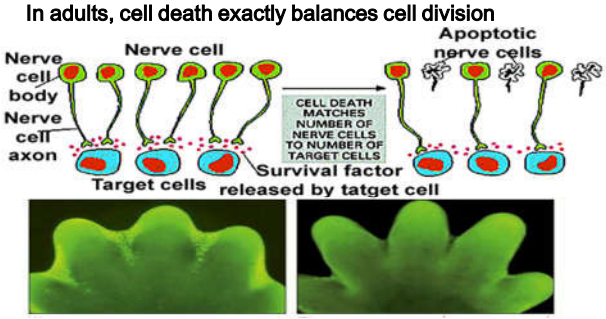


Depending on the cell type p53 then causes either cell-cycle arrest or apoptosis.

Lecture 146

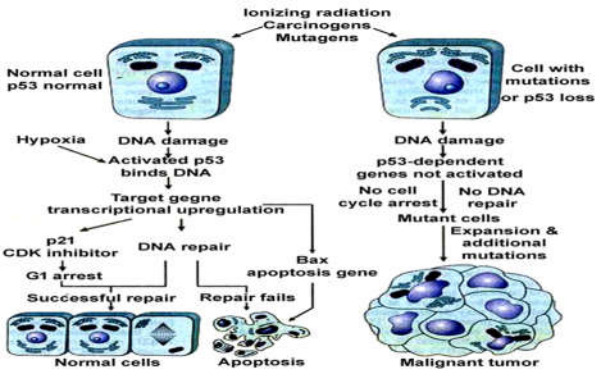
Apoptosis Genes:

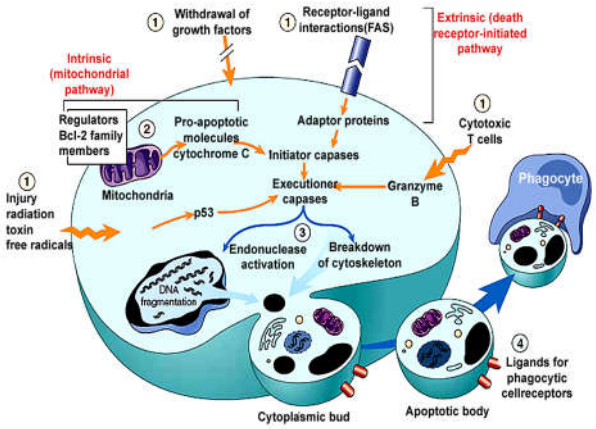
Billions of bone cells die in a healthy adult human
In adults, cell death exactly balances cell division

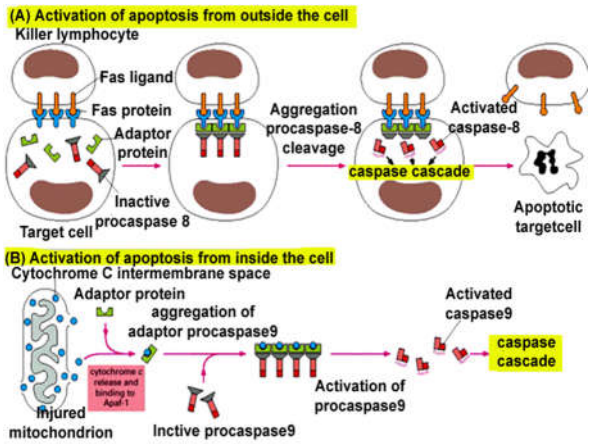


Sculpting the digits in the developing mouse paw by apoptosis)

One of the P53 alleles is mutated which causes 25 fold increased chance of developing a malignant tumor by age 50







Lecture 147

Protein Sorting:

Protein Sorting Essential for Sub-cellular Specialization :

Cytoplasm elaborately subdivided into functionally distinct, membrane-enclosed compartments.

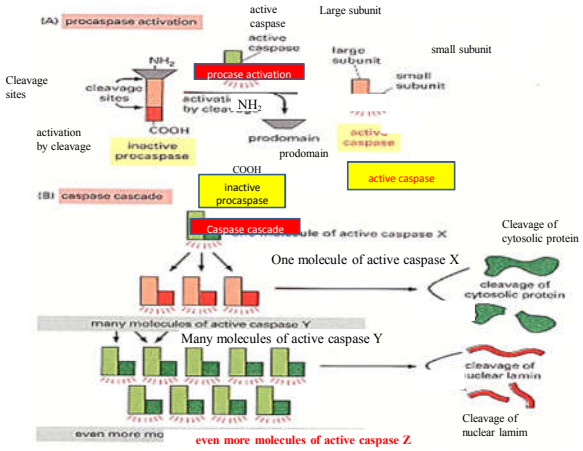
All most all the organelles are membrane enclosed & membrane-enclosed organelles are present in all eukaryotes.

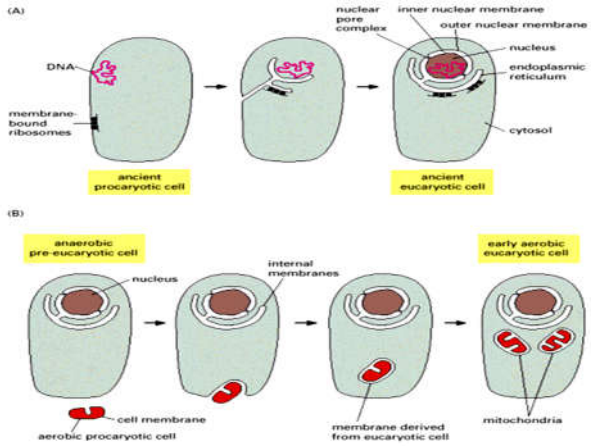
Each organelle, contains its own characteristic set of enzymes and other specialized molecules.

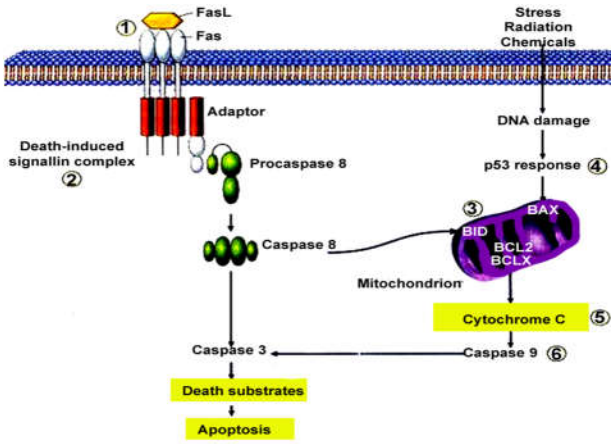
Proteins confer upon each compartment its characteristic structural & functional properties.

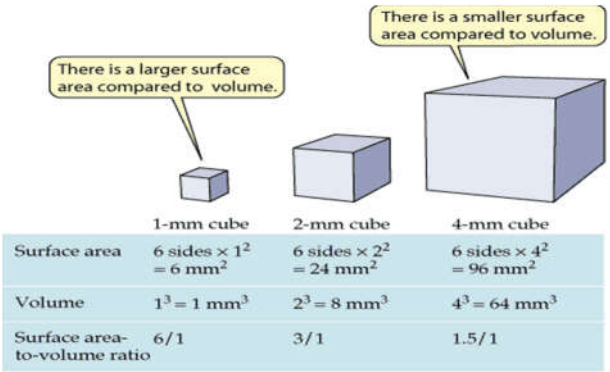
Specialized Compartments Division of Labor

Vital biochemical processes take place in or on membrane









	1-mm cube	2-mm cube	4-mm cube
Surface area	6 sides × 1^2 = 6 mm^2	6 sides × 2^2 = 24 mm^2	6 sides × 4^2 = 96 mm^2
Volume	1^3 = 1 mm^3	2^3 = 8 mm^3	4^3 = 64 mm^3
Surface area-to-volume ratio	6/1	3/1	1.5/1

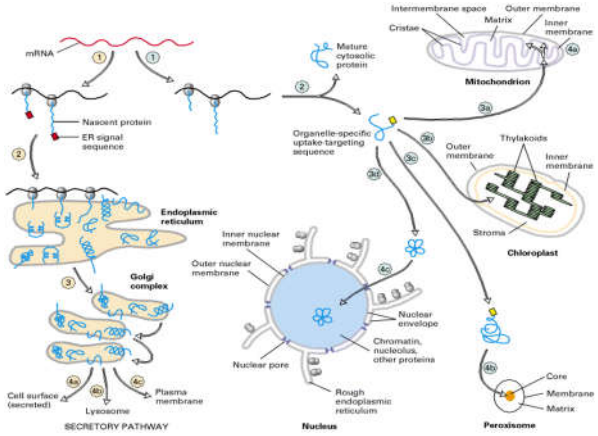
Greater Surface Area for Chemical Reactions Allows Cells to Have Bigger Size

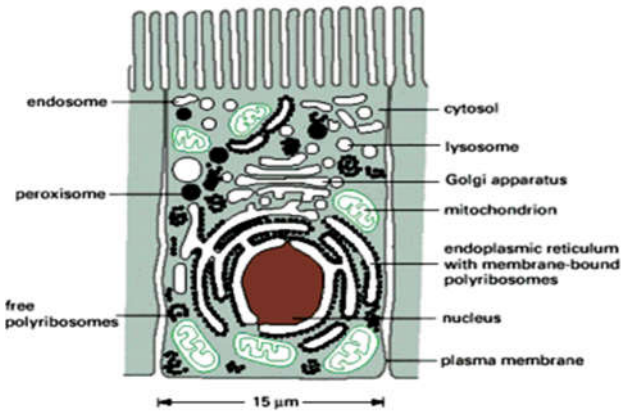
To understand the eucaryotic cell, it is essential to know how molecules move between them, and how the compartments are maintained.

An animal cell contains about 10 billion (10^{10}) protein molecules 10,000–20,000 kinds, synthesis of almost all of begins in cytosol.

Each newly synthesized protein is delivered specifically to compartment that requires it.

THE CHALLENGE: Lipid bilayer of organelle membranes is impermeable to most hydrophilic molecules





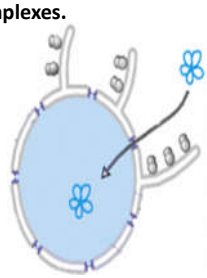
Lecture 148

Protein Sorting Mechanisms:

Gated Transport:

The protein traffic between the cytosol and nucleus occurs between topologically equivalent spaces, which are in continuity through the nuclear pore complexes.

The nuclear pore complexes function as selective gates that actively transport specific macromolecules and macromolecular assemblies, although they also allow free diffusion of smaller molecules 5000 daltons or <



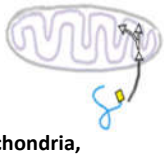
Membrane of each organelle contains membrane transport proteins that are responsible for the import and export of specific metabolites

Each organelle membrane has a mechanism for importing, & incorporating specific proteins that makes the organelle unique.

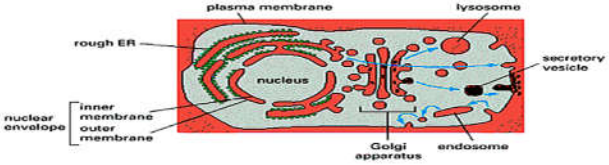
All proteins synthesis initiates in the cytosol, (mitochondria & plastids), their subsequent fate depends on their amino acid sequence, which can contain sorting signals that direct their delivery to locations outside the cytosol, no sorting signal cytosol permanent residents.

Transmembrane Transport

Membrane-bound protein translocators directly transport specific proteins across a membrane from the cytosol into a space that is topologically distinct



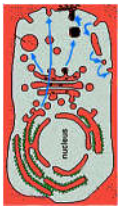
The transported protein molecule usually must unfold to snake through the translocator (initial transport of selected proteins from the cytosol into the ER lumen or from the cytosol into mitochondria, for example, occurs in this way)



Vesicular Transport

Membrane-enclosed transport intermediates—may be small, spherical transport vesicles or larger, irregularly shaped organelle fragments—ferry proteins from one compartment to another.

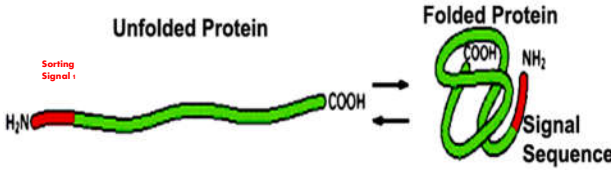
The transport vesicles become loaded with a cargo of molecules derived from the lumen of one compartment as they pinch off from its membrane; they discharge their cargo into a second compartment by fusing with that compartment.



The transfer of soluble proteins from the ER to the Golgi apparatus, transported proteins do not cross a membrane, vesicular transport moves proteins between topologically equivalent compartments

FUNCTION OF SIGNAL SEQUENCE	EXAMPLE OF SIGNAL SEQUENCE
Import into nucleus	-Pro-Pro-Lys-Lys-Lys-Arg-Lys-Val-
Export from nucleus	-Leu-Ala-Leu-Lys-Leu-Ala-Gly-Leu-Asp-Ile-
Import into mitochondria	$^{+}H_3N$-Met-Leu-Ser-Leu-Arg-Gln-Ser-Ile-Arg-Phe-Phe-Lys-Pro-Ala-Thr-Arg-Thr-Leu-Cys-Ser-Ser-Arg-Tyr-Leu-Leu-
Import into plastid	$^{+}H_3N$-Met-Val-Ala-Met-Ala-Met-Ala-Ser-Leu-Gln-Ser-Ser-Met-Ser-Ser-Leu-Ser-Leu-Ser-Ser-Asn-Ser-Phe-Leu-Gly-Gln-Pro-Leu-Ser-Pro-Ile-Thr-Leu-Ser-Pro-Phe-Leu-Gln-Gly-
Import into peroxisomes	-Ser-Lys-Leu-COO^{-}
Import into ER	$^{+}H_3N$-Met-Met-Ser-Phe-Val-Ser-Leu-Leu-Leu-Val-Gly-Ile-Leu-Phe-Trp-Ala-Thr-Glu-Ala-Glu-Gln-Leu-Thr-Lys-Cys-Glu-Val-Phe-Gln-
Return to ER	-Lys-Asp-Glu-Leu-COO^{-}

Some characteristic features of the different classes of signal sequences are highlighted in color. Where they are known to be important for the function of the signal sequence, positively charged amino acids are shown in *red* and negatively charged amino acids are shown in *green*. Similarly, important hydrophobic amino acids are shown in *yellow* and hydroxylated amino acids are shown in *blue*. $^{+}H_3N$ indicates the N-terminus of a protein; COO^{-} indicates the C-terminus.



Sorting signal1 (SS) : Continuous stretch of amino acid sequence, 15–60 residues. Some SS removed from finished protein by specialized signal peptidases after sorting is complete.

SS1 direct proteins from the cytosol into the ER, mitochondria, chloroplasts, and peroxisomes.

Lecture 149

Nuclear Transport:

Nuclear envelope encloses DNA & defines nuclear compartment

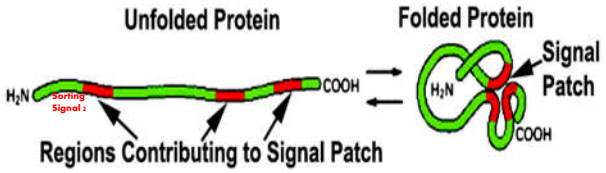
Envelope consists of 2 concentric membranes that are penetrated by nuclear pore complexes.

The inner & outer membranes are continuous & maintain distinct protein compositions.

The inner nuclear membrane contains specific proteins that act as binding sites for chromatin & protein meshwork of the nuclear lamina that provides structural support.

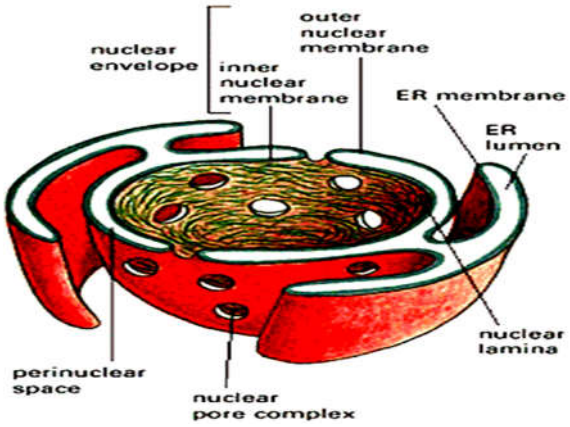
The inner membrane is surrounded by the outer nuclear membrane, which is continuous with the membrane of the ER

The outer nuclear membrane is studded with ribosomes engaged in protein synthesis



Sorting signal 2 (SS) : Consists of a specific three-dimensional arrangement of atoms on the protein's surface that forms when protein folds.

The amino acid residues that comprise this signal patch can be distant from one another in the linear amino acid sequence, and they generally persist in the finished protein



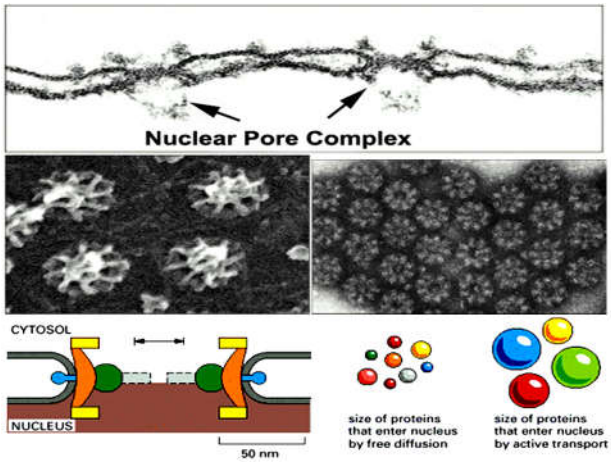
Many proteins that function in nucleus—including histones, DNA and RNA polymerases, gene regulatory proteins, & RNA-processing proteins—are selectively imported into the nuclear compartment from the cytosol,

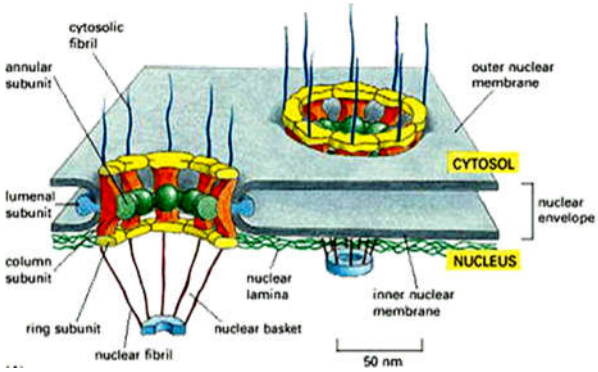
At the same time, tRNAs and mRNAs are synthesized in the nuclear compartment and then exported to cytosol

Bidirectional traffic between the cytosol and the nucleus

Like the import process, the export process is selective; mRNAs, for example, are exported only after they have been properly modified by RNA-processing reactions in the nucleus

Ribosomal proteins, are made in the cytosol, imported into the nucleus—where they assemble with newly made ribosomal RNA into particles—and are then exported again to the cytosol as part of a ribosomal subunit





Eukaryotic nuclear envelope is perforated by large, elaborate structures called nuclear pore complexes which selectively facilitate nuclear transport

equivalent to a water-filled cylindrical channel

Cells synthesizing DNA, need to import about 10^6 histone molecules from the cytosol every 3 minutes to package DNA into chromatin, ie each pore complex needs to transport about 100 histone molecules/minute

Rapidly growing cell need to transport about 6 large & small ribosomal subunits/minute from nucleus, where they are produced, to the cytosol.

Each pore complex contains one or more open aqueous channels through which small water-soluble molecules can passively diffuse.

Each complex has a molecular mass of about 125 million & is composed of aprox 50 different proteins, called nucleoporins, arranged octagonal symmetry

Small molecules (5000 daltons or less) diffuse in so fast that the nuclear envelope can be considered to be freely permeable to them, a protein of 17,000 daltons takes 2 minutes to proteins larger than 60,000 daltons are unable to enter the nucleus.

(injecting labeled water-soluble molecules)

Nuclear pore contains a pathway for free diffusion 9 nm in diameter & 15 nm long, this channel occupies a small fraction of total volume of the pore complex.The more active the nucleus is in transcription, the greater the number of pore complexes.

The nuclear envelope of a typical mammalian cell contains 3000–4000 pore complexes.

Lecture 150

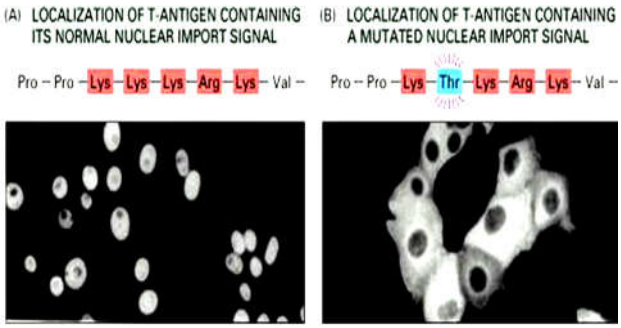
Nuclear Localization Signals (NLSs) & Transport:

Import into nucleus -Pro-Pro-Lys-Lys-Lys-Arg-Lys-Val-

When proteins are experimentally extracted from the nucleus and reintroduced into the cytosol even the very large ones re-accumulate efficiently in the nucleus.

The selectivity of this nuclear import process resides in nuclear localization signals (NLSs), which are present only in nuclear proteins.

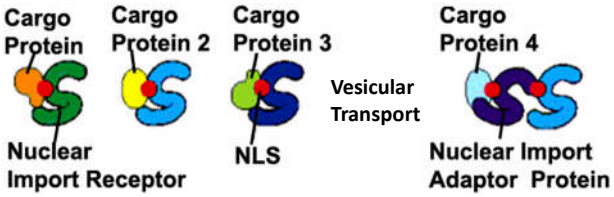
The signals have been precisely defined in numerous nuclear proteins by using recombinant DNA technology



Immunofluorescence micrographs showing the cellular location of SV40 virus T-antigen containing or lacking a short peptide that serves as a nuclear localization signal

Lecture 151

Nuclear Receptors.



To initiate nuclear import, NLS must be recognized by nuclear import receptors, encoded by family of related genes.

Each family member encodes a receptor protein that is specialized for the transport of a group of nuclear proteins sharing structurally similar nuclear localization signals.

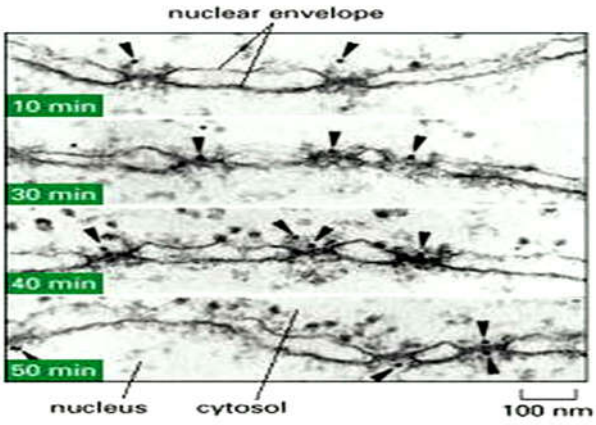
The mechanism of macromolecular transport across nuclear pore complexes is different from the transport mechanisms involved in protein transfer across the membranes of other organelles, because it occurs through a large aqueous pore rather than through a protein transporter spanning one or more lipid bilayers.

Nuclear proteins can be transported through a pore complex while they are in a fully folded conformation

A newly formed ribosomal subunit is transported out of the nucleus as an assembled particle

By contrast, proteins have to be extensively unfolded during their transport into most other organelles Very large particles traversing the pore seem to constrict as they squeeze through the nuclear pore complex, indicating that at least some of them must undergo restructuring during transport e.g. very large mRNAs

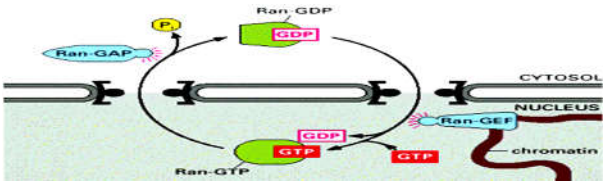
- **FG-repeats line the path through the nuclear pore complexes taken by the import receptors and their bound cargo proteins.**
- **These protein complexes move along the path by repeatedly binding, dissociating, and then re-binding to adjacent repeat sequences.**
- **Once in the nucleus, the import receptors dissociate from their cargo and are returned to the cytosol .**
- **Nuclear import receptors do not always bind to nuclear proteins directly.**
- **Additional adaptor proteins are sometimes used that bridge between the import receptors and the nuclear localization signals on the proteins to be transported.**
- **The combined use of import receptors and adaptors allows a cell to recognize the broad repertoire of nuclear localization signals that are displayed on nuclear proteins .**



Nuclear export like nuclear import, but in reverse.Nuclear export of large molecules, e.g. ribosome subunit & RNAoccurs through nuclear pore & depends on a selective transport system.Nuclear export signals are present on molecules to be exported, & on nuclear export receptors.These receptors bind both the export signal and nucleoporins to guide their cargo through the pore complex to the cytosol.Export & import receptors are structurally related encoded by the same gene family karyopherins . Ran GTPase Drives Directional Transport.

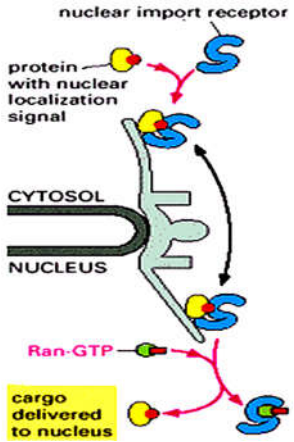
The import of nuclear proteins through the pore complex concentrates specific proteins in the nucleus, thereby increasing order (decreasing entropy) in the cell, which consumes energy.

The energy is provided by the hydrolysis of GTP by the monomeric GTPase Ran.Ran is found in both the cytosol and the nucleus, and it is required for nuclear import & export.Ran is a molecular switch that can exist in 2 states, depending on whether GDP or GTP is bound



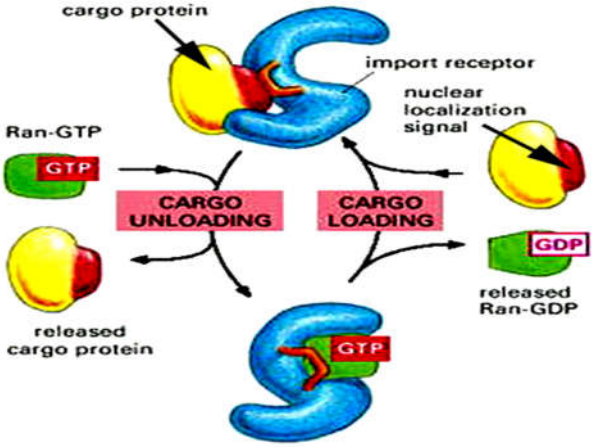
Conversion between the two states is triggered by two Ran-specific regulatory proteins: a cytosolic GTPase-activating protein (GAP) that triggers GTP hydrolysis and thus converts Ran-GTP to Ran-GDP, and a nuclear guanine exchange factor (GEF) that promotes the exchange of GDP for GTP converts Ran-GDP to Ran-GTP.Ran-GAP is located in the cytosol and Ran-GEF is located in the nucleus, the cytosol primarily contains Ran-GDP, & the nucleus Ran-GTP.

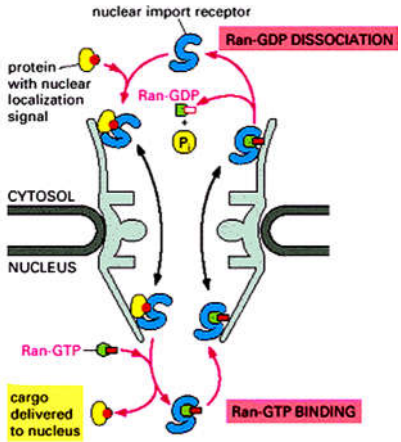
This gradient of the 2 conformational forms of Ran drives nuclear transport in appropriate direction



By favoring cargo-dependent loading of import receptors onto the FG-repeat track in the cytosol and Ran-GTP-dependent cargo release in the nucleus, the nuclear localization of Ran-GTP imposes directionality.

Having discharged its cargo in the nucleus, the empty import receptor with Ran-GTP bound is transported back through the pore complex to the cytosol.





The Ran-GDP then dissociates from the Ran Binding Protein and is reimported into the nucleus, completing the cycle.

Lecture 152

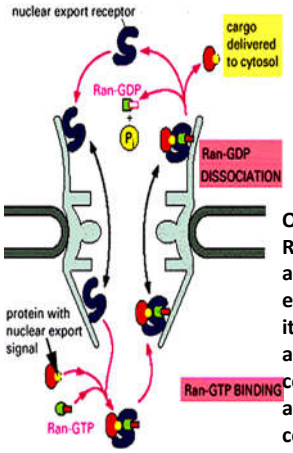
The Ran Cycle.

Docking of nuclear import receptors to FG-repeats on the cytosolic side of the nuclear pore complex, occurs only when these receptors are loaded with an appropriate cargo.

Import receptors with their bound cargo then move along tracks lined by FG-repeat sequences until they reach the nuclear side of the pore complex, where Ran-GTP binding causes the import receptors to release their cargo .

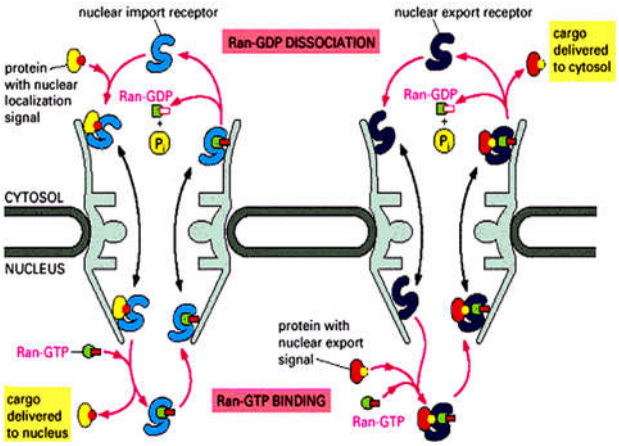
There, two cytosolic proteins, Ran Binding Protein and Ran-GAP collaborate to convert Ran-GTP to Ran-GDP.

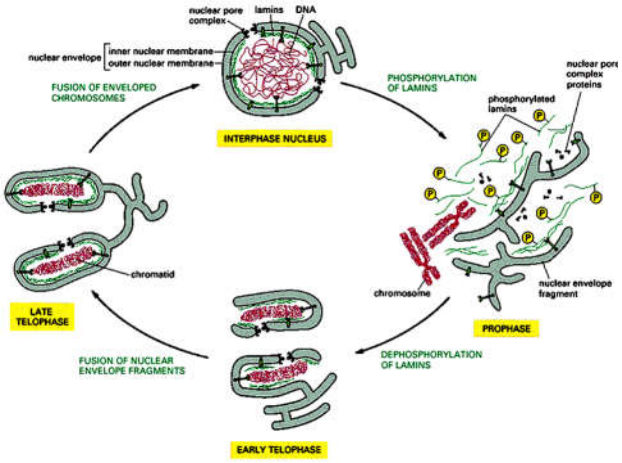
The Ran Binding Protein first displaces Ran-GTP from the import receptor, which allows Ran-GAP to trigger Ran to hydrolyze its bound GTP.



Nuclear export occurs by a similar mechanism, except that Ran-GTP in the nucleus promotes cargo binding to the export receptor and the binding of the loaded receptor to the nuclear side of the pore complex.

Once in the cytosol, Ran encounters Ran-GAP and Ran Binding Protein and hydrolyses its bound GTP. The export receptor then releases both its cargo and Ran-GDP in the cytosol and dissociates from the pore complex, and free export receptors are returned to the nucleus to complete the cycle .





Lecture 153

Control Of Import And Export.

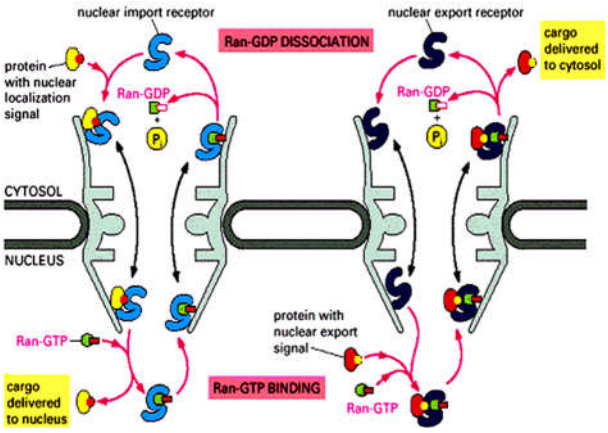
The activity of some gene regulatory proteins is controlled by keeping them out of the nuclear compartment until they are needed there.

In many cases, this control depends on the regulation of nuclear localization and export signals; these can be turned on or off, often by phosphorylation of adjacent amino acids.

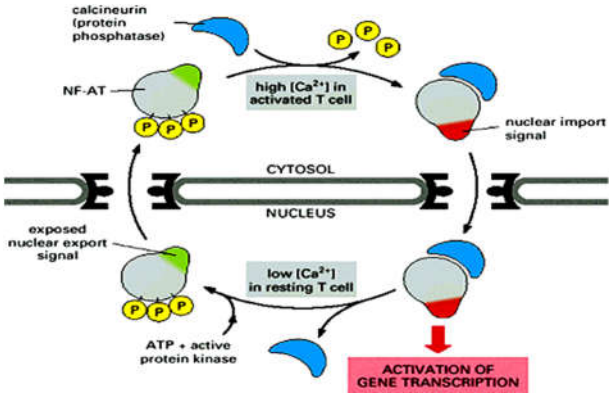
Gene regulatory proteins are bound to inhibitory cytosolic proteins that either anchor them in the cytosol or mask their NLSs so that they are unable to interact with nuclear import receptors.

When the cell receives a signal the gene regulatory protein is released & transported into the nucleus .

The transport is stringently controlled.



END



The activity of some gene regulatory proteins is controlled by keeping them out of the nucleus

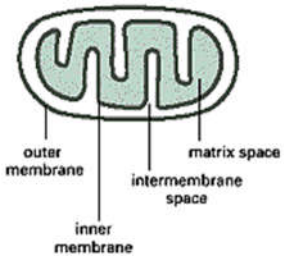
Lecture 154

Protein Transport into Mitochondria:

Mitochondria is double-membrane-enclosed organelle.

It contains its own DNA, ribosomes, and other components required for protein synthesis, most of its proteins are encoded in the cell nucleus and imported from the cytosol

There are two subcompartments in mitochondria: the internal matrix space and the intermembrane space



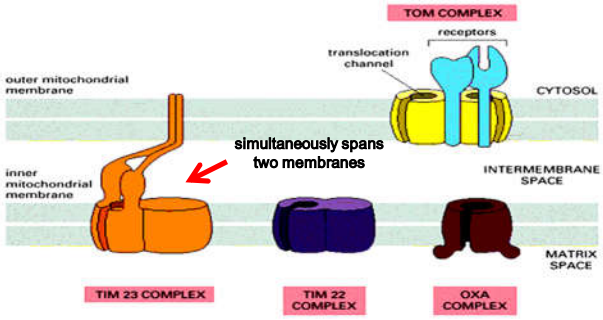
Proteins imported into the matrix of mitochondria are usually taken up from the cytosol within seconds or minutes of their release from ribosomes.

Mitochondrial proteins are first fully synthesized as precursor proteins in the cytosol & then translocated into mitochondria by a posttranslational mechanism .
Mitochondrial precursor proteins have a signal sequence at their N terminus that is rapidly removed after import by a protease (the signal peptidase) in the mitochondrial matrix .
The signal sequences are both necessary and sufficient for import of the proteins that contain them.
genetic engineering signals linked to cytosolic protein to direct the protein into the mitochondrial matrix
Matrix signal sequences have propensity to fold into an amphipathic α helix, in which positively charged residues are clustered on one side of the helix, while uncharged hydrophobic residues are clustered on the opposite side.
This configuration—rather than a precise amino acid sequence—is recognized by specific receptor proteins that initiate protein translocation.

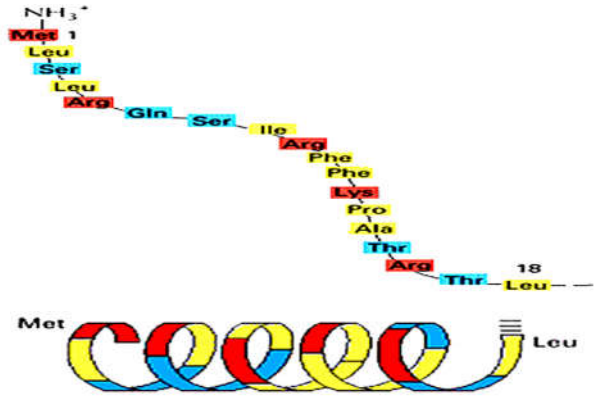
Lecture 155

The Translocators:

Multimeric membrane proteins, catalyze transport across mitochondrial membranes .



A signal sequence for mitochondrial protein import



TOM and TIM complexes functions are coupled to allow protein transport across both membranes simultaneously, both translocators can work independently.

TIM22 complex mediates the insertion inner membrane proteins.
OXA complex, mediates insertion of inner membrane proteins synthesized within mitochondria.

The N-terminal signal sequence of precursor protein is recognized by receptors of the TOM complex.

The protein is translocated across both mitochondrial membranes at or near contact sites .

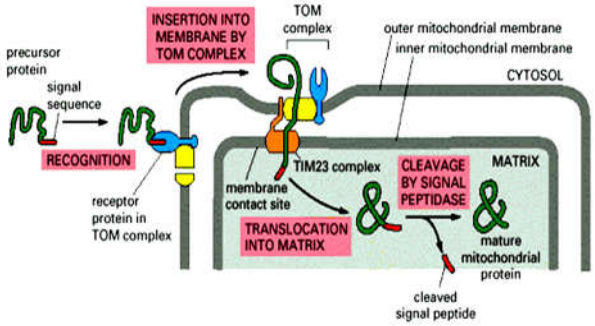
When the signal sequence is folded as an α helix, the positively charged residues (red) are seen clustered on one face of helix, while the nonpolar residues (yellow) are clustered on opposite face.

amphipathic α helix, is recognized by receptor proteins on mitochondrial surface.

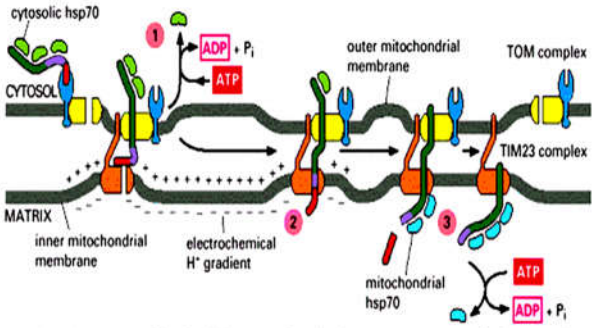
Mitochondrial precursor proteins do not fold into their native structures after they are synthesized; instead, they remain unfolded through interactions with other proteins in the cytosol.

chaperone proteins belonging to the hsp70 family

END

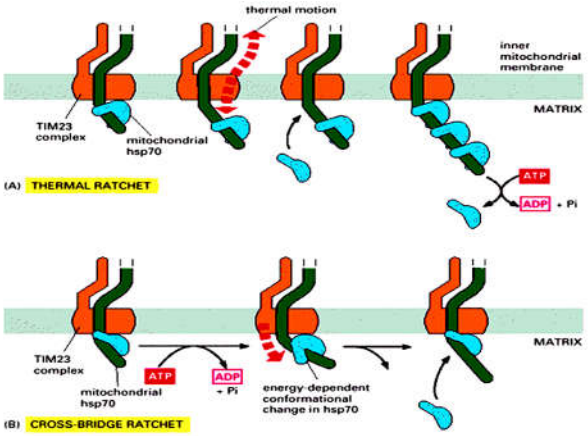


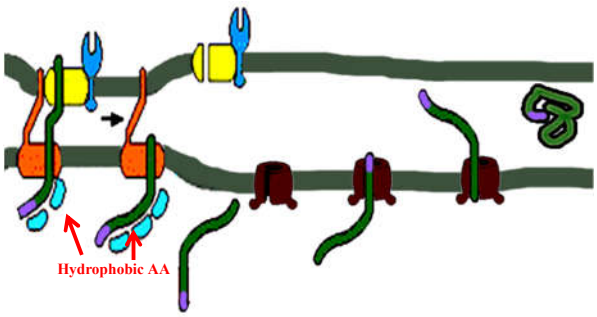
The N-terminal signal sequence of precursor protein is recognized by receptors of the TOM complex
The protein is translocated across both mitochondrial membranes at or near contact sites.



Protein import is fueled by ATP hydrolysis at 2 sites, (1) outside mitochondria & (2) in the matrix + (3) H^+ gradient across the inner mitochondrial membrane.

Electrochemical --- Unfolded proteins with SS can be translocated by TOM without energy





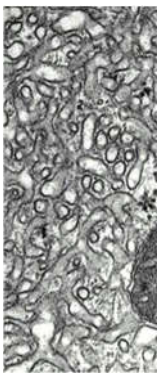
Protein Transport into the Intermembrane Space Requires Two Signal Sequences.

OXA complex is also used to insert proteins encoded in the mitochondrion into the inner membrane.

Lecture 156

ER Transport Introduction:

Smooth ER



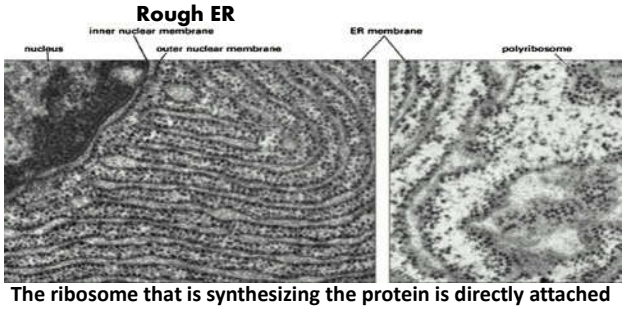
Regions of ER that lack bound ribosomes are called smooth endoplasmic reticulum, or smooth ER in the great majority of cells, such regions are scanty and are often partly smooth and partly rough.

Smooth ER contains ER exit sites from which transport vesicles carrying newly synthesized proteins and lipids bud off for transport to the Golgi apparatus

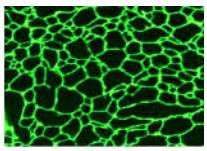
In hepatocytes smooth ER doubles in surface area during detoxification of both lipid-soluble drugs and various harmful compounds produced by metabolism e.g. phenobarbital

ER in most eucaryotic cells stores $Ca2^+$ e.g. Muscle cells, have an abundant specialized smooth ER, "sarcoplasmic reticulum", which sequesters Ca2+ from the cytosol.

Rough ER



The ribosome that is synthesizing the protein is directly attached to the ER membrane creating regions termed rough endoplasmic reticulum.



All eucaryotic cells have an endoplasmic reticulum (ER).

which is organized into a netlike branching tubules and flattened sacs throughout the cytosol.

The tubules and sacs are interconnected the ER membrane encloses a single internal space. This highly convoluted space is called the ER lumen or the ER cisternal space.

The ER membrane separates the ER lumen from the cytosol, & it mediates the selective transfer of molecules b/w these two compartments.

The ER captures selected proteins from the cytosol as they are being synthesized .

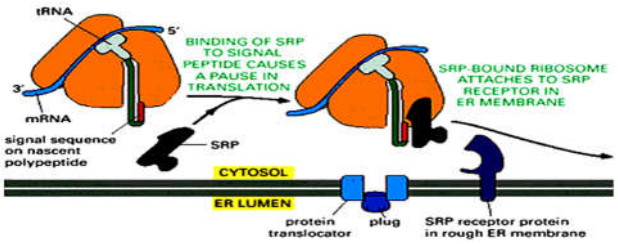
Two types: 1)transmembrane proteins, partly translocated across the ER membrane and become embedded in it,

2)water-soluble proteins, which are fully translocated across the ER membrane and are released into the ER lumen.

Some of the transmembrane proteins function in the ER, but many are destined to reside in the plasma membrane or the membrane of another organelle, water-soluble proteins are destined either for the lumen of an organelle or for secretion.

All of these proteins, have same kind of signal sequence & are translocated by similar mechanisms
ER protein import is a co-translational process the import of proteins into mitochondria, chloroplasts, nuclei, and peroxisomes, is posttranslational processes.
One end of the protein is usually translocated into the ER as the rest of the polypeptide chain is being made, the protein is never released into the cytosol and therefore is never in danger of folding up before reaching the translocator in the ER membrane.

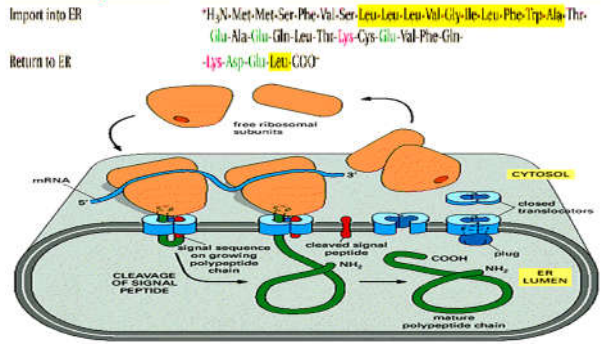
The SRP binds to the ER signal sequence as soon as the peptide has emerged from the ribosome causes a pause in protein synthesis, ensuring that the protein is not released into the cytosol.
This is safety mechanisim especially for lysosomal hydrolases that could wreak havoc in the cytosol.



the pause presumably gives the ribosome enough time to bind to the ER

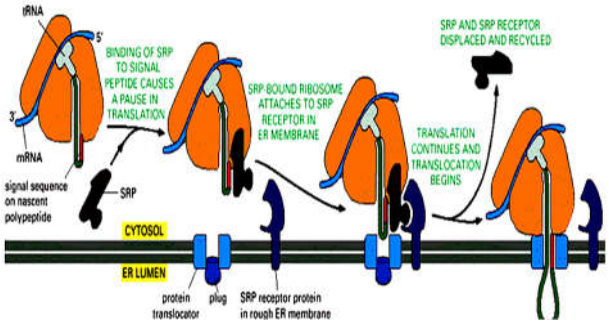
Lecture 157

ER Transport:

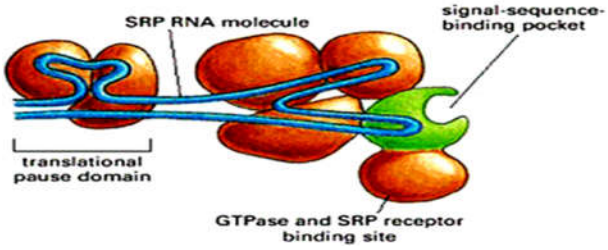


Signal Sequences Were First Discovered in Proteins Imported into the Rough ER

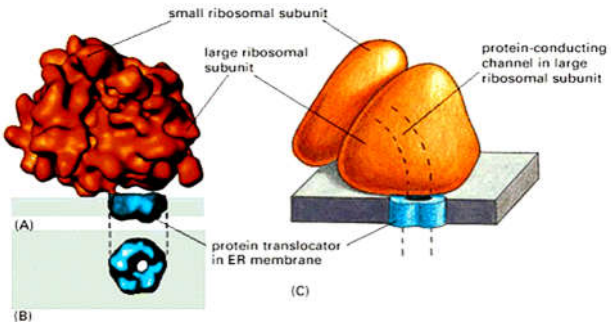
SRP-ribosome complex binds to the SRP receptor, this interaction brings the SRP-ribosome complex to a protein translocator SRP & SRP receptor are then released, and the growing polypeptide chain is transferred across the membrane.



The ER signal sequence is guided to the ER by 2 components:
1) SRP which cycles between the ER membrane and the cytosol & binds to the signal sequence, &
2) SRP receptor in the ER membrane
The SRP consisting of six different polypeptide chains bound to a single small RNA molecule

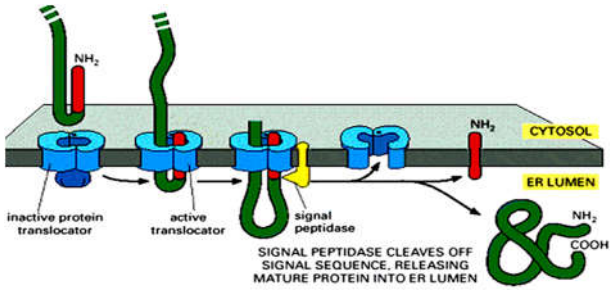


The translocator Sec61 complex forms a water-filled pore in the membrane through which the polypeptide chain traverses the membrane, it consists of 4 protein which assemble into a donut like structure.



When a ribosome binds, the central hole in the translocator lines up with a tunnel in the large ribosomal subunit through which the growing polypeptide chain exits from the ribosome .
The bound ribosome forms a tight seal with the translocator, such that the space inside the ribosome is continuous with the lumen of the ER and no molecules can escape from the ER.
The pore in the translocator cannot be open permanently, however; if it were, Ca2+ would leak out of the ER when the ribosome detaches.

A lumenal ER protein serves as a plug or that the translocator itself can rearrange to close the pore when no ribosome is bound Thus, the pore is a dynamic structure that opens only transiently when a ribosome with a growing polypeptide chain attaches to the ER membrane.



Once the C-terminus of the protein has passed through the membrane, the translocated protein is released into the ER lumen the signal sequence is released from the pore and rapidly degraded to amino acids by other proteases in the ER.
signal sequences contact with Sec61 &hydrophobic lipid core of the membrane .

The signal sequence in the growing polypeptide chain triggers the opening of the pore: after the signal sequence is released from the SRP and the growing chain has reached a sufficient length, the signal sequence binds to a specific site inside the pore itself, thereby opening the pore.

An ER signal sequence is therefore recognized twice: first, by an SRP in the cytosol, and then by a binding site in the ER protein translocator.
This may help to ensure that only appropriate proteins enter the lumen of the ER.
Some proteins, however, are imported into the ER after their synthesis has been completed, demonstrating that translocation does not always require ongoing translation.

An N-terminal signal sequence initiates translocation, just as for a soluble protein, but an additional hydrophobic segment in the polypeptide chain stops the transfer process before the entire polypeptide chain is translocated.

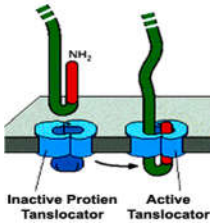
This stop-transfer signal anchors the protein in the membrane after the ER signal sequence (the start-transfer signal) has been released from the translocator and has been cleaved off .

The stop-transfer sequence is transferred into the bilayer by the lateral gating mechanism, and it remains there as a single α-helical membrane-spanning segment.

Lecture 158

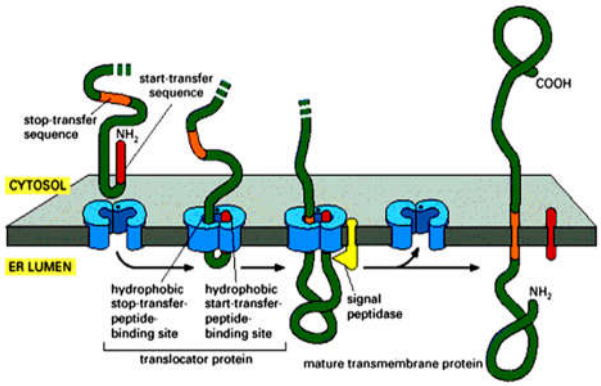
Translocation

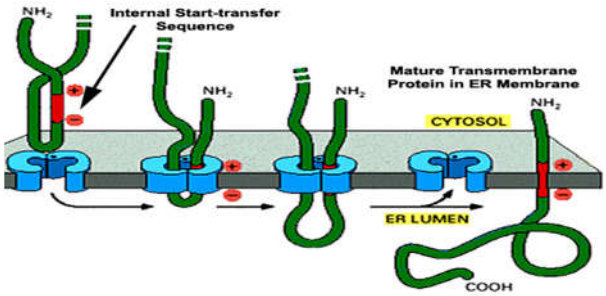
The N-terminal ER signal sequence of a soluble protein has two signaling functions 1) It directs the protein to the ER membrane 2) It serves as a start-transfer signal (or start-transfer peptide) that opens the pore.



Even after it is cleaved off by signal peptidase, the signal sequence remains bound to the translocator while the rest of the protein is threaded continuously through the membrane as a large loop.

N-terminus lumenal side & C-terminus on the cytosolic side.





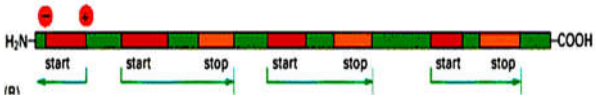
If there are more +ly charged AAs immediately preceding the hydrophobic core of start-transfer sequence (STS) than there are following it, the STS is inserted in the orientation shown here.

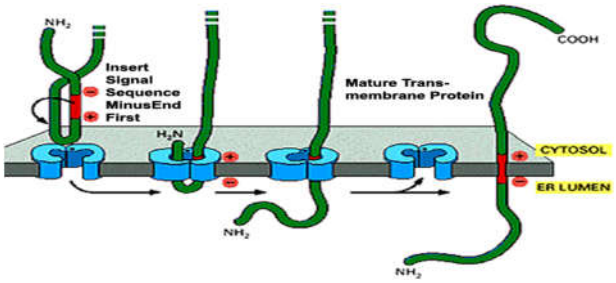
C-terminal to the start-transfer sequence will therefore be passed across the membrane



Whether a given hydrophobic signal sequence functions as a start-transfer or stop-transfer sequence depends on its location in a polypeptide chain, since its function can be switched by changing its location in the protein using recombinant DNA techniques.

The distinction between start-transfer and stop-transfer sequences results mostly from their relative order in the growing polypeptide chain.

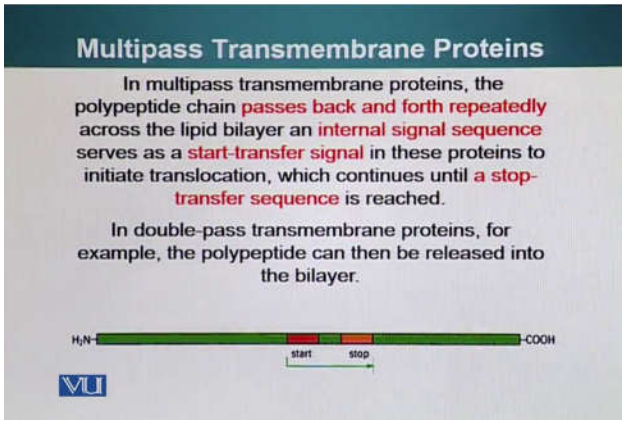




If there are more +ly charged amino acids immediately following the hydrophobic core of the STS than there are preceding it, the start-transfer sequence is inserted into the translocator in the orientation shown here

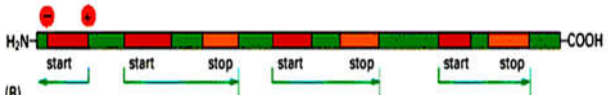
N-terminal to the start-transfer sequence will therefore be passed across the membrane

Lecture 159

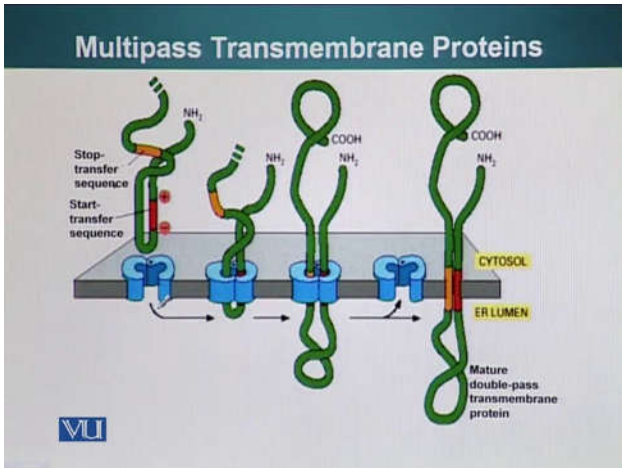


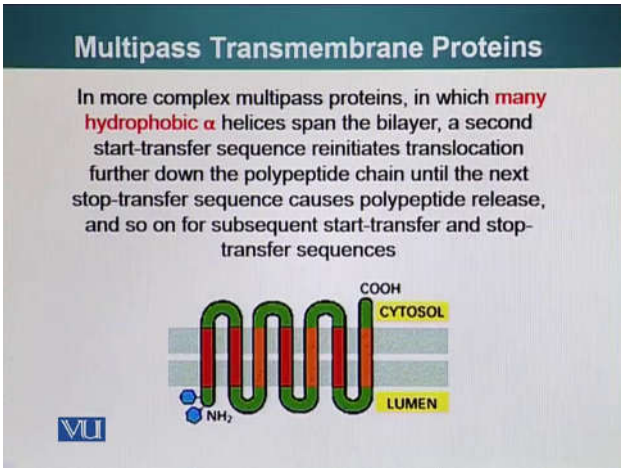
In multipass transmembrane proteins, the polypeptide chain passes back and forth repeatedly across the lipid bilayer an internal signal sequence serves as a start-transfer signal in these proteins to initiate translocation, which continues until a stop-transfer sequence is reached.

In double-pass transmembrane proteins, for example, the polypeptide can then be released into the bilayer.



In more complex multipass proteins, in which many hydrophobic α helices span the bilayer, a second start-transfer sequence reinitiates translocation further down the polypeptide chain until the next stop-transfer sequence causes polypeptide release, and so on for subsequent start-transfer and stop-transfer sequences.





Lecture 160

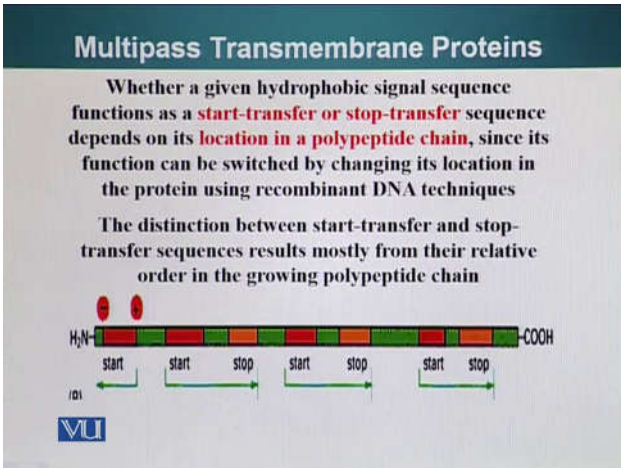
Vesicle Transport.

The Golgi complex is a major site of carbohydrate synthesis, as well as a sorting and dispatching station for the products of the ER.

Many of the cell's polysaccharides are made in the Golgi, including the pectin and hemicellulose of the cell wall in plants and most of the glycosamino-glycans of the extracellular matrix in animals.

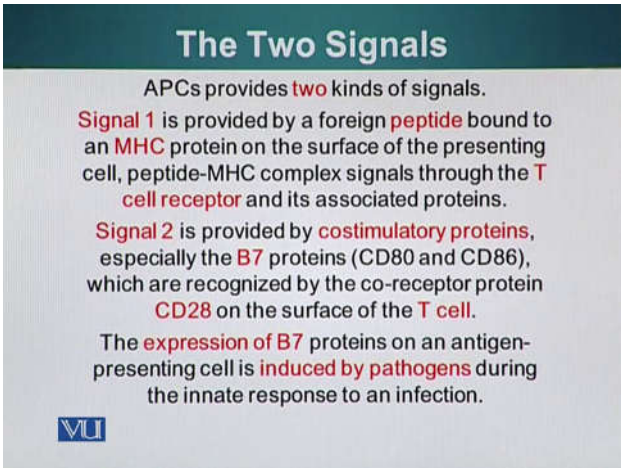
Golgi apparatus also lies on the exit route from the ER, and a large proportion of the carbohydrates that it makes are attached as oligosaccharide side chains to the many proteins & lipids that ER sends.

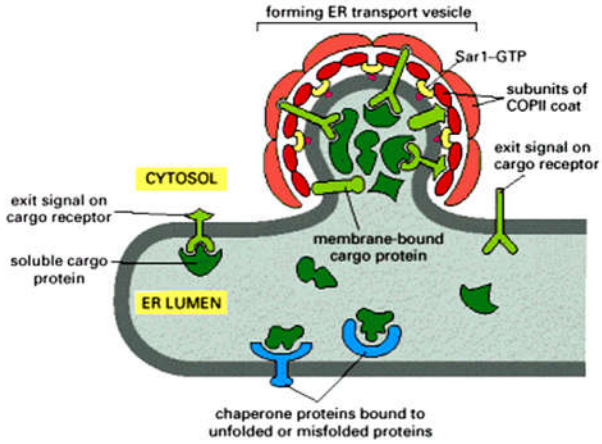
A subset of these oligosaccharide groups serve as tags to direct specific proteins into vesicles that then transport them to lysosomes



Most proteins and lipids, after acquiring their appropriate oligosaccharides in the Golgi, are recognized in other ways for targeting into the transport vesicles going to other destinations. Proteins that enter the ER and are destined for the Golgi apparatus are packaged into small COPII-coated transport vesicles which, bud from specialized regions of the ER called ER exit sites whose membrane lacks bound ribosomes.

Some cargo proteins are actively recruited into such vesicles, where they become concentrated, these cargo proteins display exit (transport) signals on their surface that are recognized by complementary receptor proteins that become trapped in the budding vesicle by interacting with components of the COPII coat Cargo proteins display exit (transport) signals on their surface that are recognized by complementary receptor proteins that become trapped in the budding vesicle by interacting with components of the COPII coat, some ER resident proteins without such exit signals can also get packaged in vesicles .





To exit from the ER, proteins must be properly folded and completely assembled.

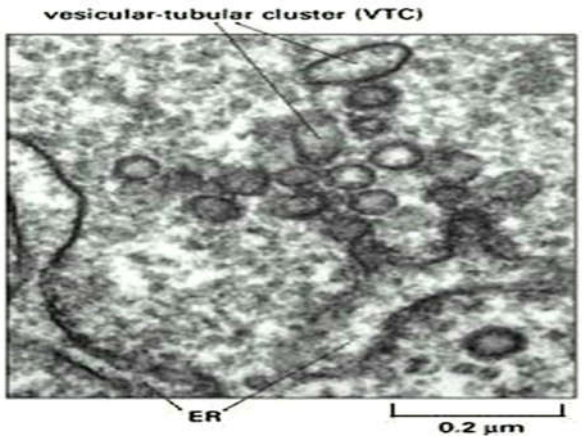
Misfolded or incompletely assembled are retained in the ER, where they are bound to chaperone proteins.

The ERGIC53 protein serves as a receptor for packaging some secretory proteins into COPII-coated vesicles.

Its role in protein transport was identified because humans who lack it owing to an inherited mutation have lowered serum levels of two secreted blood-clotting factors (Factor V and Factor VIII) and therefore bleed excessively.

The ERGIC53 protein is a lectin that binds mannose on Factor V and Factor VIII proteins, thereby packaging the proteins into transport vesicles in the ER.

END

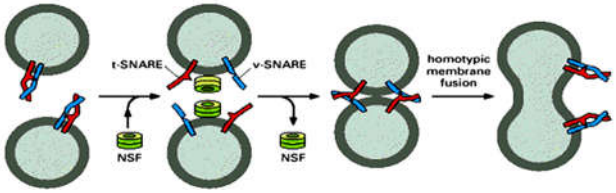


Lecture 161

COPI.

After transport vesicles have budded from an ER exit site they shed their coat, and then they begin to fuse with one another.

This fusion of membranes from the same compartment is called homotypic fusion, to distinguish it from heterotypic fusion, in which a membrane from one compartment fuses with the membrane of a different compartment.

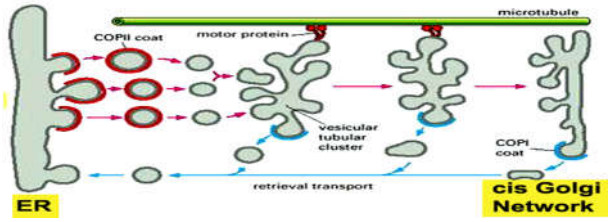


As with heterotypic fusion, homotypic fusion requires a set of matching SNAREs.

The interaction is symmetrical, with v-SNAREs and t-SNAREs contributed by both membranes .

The structures formed when ER-derived vesicles fuse with one another are called vesicular tubular clusters .

These clusters constitute a new compartment that is separate from the ER and lacks many of the proteins that function in the ER.



Lecture 162

As soon as vesicular tubular clusters form, they begin budding off vesicles of their own.

Unlike the COPII-coated vesicles that bud from the ER, these vesicles are COPI-coated.

They carry back the ER resident proteins that have escaped, as well as proteins that participated in the ER budding reaction and are being returned.

This retrieval process demonstrates the exquisite control mechanisms that regulate coat assembly.

The COPI coat assembly begins only seconds after the COPII coats have been shed.

The clusters continuously mature, gradually changing their composition as selected proteins are returned to the ER a similar retrieval process continues from the Golgi apparatus, after the vesicular tubular clusters have delivered their cargo.

Resident ER membrane proteins, for example, contain signals that bind directly to COPI coats and are thus packaged into COPI-coated transport vesicles for retrograde delivery to the ER.

Signal of this type consists of two lysines, followed by any two other amino acids, at the extreme C-terminal end of the ER membrane protein.

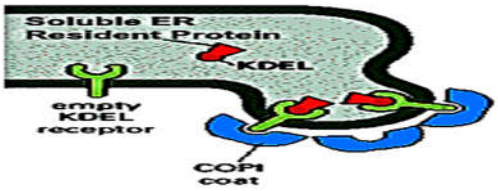
It is called a KKXX sequence, based on the AA code.

Soluble ER resident proteins, such as BiP, also contain a short retrieval signal at their C-terminal end, but it is different: it consists of a Lys-Asp-Glu-Leu or similar sequence.

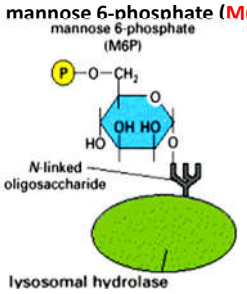
If this signal (called the KDEL sequence) is removed from BiP by genetic engineering, the protein is slowly secreted from the cell.

If he signal is transferred to a protein that is normally secreted, the protein is now efficiently returned to the ER, where it accumulates

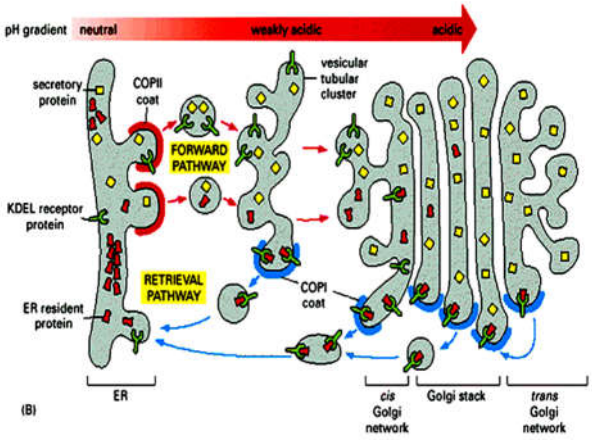
ER resident proteins must bind to specialized receptor proteins such as the KDEL receptor—a multipass transmembrane protein that binds to the KDEL sequence and packages any protein displaying it into COPI-coated retrograde transport vesicles.
ER resident proteins bind to one another, thus forming complexes that are too big to enter transport vesicles
Aggregation of proteins that function in the same compartment—called kin recognition—is a general mechanism that compartments use to organize and retain their resident proteins.

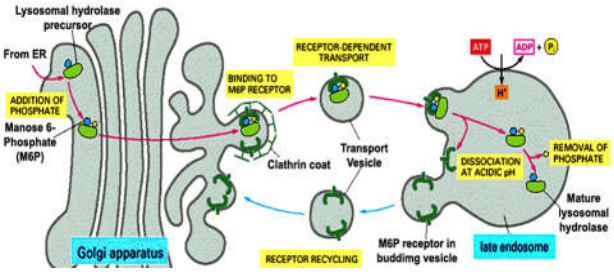


These vesicles incorporate the lysosomal proteins & exclude other proteins being packaged into different transport vesicles for delivery elsewhere.Lysosomal proteins are recognized and selected in the trans Golgi network with the required accuracy.
Lysosomal hydrolases carry a unique marker in the form of mannose 6-phosphate (M6P) groups,.



M6P is added to N-linked oligosaccharides of the soluble lysosomal enzymes in the lumen of the C-Golgi.The M6P groups are recognized by trans-membrane M6P receptor proteins, which are present in the T-Golgi
Receptors bind to lysosomal hydrolases on the lumenal side of the membrane and to adaptins to assemble clathrin coats on the cytosolic side



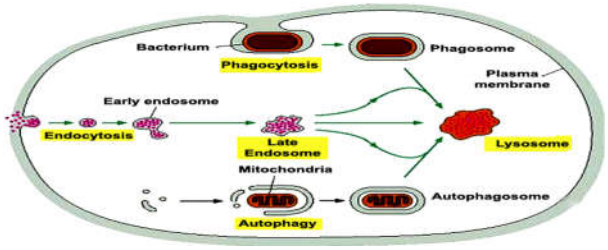


Receptors help package the hydrolases into clathrin-coated vesicles that bud from the T-Golgi network.
Vesicles contents are subsequently delivered to Lysosomes.
M6P receptor binds its specific oligosaccharide at pH 6.5 in the T-Golgi network & releases it at pH 6, late endosomes M6P receptors are retrieved into transport vesicles from late endosomes the receptors are returned to the T-Golgi for reuse.

Lecture 163

Lysosomal Transport.

Lysosomes are membrane-enclosed compartments filled with hydrolytic enzymes that are used for the controlled intracellular digestion of macromolecules.
They contain 40 types of hydrolytic enzymes, including proteases, nucleases, glycosidases, lipases, phospholipases, phosphatases, and sulfatases they are all are acid hydrolases.
For optimal activity they require an acid environment, and the lysosome provides this by maintaining a pH of about 5.0 in its interior.The contents of the cytosol are doubly protected against attack by the cell's own digestive system.
Lysosomal hydrolases and membrane proteins are synthesized in the rough ER and transported through the Golgi apparatus to the lysosome.The transport vesicles that deliver these proteins to lysosomes bud from the trans Golgi network



In this disease almost all of the hydrolytic enzymes are missing from the lysosomes of fibroblasts, and their undigested substrates accumulate in lysosomes, which consequently form large "inclusions" in the cells .
In I-cell disease patients, all the hydrolases missing from lysosomes are found in the blood. Because they fail to be sorted properly in the Golgi apparatus, the hydrolases are secreted rather than transported to lysosomes.

Lecture 164

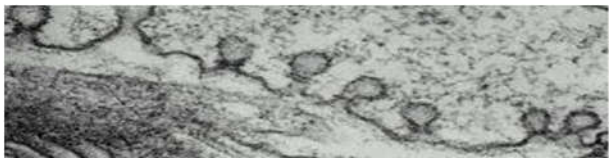
Endocytosis.

The material to be ingested is progressively enclosed by a small portion of the plasma membrane, which first invaginates and then pinches off to form an endocytic vesicle containing the ingested substance or particle.

Phagocytosis involves the ingestion of large particles microorganisms or dead cells via large vesicles called phagosomes. (>250 nm in diameter)

Pinocytosis involves ingestion of fluid & solutes via small pinocytic vesicles .(about 100 nm in diameter)

Most eucaryotic cells are continually ingesting fluid and solutes by pinocytosis; large particles are most efficiently ingested by specialized phagocytic cells.



In mammals, three classes of white blood cells act as professional phagocytes macrophages, neutrophils, and dendritic cells.

These cells develop from hemopoietic stem cells and they defend us against infection by ingesting invading microorganisms.

Macrophages also scavenge senescent cells and cells that have died by apoptosis.

In quantitative terms, the latter function is by far the most important: our macrophages phagocytose more than 10^{11} senescent red blood cells every day.

Particles first bind to the surface of the phagocyte.

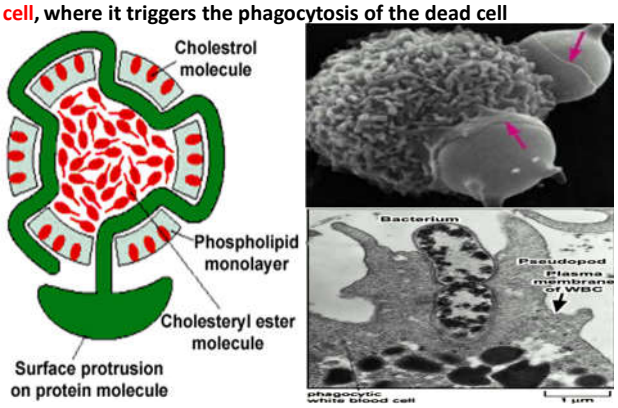
Phagocytes have a variety of specialized surface receptors that are functionally linked to the phagocytic machinery of the cell.

Phagocytosis is a triggered process, requiring that receptors be activated that signal the cell interior and initiate the response.

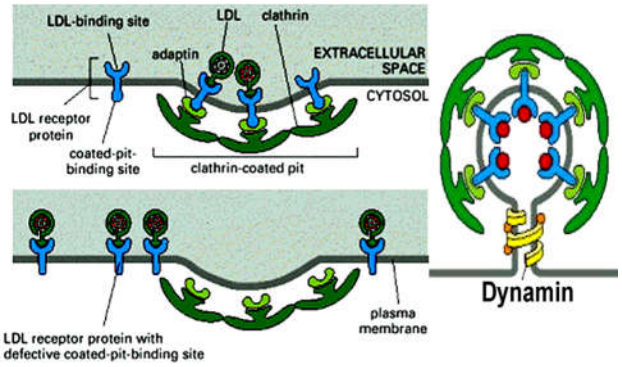
Best-characterized triggers are antibodies, which protect us by binding to the surface of infectious microorganisms to form a coat in which the tail region of each antibody molecule, called the Fc region, is exposed on the exterior.

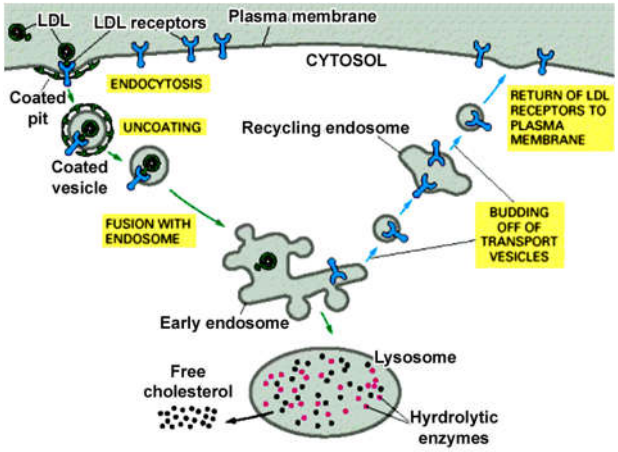
This antibody coat is recognized by specific Fc receptors on the surface of macrophages and neutrophils, whose binding induces the phagocytic cell to extend pseudopods that engulf the particle and fuse at their tips to form a phagosome.Apoptotic cells lose the asymmetric distribution of phospholipids in their plasma membrane as a consequence, negatively charged.

phosphatidylserine, which is normally confined to the cytosolic leaflet of the lipid bilayer, is now exposed on the outside of the cell, where it triggers the phagocytosis of the dead cell



Each particle contains a core of about 1500 cholesterol molecules esterified to long-chain fatty acids.





Lecture 165

Tissue Replacement :

A Natural Phenomenon

Bones regenerate every 10 years

Skin regenerates every two weeks

Macrophages scavenge senescent RBCs (10^{11}/day)

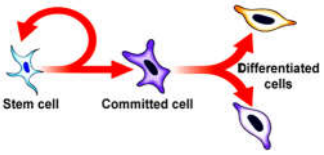
Tissue repair in response to minor injury

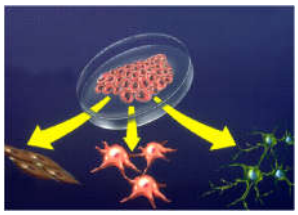
All organs have limited ability to regenerate (1 cm)

Every 30 Seconds a Person Dies from a Disease that can be Treated with
Tissue Replacement

Lecture 166

Organ Repair via Stem Cells:





1. Stem cells are unspecialized cells that renew themselves.
2. Under certain physiologic or experimental conditions, Stem cells can be induced to become cells with special functions.

Embryonic & Adult Stem Cells

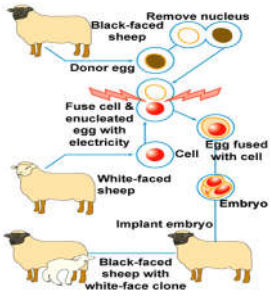
Blood Transfusion

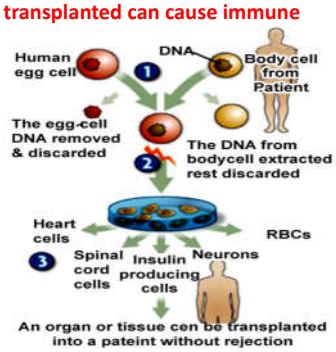


- •1492: Pope Innocent VIII
- •1901: Human blood groups discovery Austrian Karl Landsteiner (1939 Rh)
- •1925: First transfusion center Alexander Bogdanov Moscow

Embryonic Stem Cells

Are derived from the embryos (ethical dilemma) BB Option.
Have the ability to differentiate into any cell type in the body.
Have unlimited ability to renew themselves.
Being foreign in origin, when transplanted can cause immune rejection.







Solid Tissue

- •1905: Cornea transplant Eduard Zirm
- •1954: Kidney transplant Joseph Murray
- •1998: Hand transplant France
- •2005: Partial face transplant France

Organ Repair via Stem Cells

Cytokine Induce Stem Cell Proliferation & Differentiation

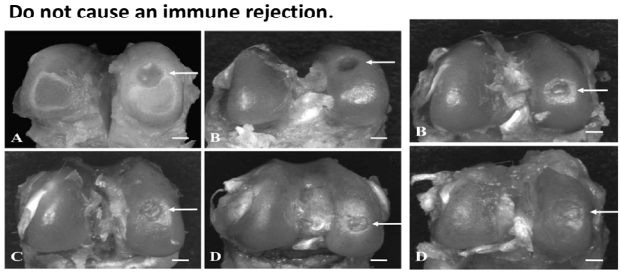
In-vitro Organ/Tissue Fabrication

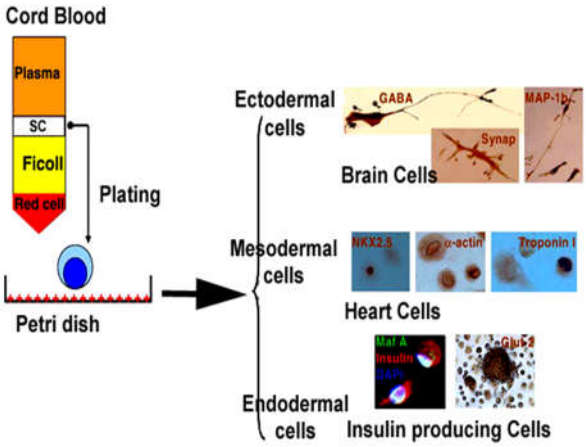
Adult Stem cells

Are harvest from there niches from a (postnatal) individual,

No ethical dilemma for their use in research,

Have a limited ability to renew themselves and differentiate in to different cell types.
Do not cause an immune rejection.

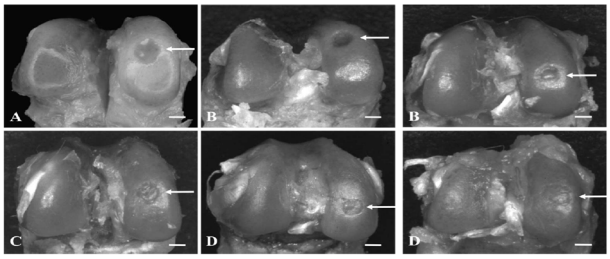




Lecture 168

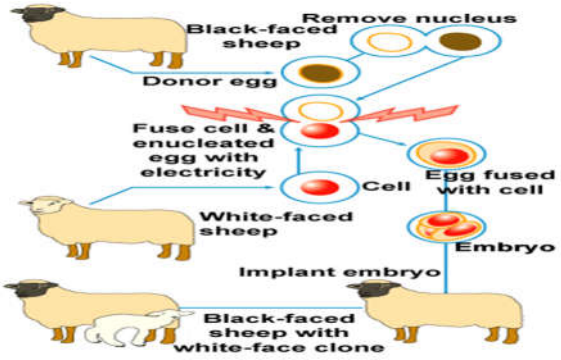
Adult Stem Cells:

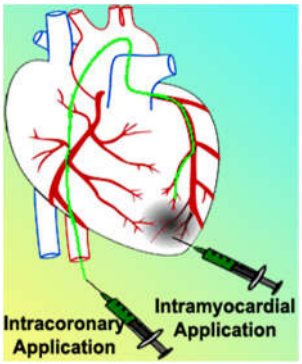
Are harvest from there niches from a (postnatal) individual,
No ethical dilemma for their use in research.
Have a limited ability to renew themselves and differentiate in to different cell types.
Do not cause an immune rejection



Lecture 167

Customized Stem Cells:

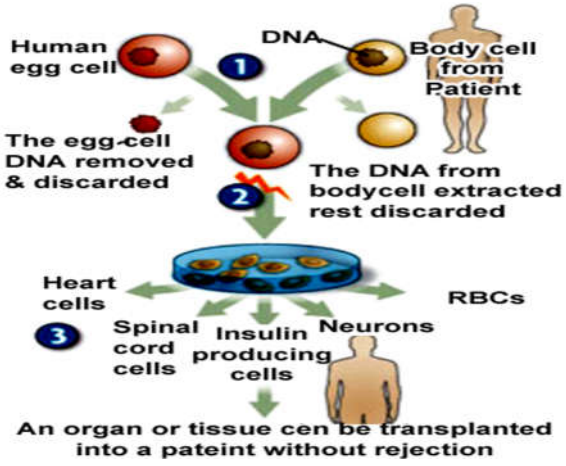


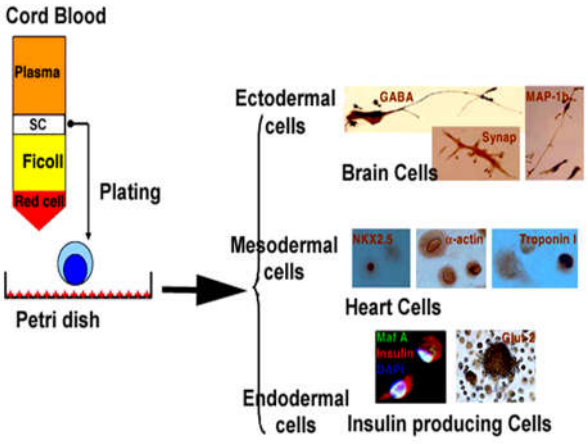


Düsseldorf group performed clinical stem cell application for the first time in 2001, treating acute myocardial infarction by intracoronary cell transfer.

Currently, the optimal time point for cardiac stem cell application in a ischemic setting seems to lay within the first month after myocardial infarction.

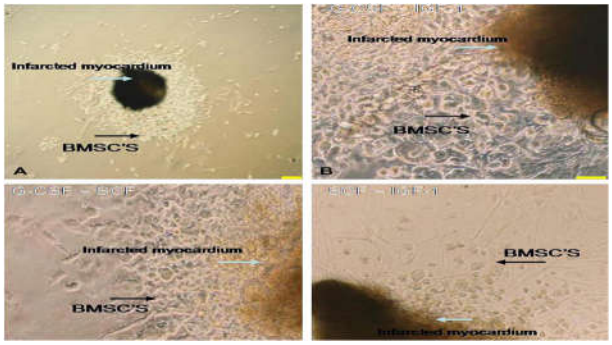
"There is growing evidence that bone marrow-derived stem/progenitor cells improve cardiac function" (Leri 2009).



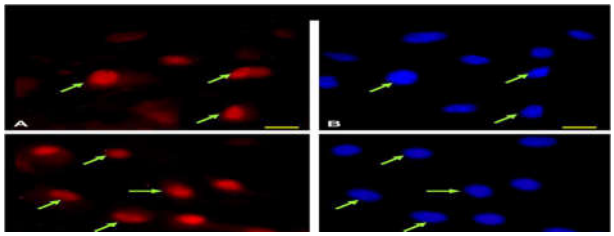


Lecture 169

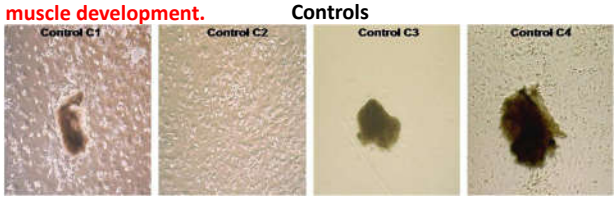
Cytokine Induced Stem Cell Proliferation & Differentiation:

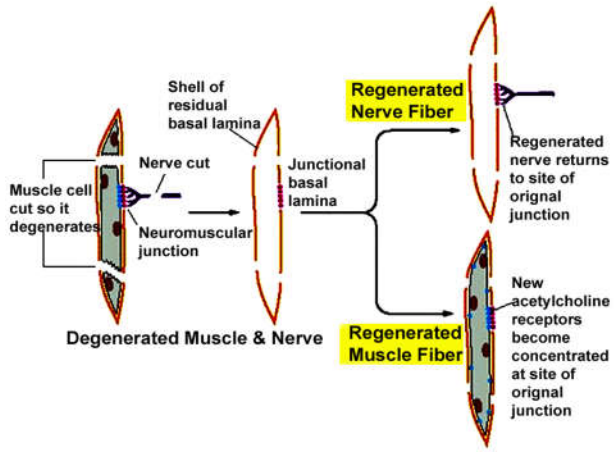


IGF-1 & G-CSF → BMSC migration infarcted myocardium



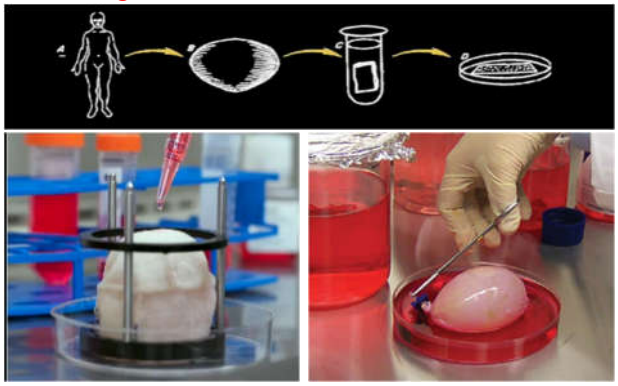
Myocyte enhancer factor 2 transcription factors regulates muscle development.

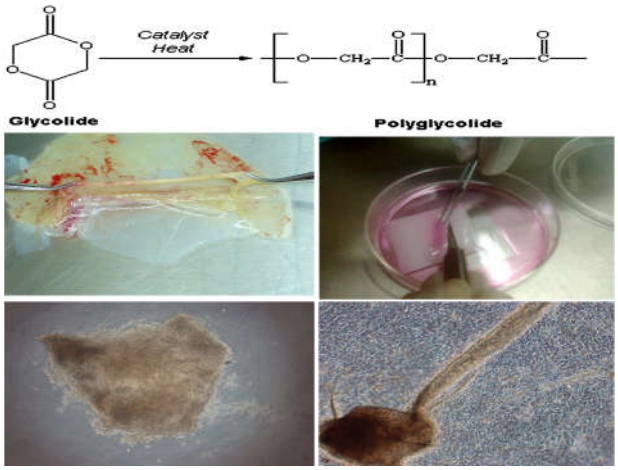


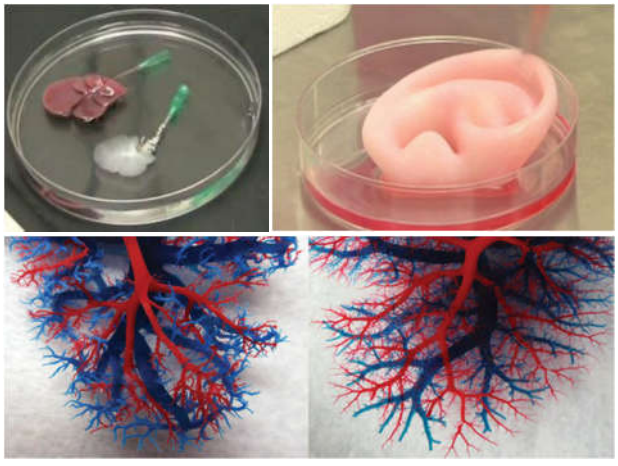


Lecture 170

In-vitro Organ/Tissue Fabrication:

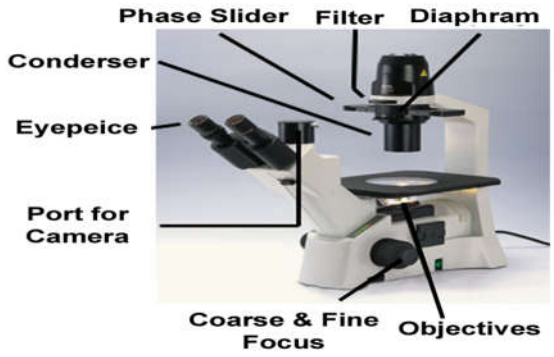






In 1924, Bogdanov started blood transfusion experiments, apparently hoping to achieve eternal youth or at least partial rejuvenation. Lenin's sister Maria Ulianova was among many who volunteered to take part in Bogdanov's experiments. After undergoing 11 blood transfusions, he remarked with satisfaction on the improvement of his eyesight, suspension of balding, & other positive symptoms. The fellow revolutionary Leonid Krasin wrote to his wife that "Bogdanov seems to have become 7, no, 10 years younger after the operation". In 1925-1926, Bogdanov founded the Institute for Haemotology and Blood Transfusions, which was later named after him. But a later transfusion cost him life, when he took the blood of a student suffering from malaria & tuberculosis. (Bogdanov died, but the student injected with his blood made a complete recovery.) Some scholars (e.g. Loren Graham) have speculated that his death may have been suicide, because Bogdanov wrote a highly nervous political letter shortly beforehand. Others, however, attribute his death to blood type incompatibility, which was poorly understood at the time. In May 2014 the media reported on a blood transfusion that contains a protein called GDF11 to possibly cure Alzheimer's disease and heart disease.

Microscope



Separating cells from the media washing

Lecture 171 & 172

Rules & Cell Culturing Equipment:

The Golden Rules:

Laboratory coats should be worn and fastened at all times when working with chemicals or biological agents.

Wear Safety Glasses when working with liquids or other material that could get into your eyes.

Footwear that properly protects the foot from chemical splashes should be worn at all times in the laboratory.

Working in laboratory is fun way to learn but first and foremost is your safety to ensure follow instructions

- **Special gloves should be worn when handling chemicals or samples.**
- **Do not touch door and/or fridge handles etc. while wearing protective gloves.**
- **Do not pipette by mouth.**
- **Do not eat, drink or smoke anywhere in the laboratory space.**

Centrifuge

Rotates sample around a fix axis simulating high gravity causing denser substances to sink to the bottom of the sample tube.



Water bath

Maintains a fix temperature.
Used to thaw samples.
Incubate samples for enzymatic digestion.



Separating cells from the media washing

Laminar Flow Hood

Provides sterile working area where cells can be handled without worrying about contamination.



Changing media Air curtain Invisible barrier

Incubator

Maintains optimal temperature & humidity for cell survival & proliferation. Double door. CO_2 to maintain pH .

Increasing atmospheric CO2 decreases the pH of the medium and vice versa.



Lecture 173

Getting Started:

Check Equipment Settings before Usage:

- **Check the incubator Settings before you place your cells most incubators are set at 37°C .**
- **100% humid environment.**
- **Cells in Bicarbonate media should be place in CO2 incubators& 5-10% CO_2**
- **CO2 + H2O yields H+ HCO3- buffer system.**
- **Bicarbonate Maintains PH of the media.**
- **Some cells are grown in Bicarbonate free media.**

Sterilize Reagents and Equipment.

(i) UV, (ii) 70% ethanol, (iii) Bake, (iv) Autoclave, (v) Flame & (vi) Filter

Cross linking: permeability: Heat: Heat & pressure: Filter: oxidation not heat

Filter Types

- Filters come in two pore sizes 0.2 um and 0.45 um.
- 0.2 um filter to remove bacteria from the tissue culture media.
- Smaller particles i.e. Mycoplasma and viruses can pass through 0.2 um filter.
- 0.45 is primarily used to remove particulates from solutions.
- Use an appropriate filter size & type Cellulose, polycarbonate.

Media Should be filtered (prevent deactivation)

- Tissue culture hood should be cleaned with 70% ethanol and disinfected with UV.
- Plastic can be Autoclaved, UV treated or sprayed with 70% ethanol.
- Glass can be Autoclaved, UV treated, sprayed with 70% ethanol or Baked.
- Metal can be Autoclaved, UV treated, sprayed with 70% ethanol or Baked.

Why do we need to culture cells in vitro?

- To study or manipulate cells in controlled environment.

Clinical use of cell culture:

- Fertility treatment, (IVF and embryo transplant)
- Chromosomal studies, amniocentesis, CVS, Blood bone Marrow Tissue grafts, (stem cell)
- Cytotoxicity testing of therapeutic drugs tissue/organ engineering etc.

Use of cell culture in Research:

- To study cell structure and function.
- To study behaviour of cells (developmental stages, response).
- Generation of monoclonal antibodies.
- Generation of recombinant proteins.
- Culturing viruses for research.

- Flame the mouth of bottles when after and before closing.

- Sterilize the Tissue culture hood (UV + Ethanol)
- All manipulations must be carried out in a sterile cabinet
- Turn the UV light off (Should ordinarily be on)
- Bring materials into the hood
- Light up the flame or gas
- Begin your work
- Flame all caps and lids

Types of cell culture:

Primary cells are cells straight from the tissue with no passages. A primary cell culture may be composed of mixtures of cell types.

Cell lines have at least one passage. With each subsequent subculture. There are several means of obtaining immortalized cell lines. Some cells such as tumor cells, are inherently immortal.

Pure cultures: Use of selective media to suppress growth of unwanted cells and promoting growth of desired cells.

Explant Culture:

A piece of a tissue is placed in an appropriately coated dish containing media.

Cells typically move out o f the tissue and start to proliferate in the dish.

This is useful for sensitive tissue, which may be prone to damage by proteases or other enzymes.

Lecture 174

Mammalian Cell Culture:

What is cell culture?

- In vitro cultivation of cells from different tissues is referred to as cell culture.
- A vertebrate can have more than 100 different cell types, each cell type requires specific culturing conditions.



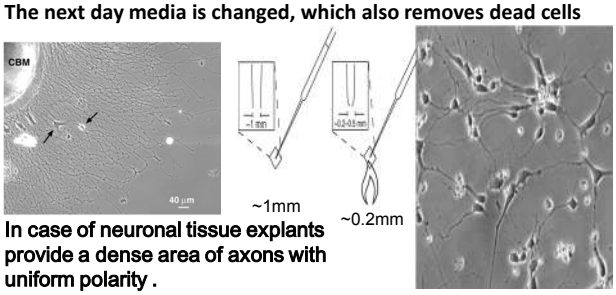
Adherent Cell Culture	Suspension Cell Culture
Appropriate for most cell types, including primary cultures	Appropriate for cells adapted to suspension culture and a few other cell lines that are nonadhesive
Requires periodic passaging, but allows easy visual inspection under inverted microscope	Easier to passage, but requires daily cell counts and viability determination to follow growth patterns;
Cells are dissociated enzymatically or mechanically	Does not require enzymatic or mechanical dissociation

Dissociated Culture:

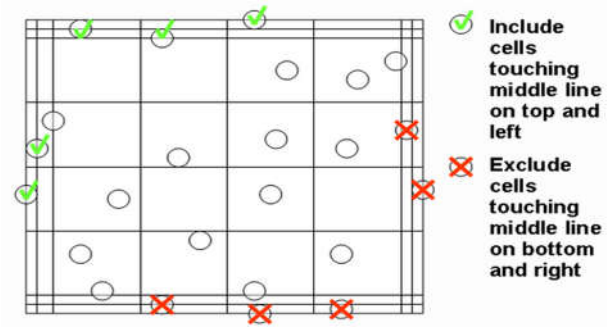
Tissue sample is dissociated, either mechanically (titration) or enzymatically, producing a cell suspension.

Cells are plated on coated tissue culture dish, cells attach to dish and start dividing.

The next day media is changed, which also removes dead cells



In case of neuronal tissue explants provide a dense area of axons with uniform polarity .



- Cells counted in a proportion of the grid squares, multiplying the total number of cells in the entire grid by 10^4 gives the number of cells per ml.

Lecture 175

Culturing Cells:

Procedure: Primary culture:

- Tissue is obtained from the animal under aseptic conditions.
- It is usually necessary to treat the tissue with enzymes e.g. trypsin collagenase.
- Serum to bock the effect of these enzymes Trituration.
- Evaluate the quality and quantity of cells.
- Plating cells on appropriately coated dish.

$C = \tilde{N} \times 10^4$ where

C = cells per ml $\tilde{N}$= average of cells counted

10^4 = volume conversion factor for 1mm2

Total Yield = C x V (V= total volume of cells (ml))

Example: -

Count1 = 182 cells/1mm2

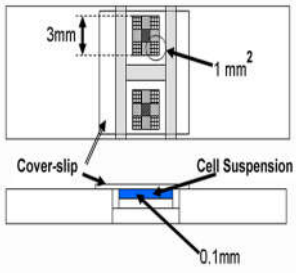
Count 2 = 175 cells/1mm2

Volume of Cells = 55 ml

Use of the Neubauer Haemocytometer

The haemocytometer is a specialised microscope slide

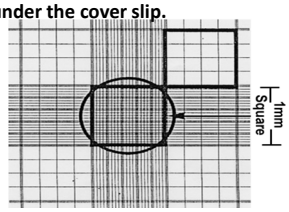
The central region is 0.1 mm lower than the rest of the slide.



Each grid comprises 25 large squares.

This creates a region of known volume (0.1 mm^3 = 0.1 x1x1= 0.1 microliter = $1x10^{-4}$ ml) when a special coverslip is placed over the central region.

Ten µl of culture are then pipetted under the cover slip.



Average cells counted =(Cnt1 + Cnt2)/2 = (182 + 175)/2 = 178.5

C = 178.5 x 10^4 = 1,785,000 cells/ml

Total yield = C x V =1,785,000 x 55 = 98,175,000 cells

- Pipet out appropriate volume of cell suspension & place it in a tissue culture dish or flask.
- Make sure the tissue culture dish or flask are coated with appropriate substrate for the cells e.g. Collagen, Fibronectin, Laminin, Poly-L-lysine etc.
- Most tissue culture dishes are pre-coated.
- Next day change media to get rid of dead cells
- Dead cells may release toxins, which may be injurious to healthy cells

Lecture 176

Sub-culturing & Cell Banking:

- Cells should be sub-cultured just before they approach confluency.
- A confluent monolayer is an adherent cell culture (dish, plate or flask) in which the cells have formed a single layer over the entire surface area available for growth.
- Once the cells have started to form clusters above the first layer or have started to lift up from the surface, the culture is past confluency stage.
- Cells that are repeatedly passaged at densities past confluency display decreased doubling times, decreased viabilities, & decreased ability to attach.

Procedure:

1. Trypsinize the cells
2. Remove cells from the container
3. Spin the cell remove trypsin
4. Block trypsin with Media + 10% serum
5. Spin the cell remove blocking media
6. Suspend cells in 5%-10% DMSO+ media+20%serum
7. Freeze cells slowly in tubes inside Styrofoam
8. Store cells in LN

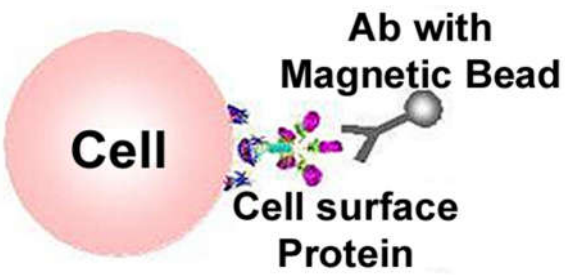
Stem Cell tissue Banking in the west advantages

- To increase the sample and keep cells healthy.
- Suspension cultures should be passaged before they reach a density of 2.0 to 2.5 x 106 cells/ml and diluted back to 0.7 to 1.0 x 106 cells/ml
- An average cell can divide 50 times before it gets unhealthy
- Cells are treated with Trypsin-EDTA
- EDTA is a Ca^{++} chelator EDTA binds Ca^{++}
- This lowers the Ca^{++} concentration in the media causes cell rounding.

Trypsin is highly active, relatively non-specific, protease, always make fresh trypsin.
Cleaves proteins on the cell surface and extracellular matrix.
This causes cells to detach from the surface .

The culture is considered to be unhealthy

Cells have surface proteins that are unique to them.



Cells are incubated with an antibody that recognizes an antigen present on the surface of cell of interest .
Antibodies are conjugated to magnetic beads

- Over trypsinization can damage cells.
- Presence of EDTA allows us to use lower concentration of Trypsin.
- Always block Trypsin with Serum to prevent it from damaging cells.

Storing cells at low temperatures:

Advantages:

1.Cells don't have to be maintained

2.Cells can be readily plated

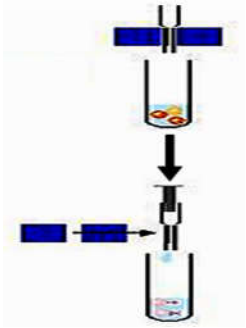
3.Cells can be transported long distances

Cryoprotectants

DMSO or Glycerol

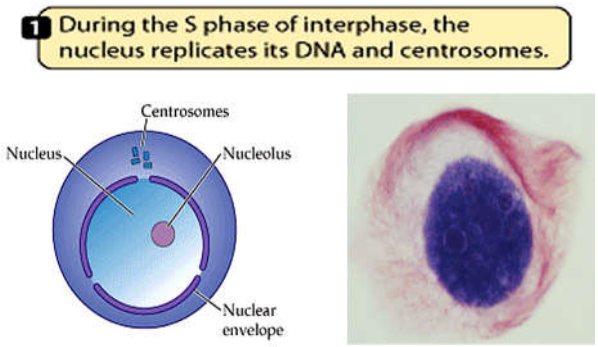
DMSO is more penetrating and is usually the agent of choice for larger, more complex cells

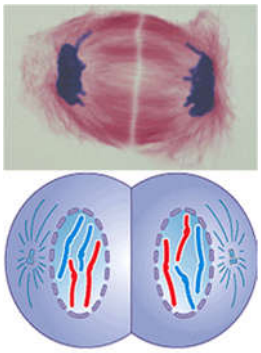
- Cells are passed through a magnetic column cells with magnetic beads are retained others cells pass through
- Magnet is remove and the column is subsequently washed
- Cells are collected from the second wash will be homogenous



Lecture 177

Identifying different stages of cell cycle:





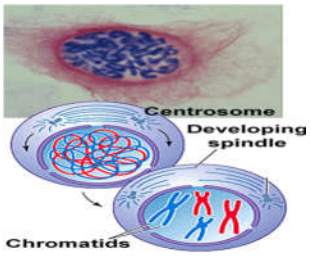
Telophase

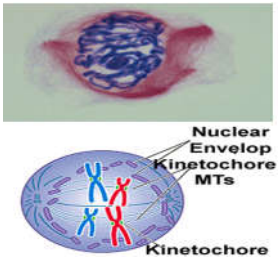
Chromosomes stop moving at the end of anaphase, the cell enters telophase.

The chromosomes begin to uncoil, the nuclear envelopes and nucleoli, reform their respective structures.

Prophase:

The beginning of mitosis. so the individual chromatids become visible.
Microtubules form b/w each pole and the chromosomes to make up a spindle.





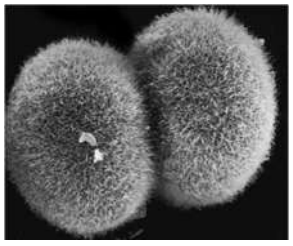
Prometaphase:

Prometaphase is marked by the disappearance of the nuclear envelope.

Chromosomes begin to move toward the poles.

Division of the Cytoplasm

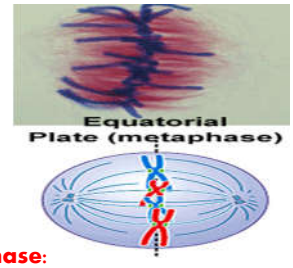
Begin before telophase ends.



Animal cells usually divide by a furrowing of the plasma membrane, as if an invisible thread were tightening between the two poles.

Metaphase:

All the centromeres arrive at the equatorial plate.
Metaphase is the best time to see the sizes and shapes of chromosomes because they are maximally condensed.





Anaphase:

Separation of the chromatids marks the beginning of anaphase, during which the two sister chromatids move to opposite ends of the spindle.
Each chromatid con-tains one double-stranded DNA molecule & is now referred to as a daughter chromosome.